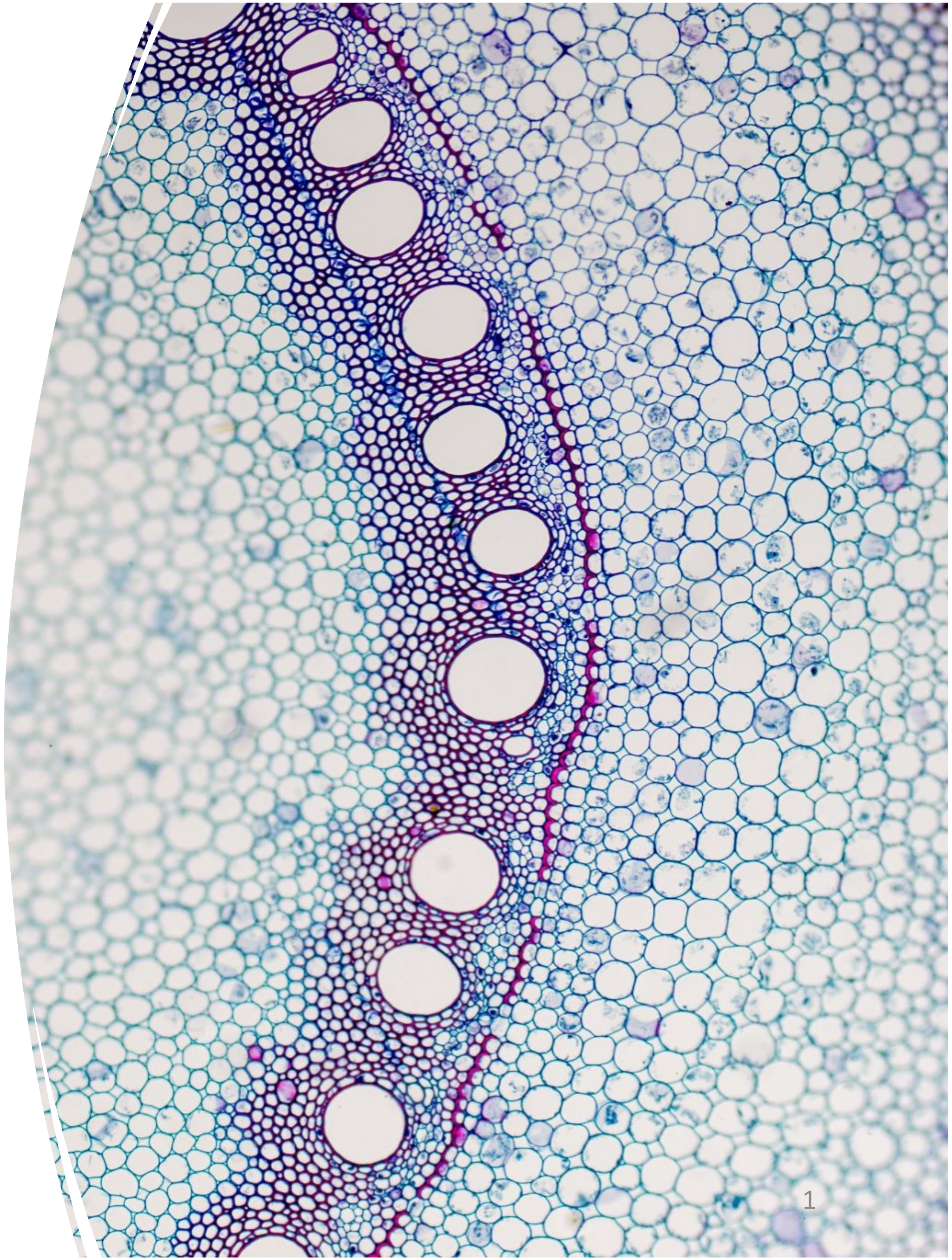
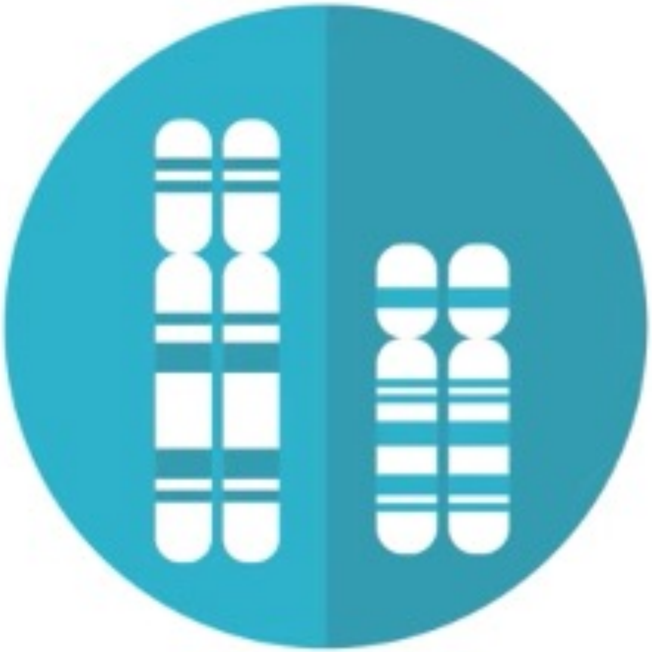
Transcriptome

Chapter18



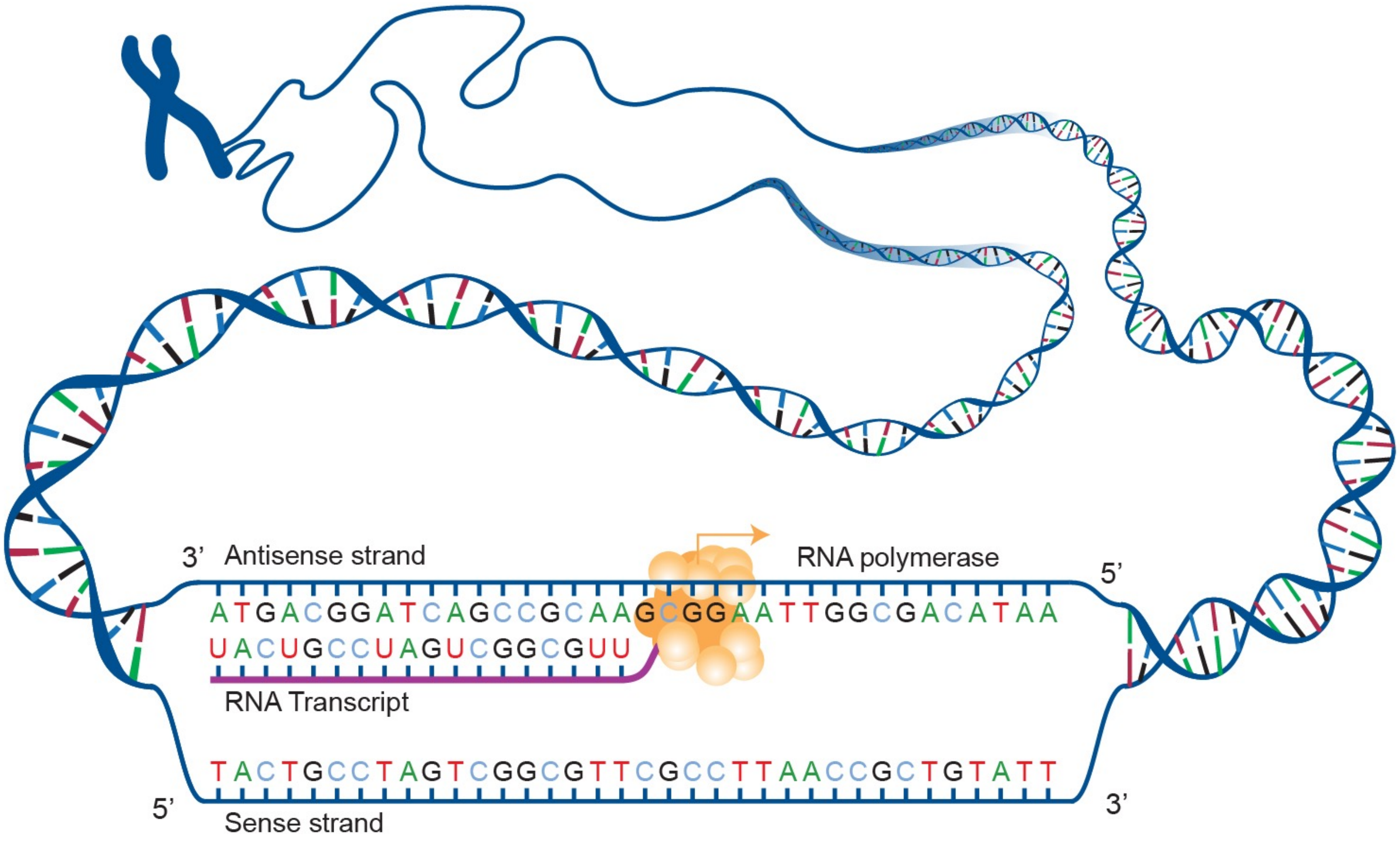
in biology we study **living organisms**

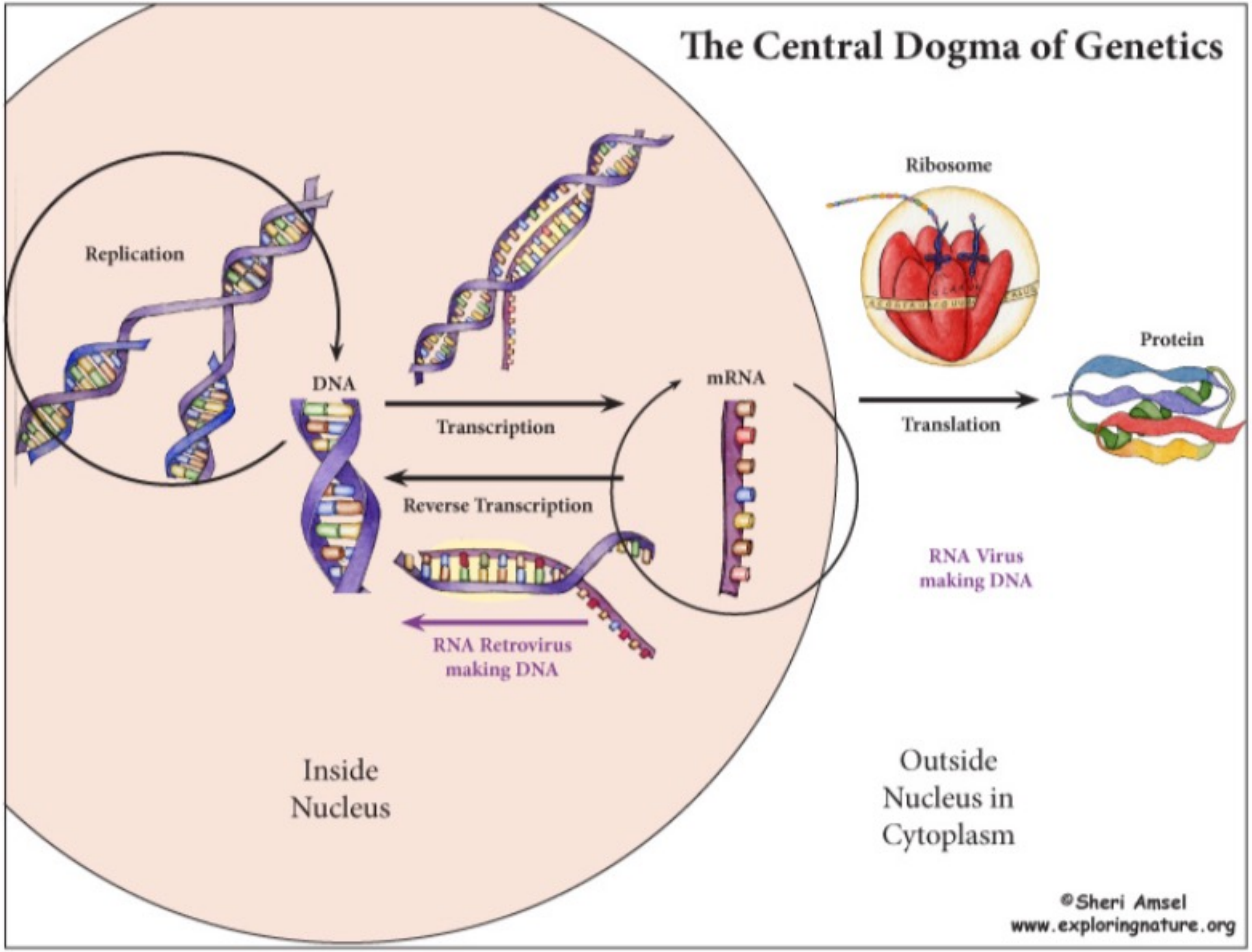


What **genes** are present?

What is happening in a particular cell?

How does a cell type perform gene expression?


3’ Antisense strand
RNA polymerase
5’
ATGACGGATCAGCCGCAAGCGGAATTGGCGACATAA
UACUGCCUAGUCGGCGUU
RNA Transcript
TACTGCCTAGTCGGCGTTCGCCTTAACCGCTGTATT
5’
3’
Sense strand


The Central Dogma of Genetics
Replication
DNA
Transcription
Reverse Transcription
mRNA
Ribosome
Translation
Protein
RNA Retrovirus
making DNA
RNA Virus
making DNA
Inside
Nucleus
Outside
Nucleus in
Cytoplasm
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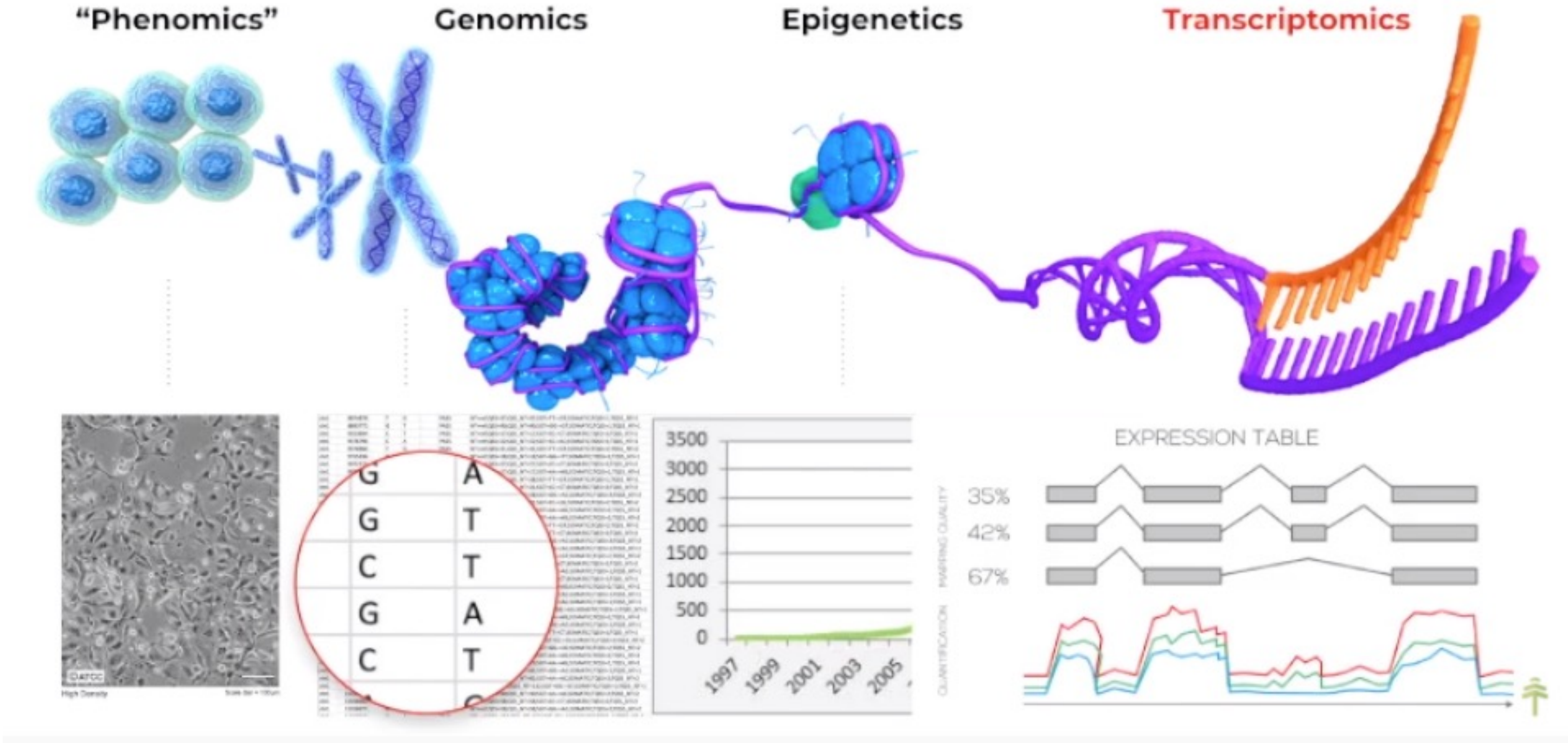
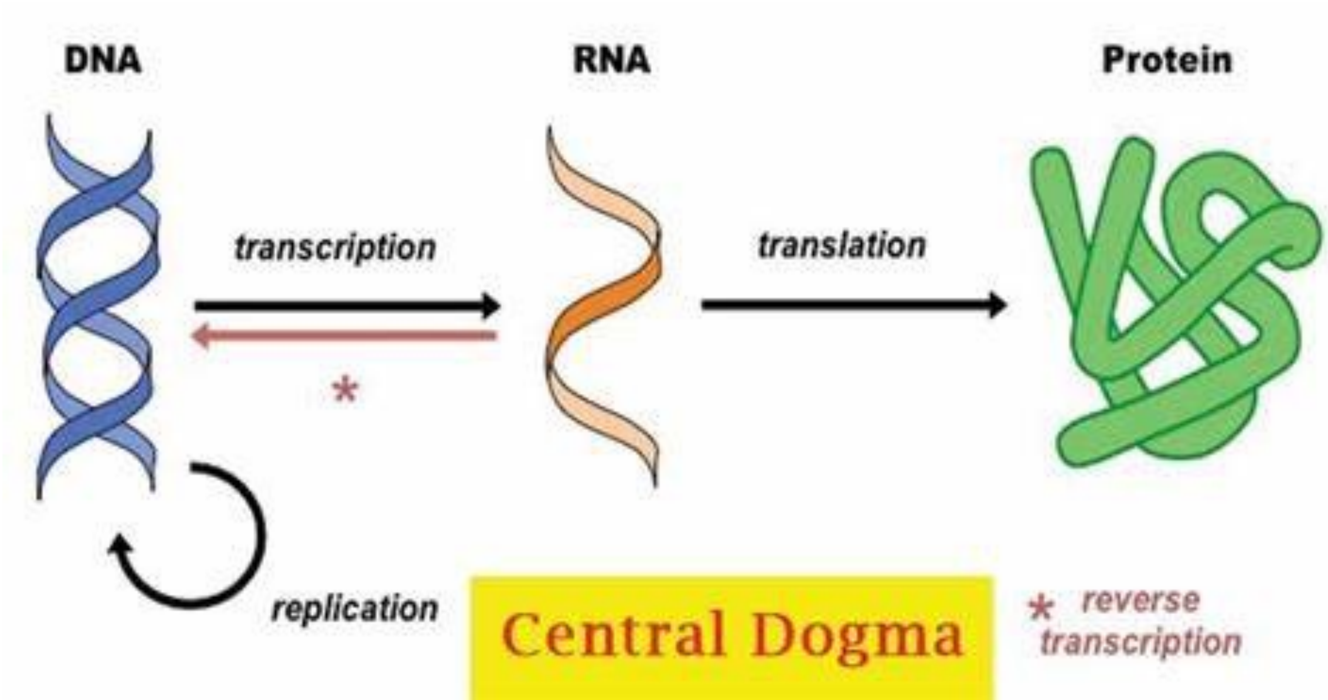
Concept

- **Transcriptome** is now widely understood to mean the complete set of all the ribonucleic acid (RNA) molecules expressed in some given entity, such as a cell, tissue, or organism.
- **Transcriptomics** encompasses everything relating to RNAs, including their transcription and expression levels, functions, locations, trafficking, and degradation. It also includes the structures of transcripts and their parent genes with regard to start sites, 5′ and 3′ end sequences, splicing patterns, and posttranscriptional modifications.
- **Transcriptomics** covers all types of transcripts, including messenger RNAs (mRNAs), microRNAs (miRNAs), and different types of long noncoding RNAs (lncRNAs).

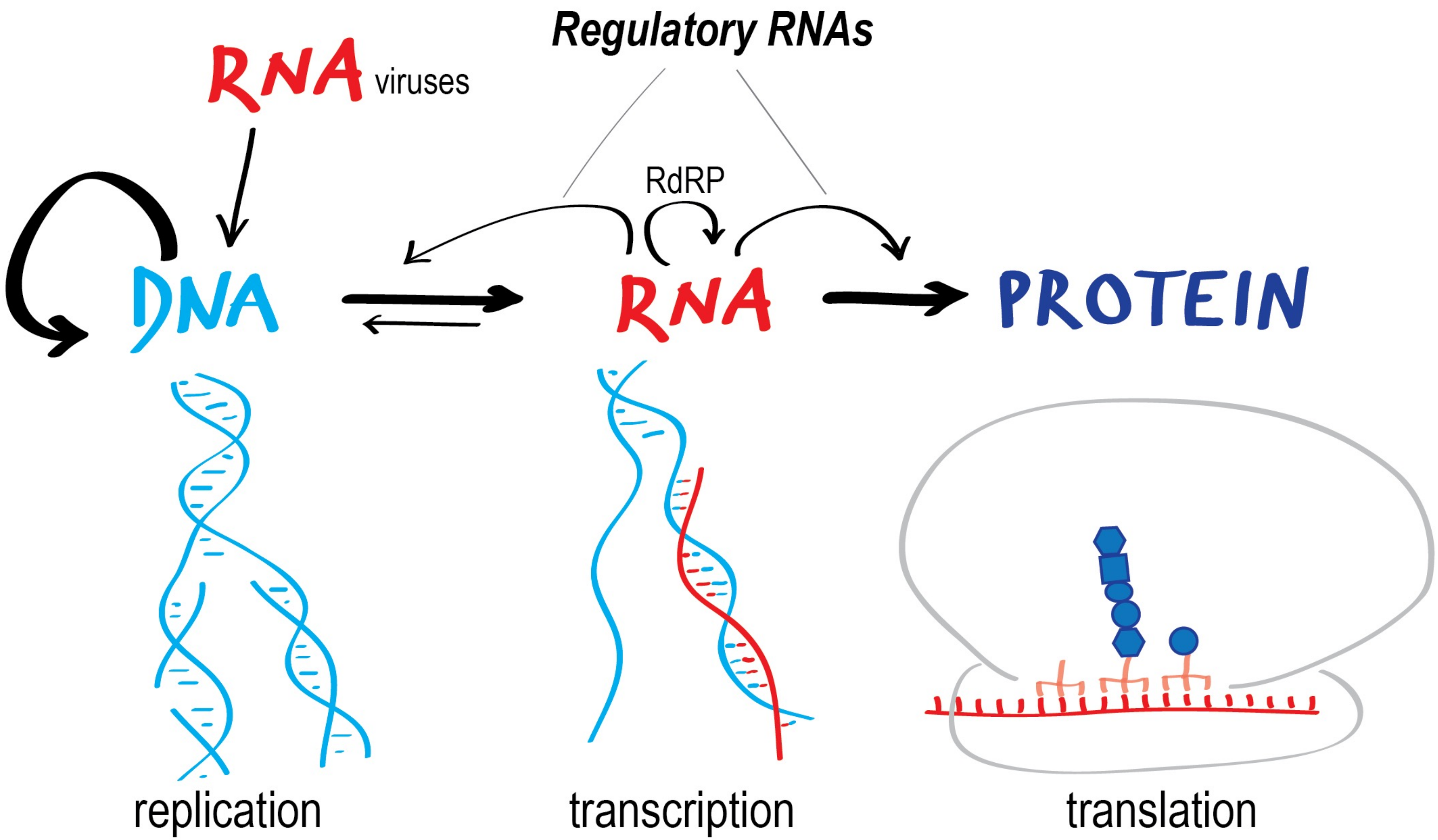
Concept

- **Modern transcriptomics** uses high-throughput methods to analyze the expression of multiple transcripts in different physiological or pathological conditions and this is rapidly expanding our understanding of the relationships between the transcriptome and the phenotype across a wide range of living entities.
- **Transcriptomic data** based on deep RNA-Seq approach can provide valuable information on differential gene and transcript expression patterns in specific cell types.


"Phenomics"
Genomics
Epigenetics
Transcriptomics
G A
G T
C T
G A
C T
3500
3000
2500
2000
1500
1000
500
0
1997
1999
2001
2003
2005
EXPRESSION TABLE
35%
42%
67%


DNA
RNA
Protein
transcription
translation
*
replication
Central Dogma
* reverse transcription

The expanded Central Dogma of Molecular Biology

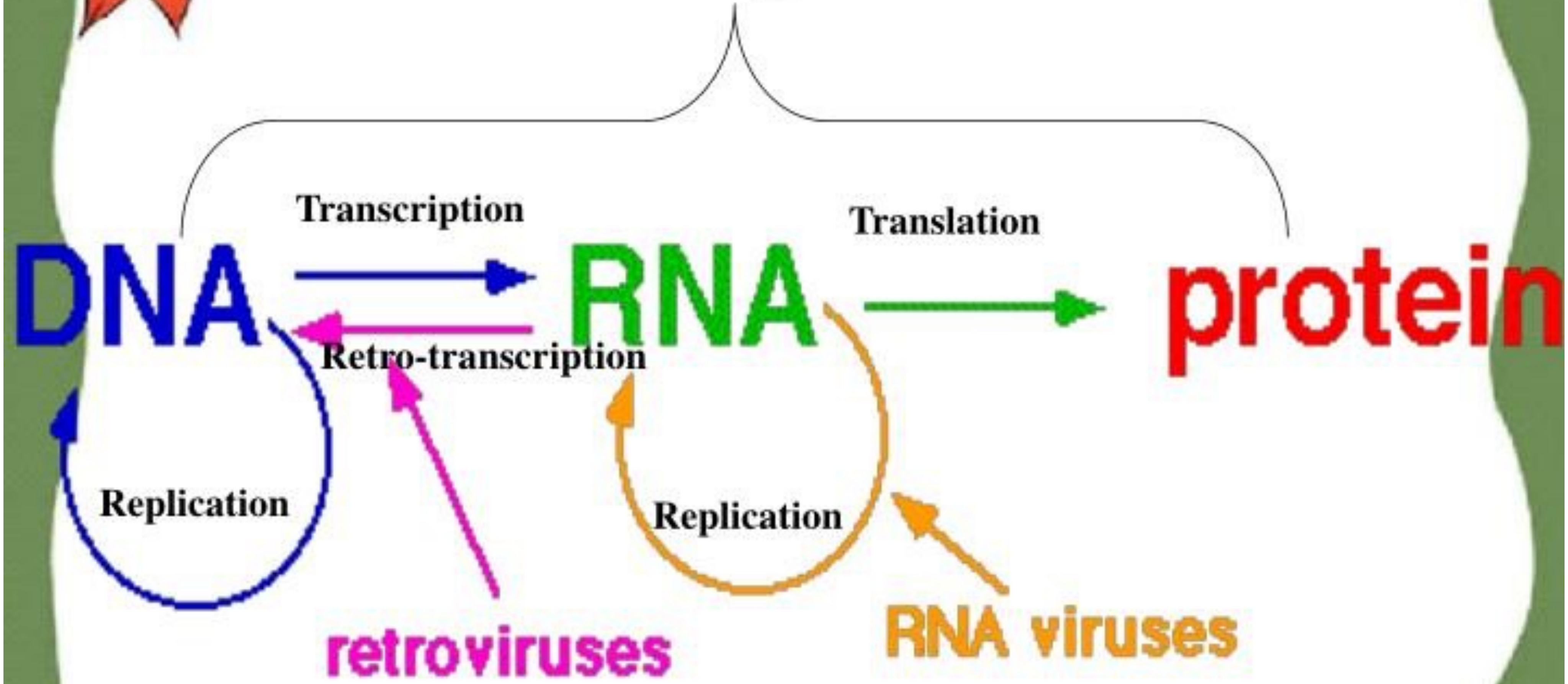


rRNA, tRNA, mRNA

Catalytic RNA (RNase P, ribosome)

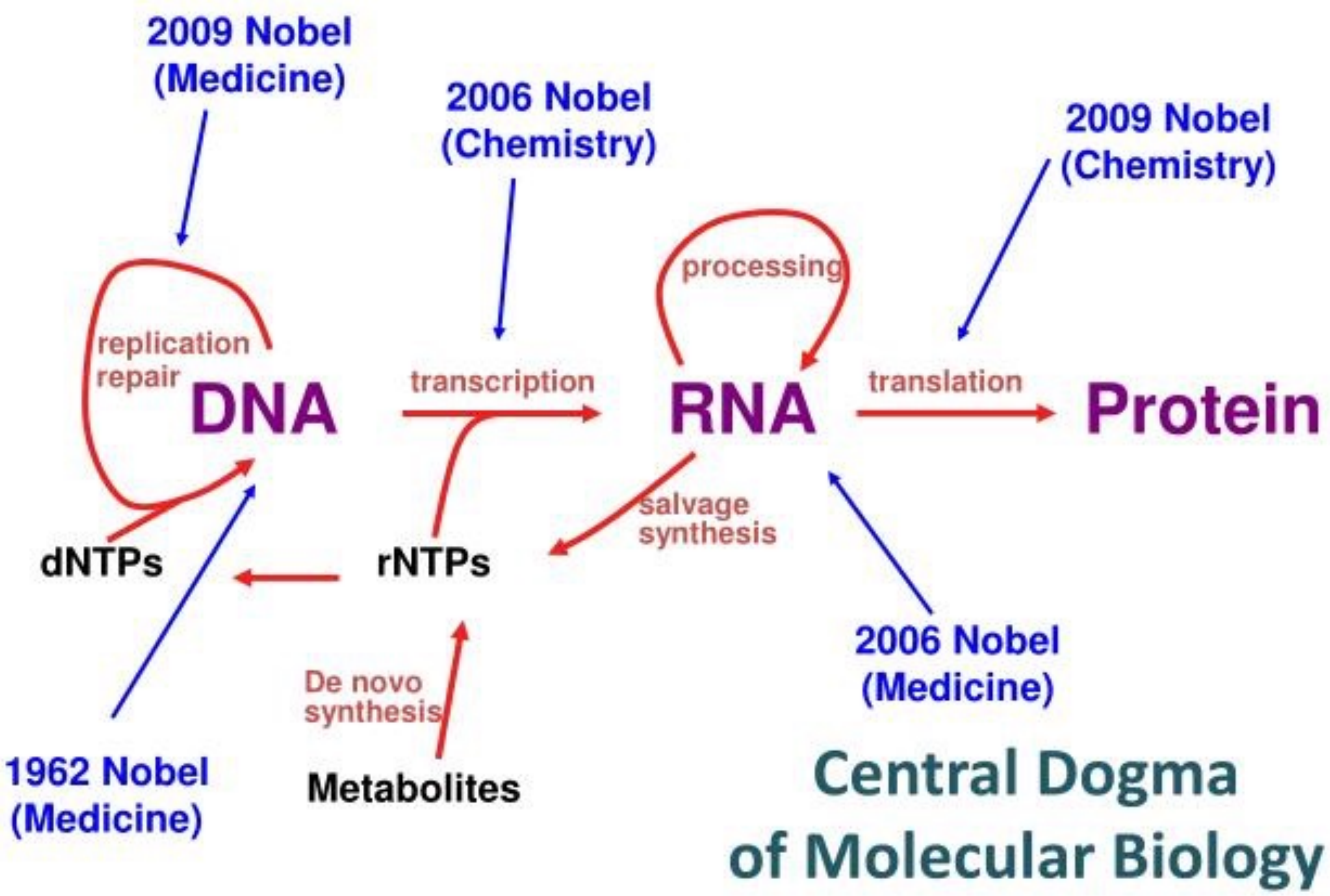
Non-coding RNA (miRNAs, piRNAs, eRNAs, lincRNAs, tRFs etc)

Gene expression



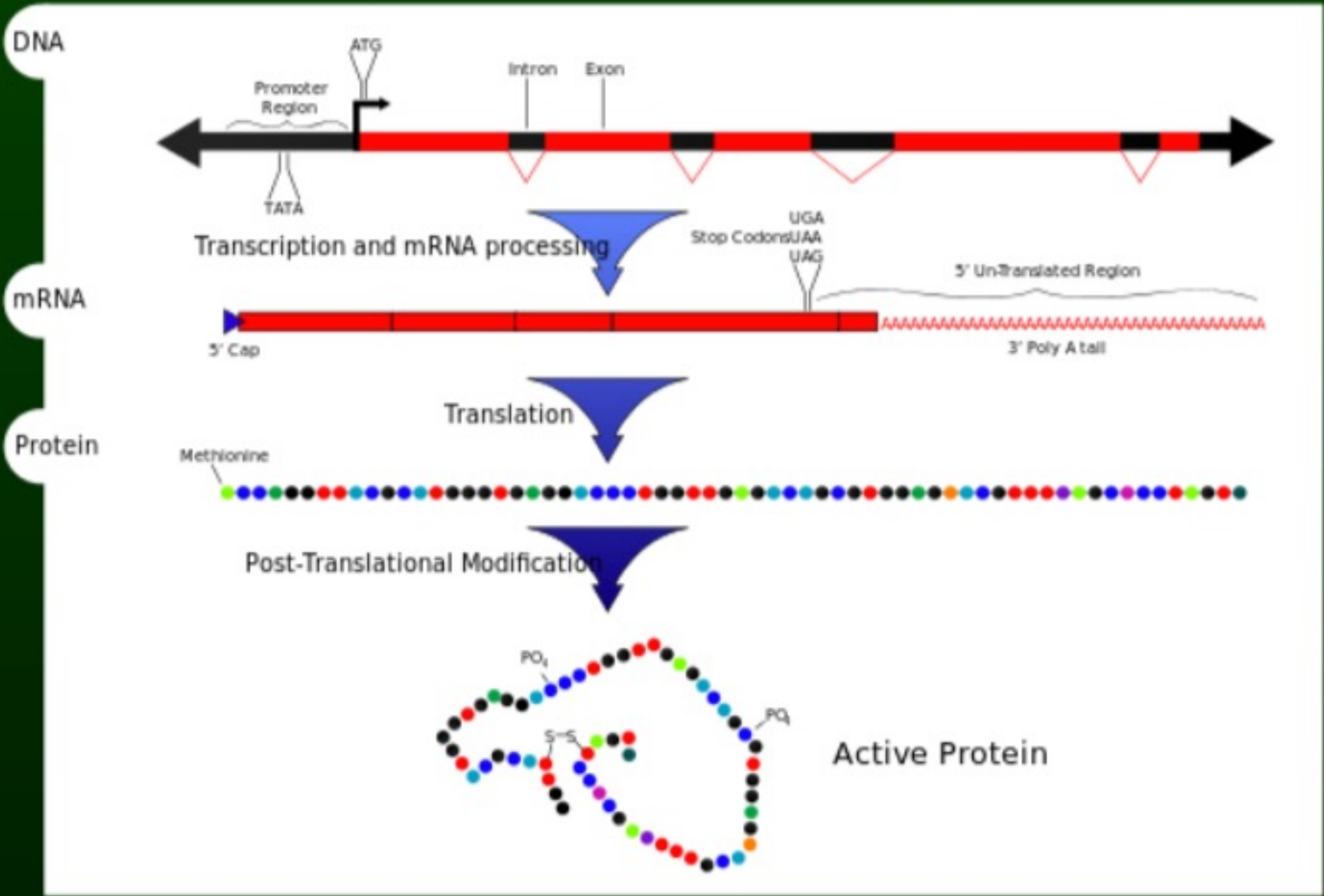
Central Dogma

153B = the hottest science in town !

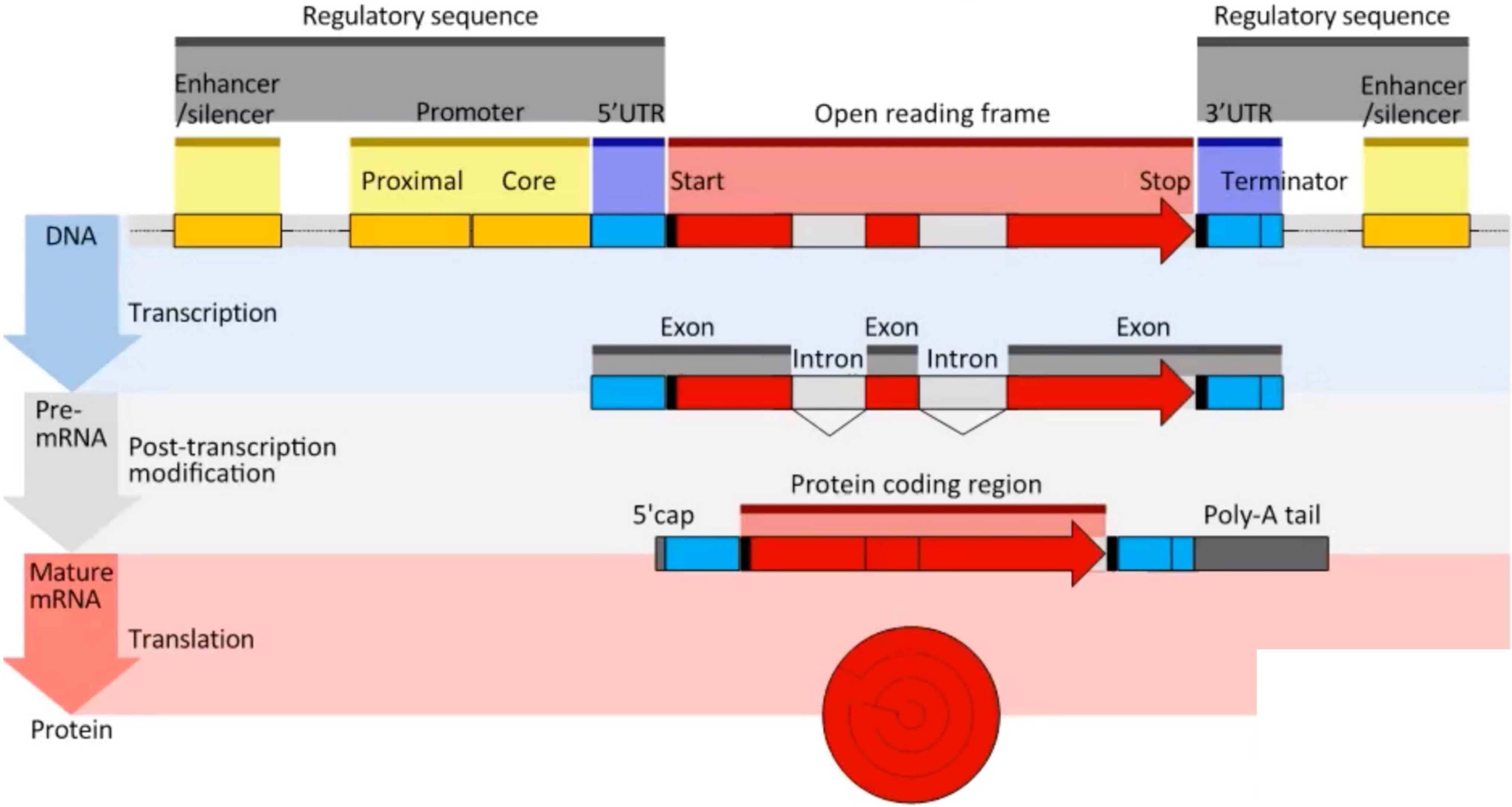


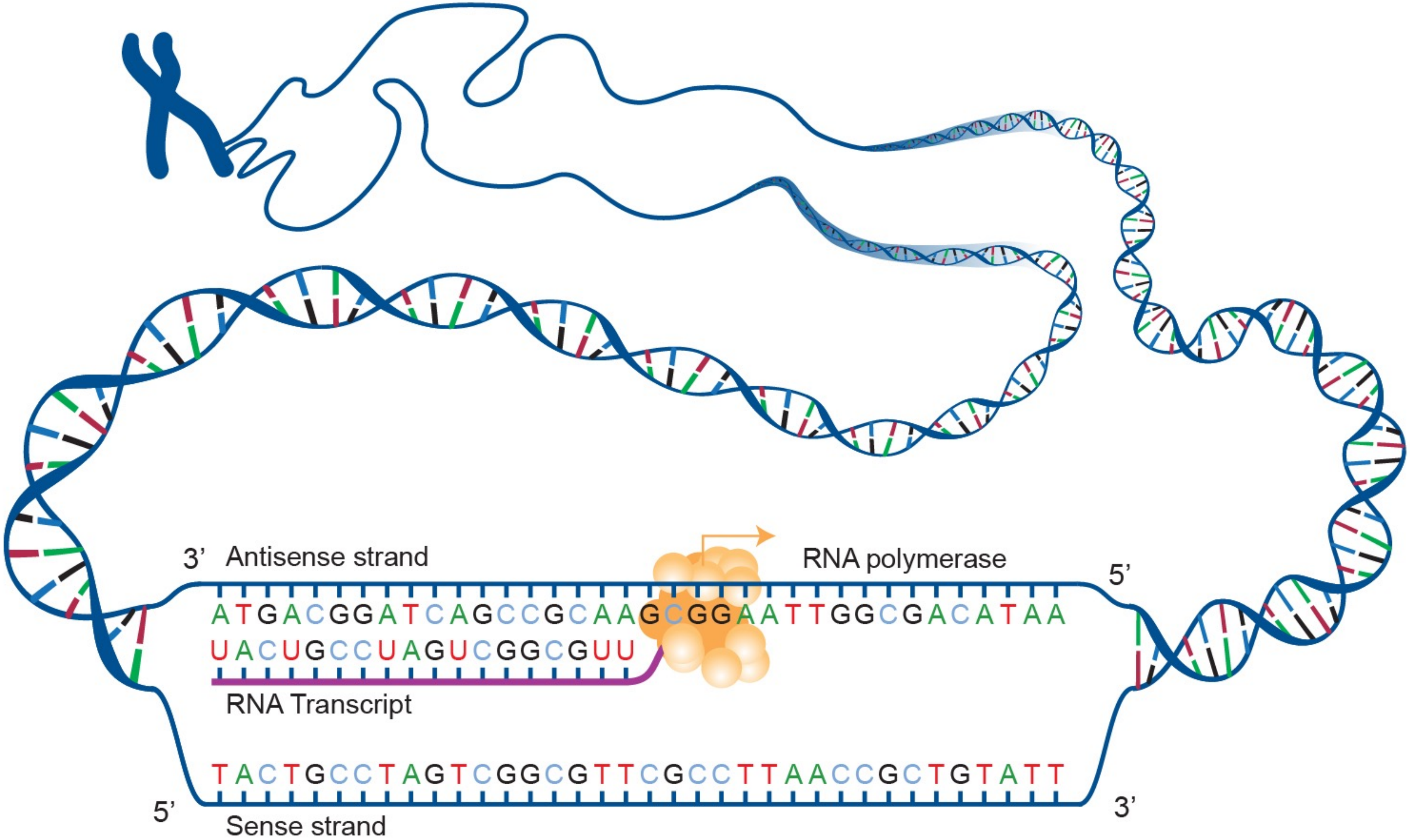
Central Dogma Song: http://www.youtube.com/watch?v=_Q2Ba2cFAew

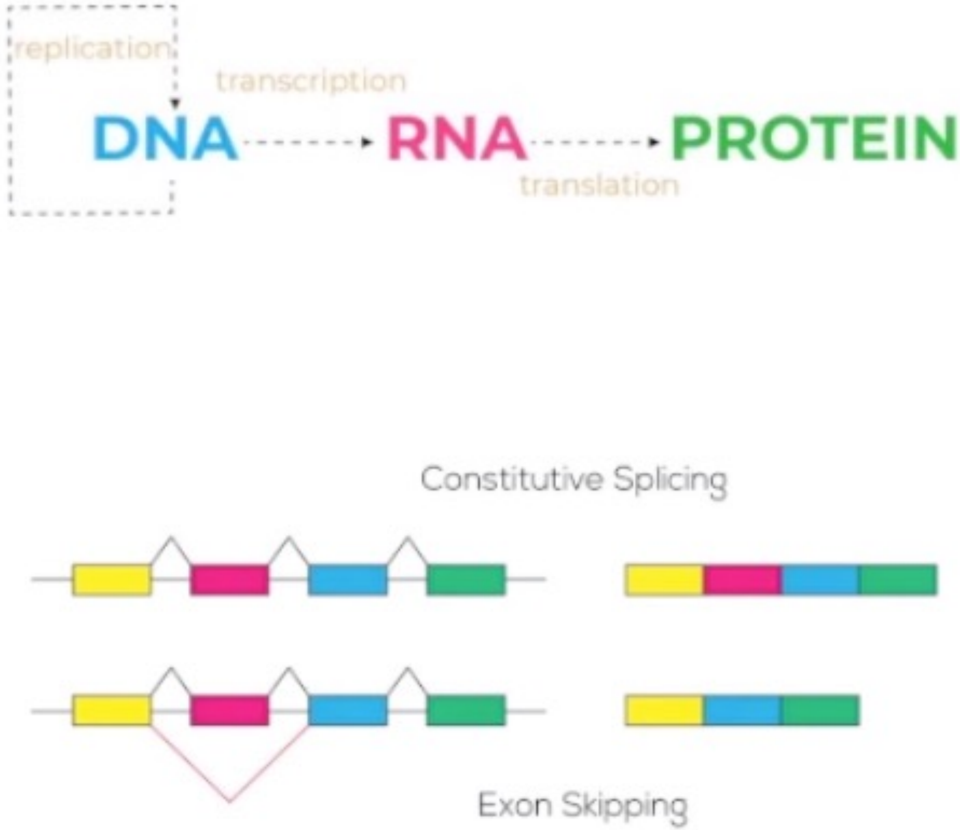
Central Dogma of Molecular Biology : Eukaryotic Mode



Eukaryotic gene expression




3’ Antisense strand
RNA polymerase
5’
ATGACGGATCAGCCGCAAGCGGAATTGGCGACATAA
UACUGCCUAGUCGGCGUU
RNA Transcript
TACTGCCTAGTCGGCGTTCGCCTTAACCGCTGTATT
5’
3’
Sense strand


replication
transcription
DNA
RNA
PROTEIN
translation
Constitutive Splicing
Exon Skipping

Kalvala *et al. Molecular Cancer* (2016) 15:47
DOI 10.1186/s12943-016-0527-1

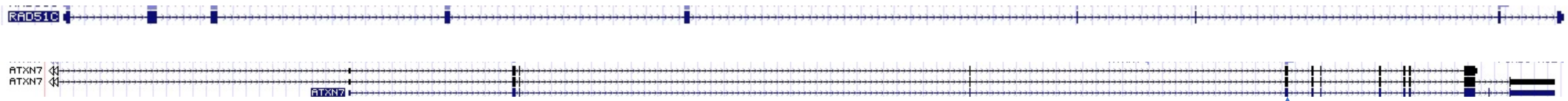
Molecular Cancer

RESEARCH Open Access

Rad51C-ATXN7 fusion gene expression in colorectal tumors

Arjun Kalvala[1], Li Gao[1], Brittany Aguila[1], Kathleen Dotts[1], Mohammad Rahman[1], Serge P. Nana-Sinkam[1,3], Xiaoping Zhou[4], Qi-En Wang[5], Joseph Amann[1,2], Gregory A. Otterson[1,2], Miguel A. Villalona-Calero[1,2,6*] and Wenrui Duan[1,2*]

exon7



exon6

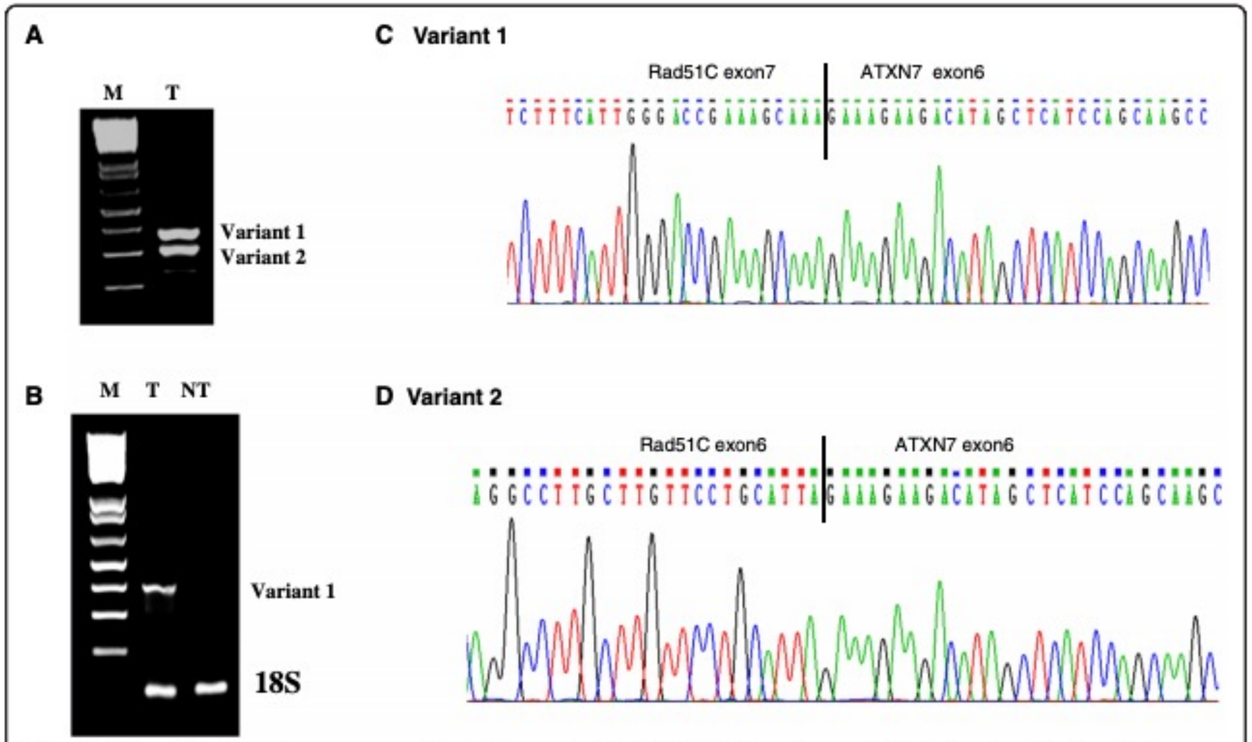


Fig. 1 RT-PCR amplification and sequencing analysis of fusion gene Rad51C-ATXN7. Colorectal tumor (T) RNA isolated by TRIzol method, was RT-PCR amplified and sequenced. The primers (Additional file 1: Table S1) that span exon-5 of Rad51C and exon -8 of ATXN7 produced two fragments of approximately 376 bp and 316 bp (**a**). The RT-PCR amplified product from a colorectal tumor (T) showed 376 bp fragment only which was absent in the corresponding non-tumor (NT) sample (**b**). The RT-PCR amplified products were Topo TA cloned and sequenced. The RT-PCR products contained a Variant 1 of Rad51C (exon 1–7) and ATXN7 (exon 6–13) fusion gene (**c**) and a Variant 2 without exon-7 Rad51C (exon 1–6) ATXN7 (exon 6–13) (**d**). The solid bar line indicates the sequence breakpoint joining Rad51C and ATXN7

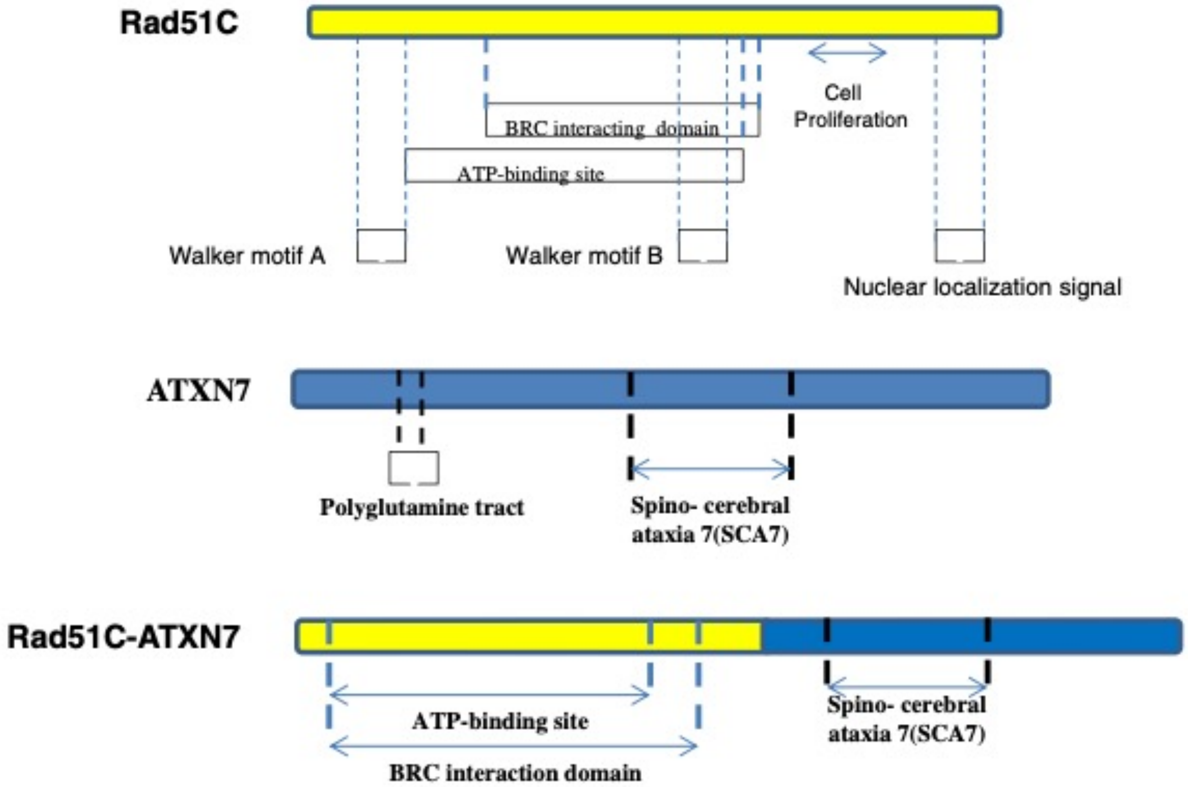


Fig. 2 Schematic representation of portion of the fusion Variant 2 of Rad51C-ATXN7. The fusion transcript and their breakpoints are indicated with exons of Rad51C and ATXN7. The Variant 2 of the fusion gene transcript is joined at 3′-end of exon-6 of Rad51C with 5′-end of the exon-6 of ATXN7. The Fusion protein has lost its nuclear localization signal located at the C-terminus end of Rad51C and CAG (Poly glutamine tract) repeat sequence of ATXN7. The ATP binding site and BRC (BRCA1) interacting domains of Rad51C are conserved at N-terminus of the fusion protein and SCA7 (Spino- cerebral ataxia 7) domain of ATXN7 conserved at middle terminus of the fusion protein

diagnostics

MDPI

Case Report

Molecular Characterization of a Novel Splicing Mutation underlying Mucopolysaccharidosis (MPS) type VI—Indirect Proof of Principle on Its Pathogenicity

Maria Francisca Coutinho [1,2,†], Marisa Encarnação [1,2,3,†], Liliana Matos [1,2], Lisbeth Silva [1,3], Diogo Ribeiro [1,2], Juliana Inês Santos [1,4], Maria João Prata [4,5], Laura Vilarinho [1,2,3] and Sandra Alves [1,2,*]

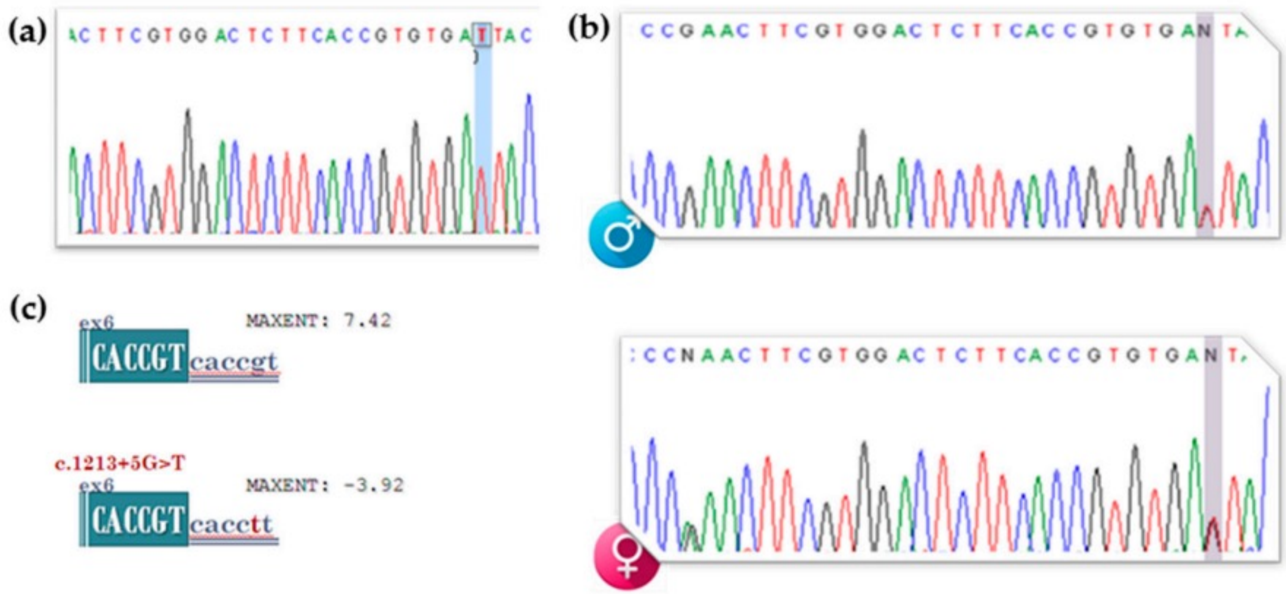


Figure 1. Molecular diagnosis of the patient by classical gDNA screening of the *ARSB* gene: detection of the novel c.1213+5G>T (IVS6+5G>T) mutation. Electropherogram highlighting the affected residue (**a**) in the patient and (**b**) parents. (**c**) *In silico* predictions on the effect of the novel mutation over the splice-site junctions. Score predictions by the MaxEntScan bioinformatic tool.

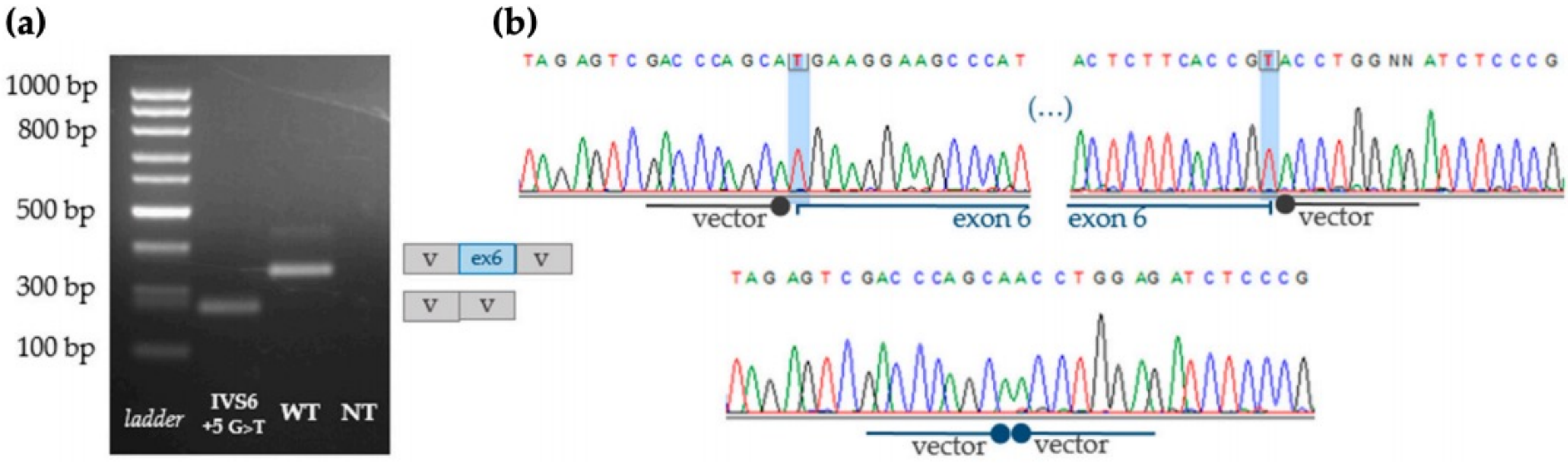
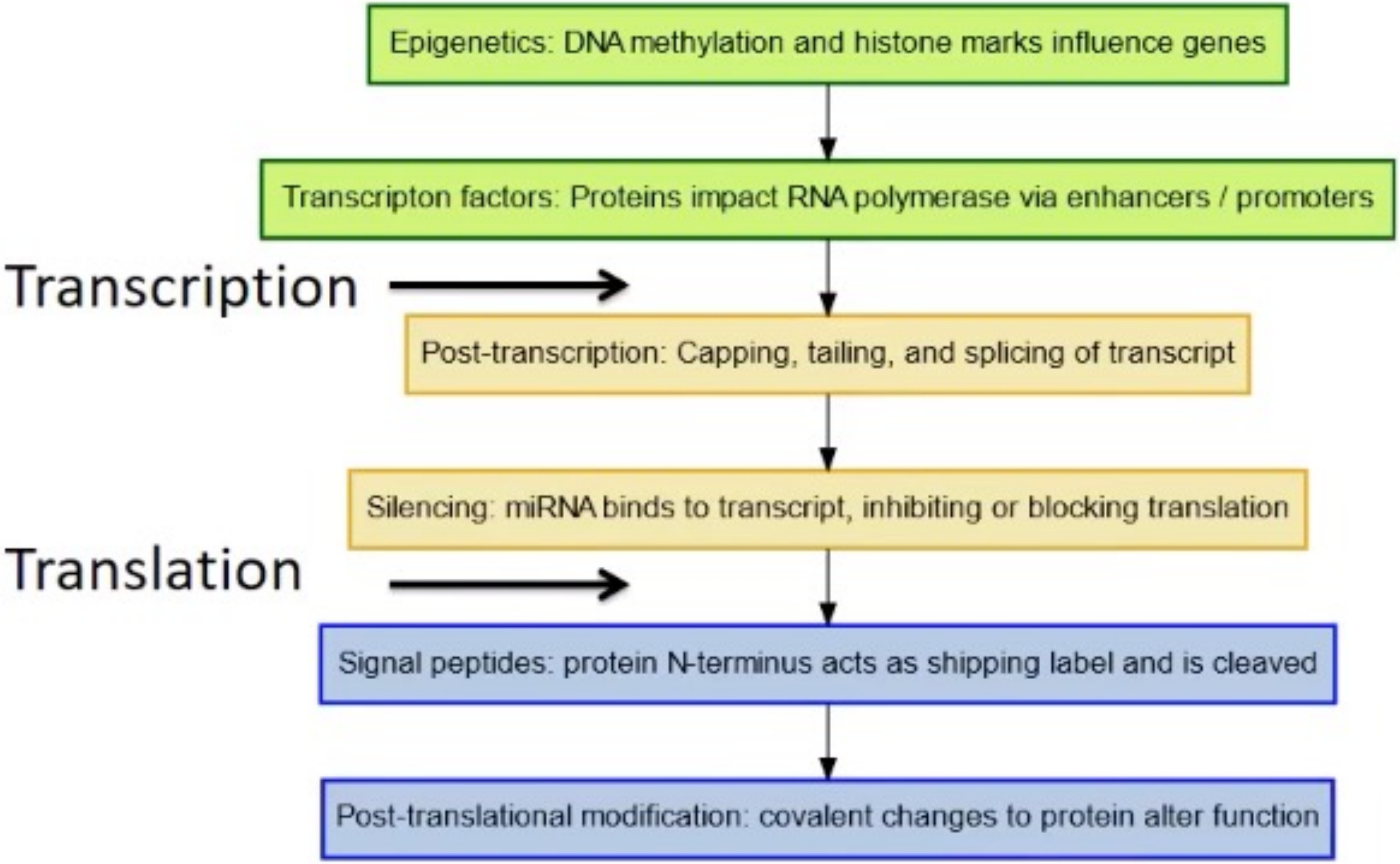
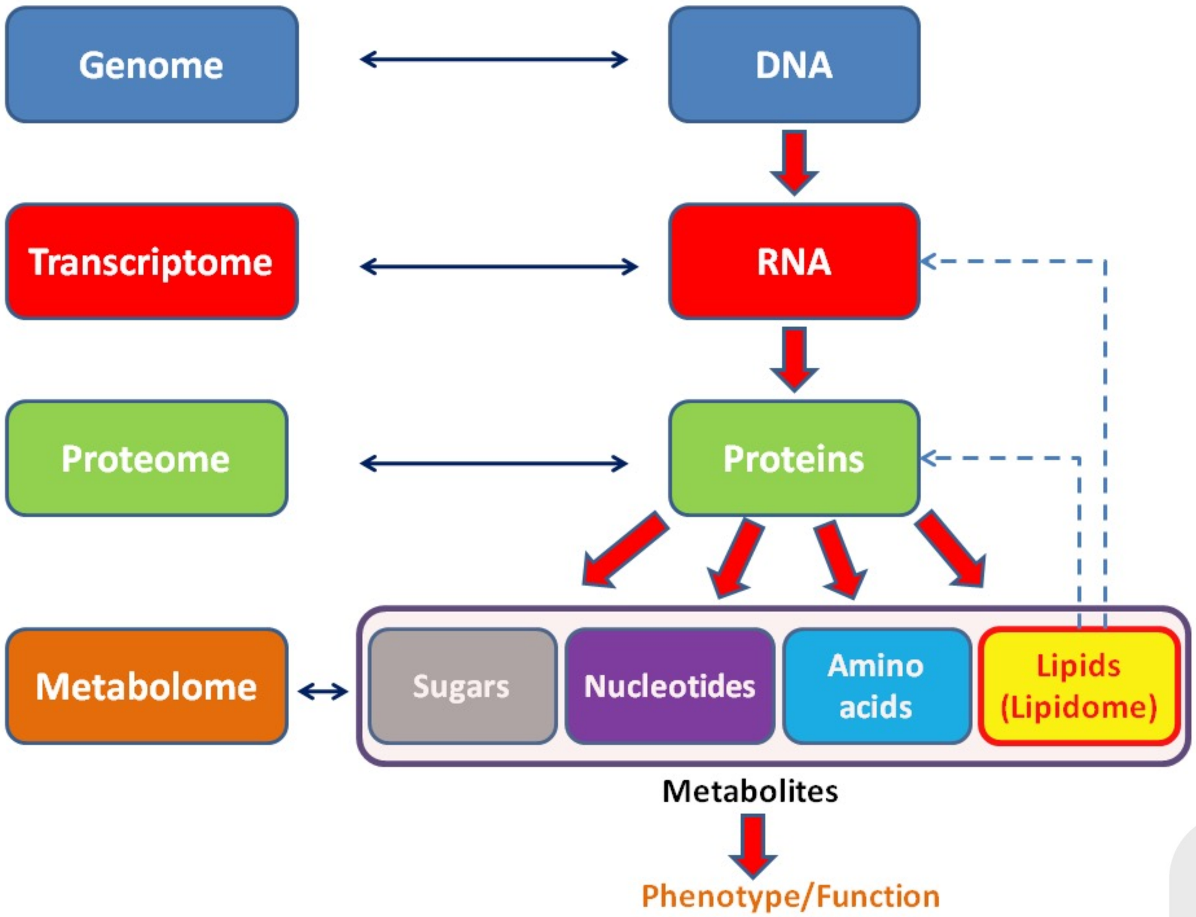
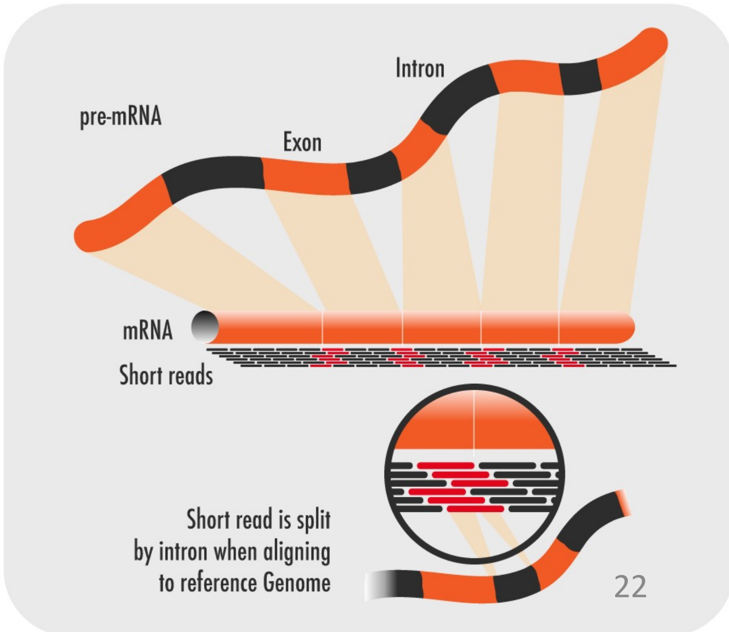


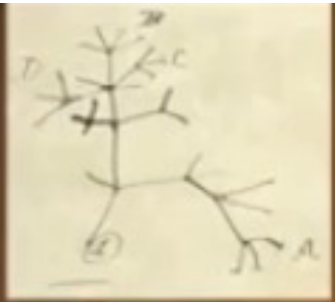
Figure 4. RT-PCR amplification and Sanger sequencing of WT and mutant splicing reporter minigenes using vector-specific primers. The results of RT-PCR analysis using vector specific primers are shown along with a diagram of the transcripts obtained in Hep3B cells after 24 h of incubation (**a**) and their sequencing results depicted in electropherograms (**b**). c.1213+5G>T [IVS6+5G>T]: mutant minigene; WT: wild-type; NT: non-transfected.

Complex control of gene expression

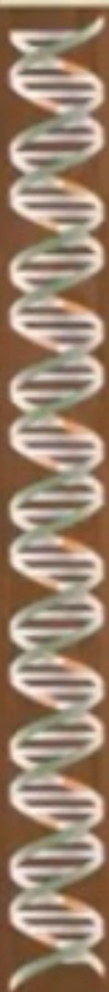



Genome
DNA
Transcriptome
RNA
Proteome
Proteins
Metabolome
Sugars
Nucleotides
Amino acids
Lipids (Lipidome)
Metabolites
Phenotype/Function


pre-mRNA
Intron
Exon
mRNA
Short reads
Short read is split by intron when aligning to reference Genome



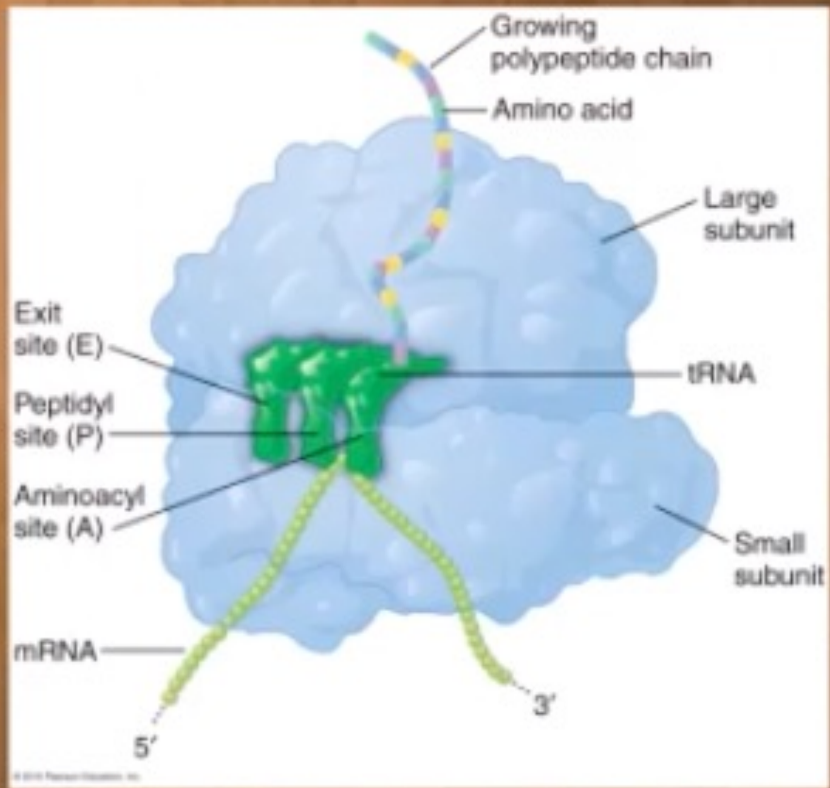
Transcriptomes



Transcriptome Sequencing

Starting material?

- rRNA
- tRNA
- mRNA



Reverse Transcriptase

Article in Cell Systems · September 2016
DOI: 10.1016/j.cels.2016.09.006

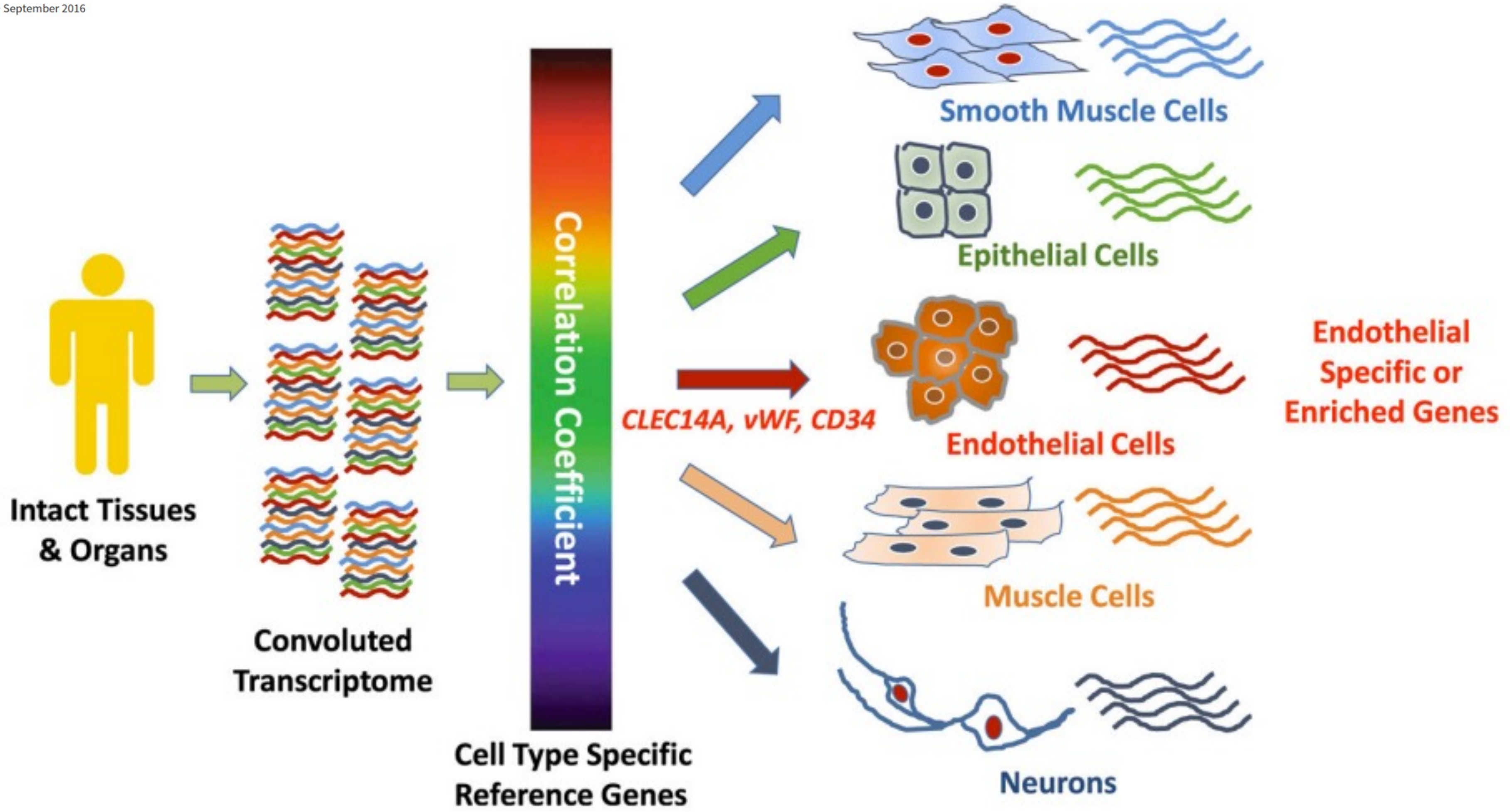


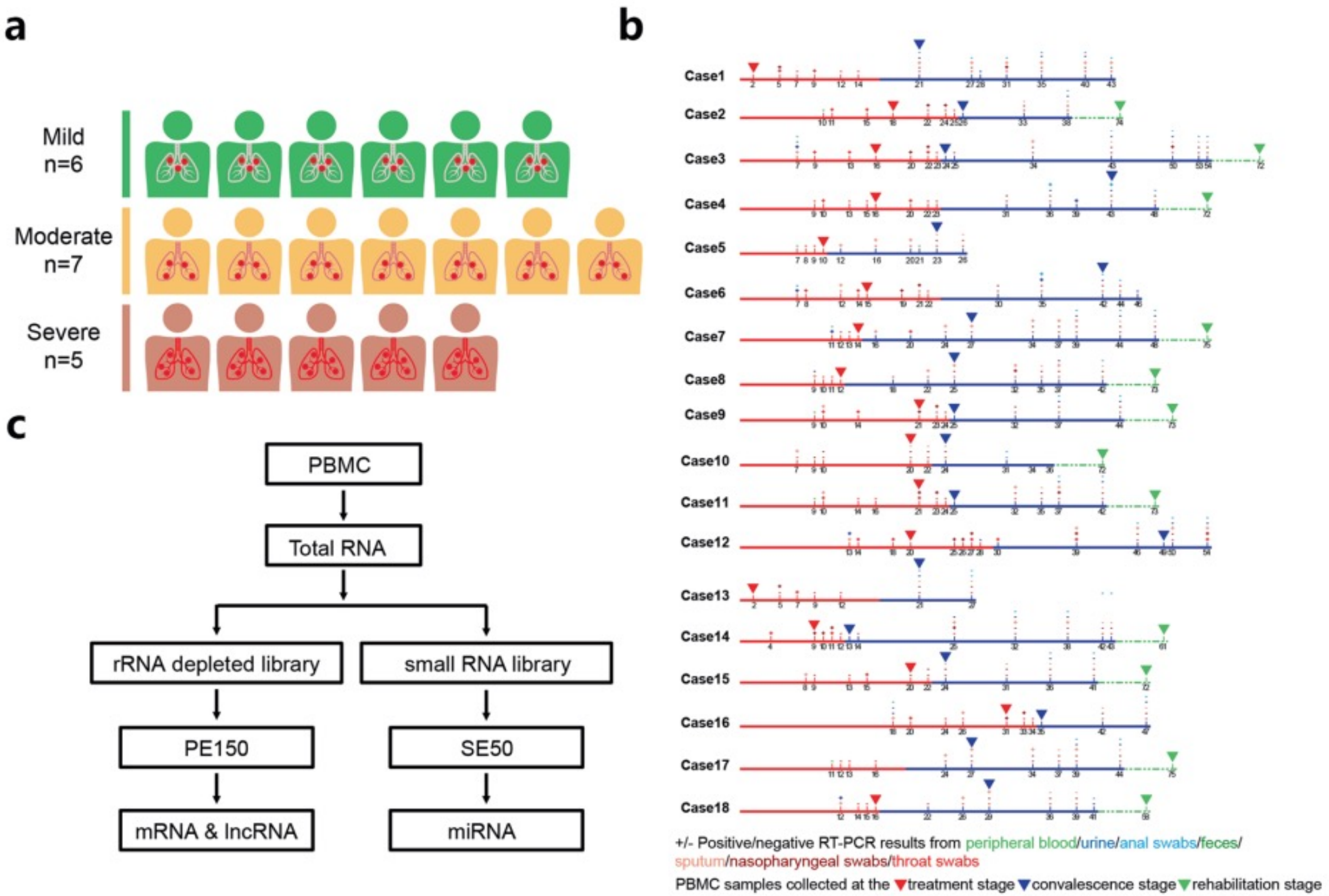
Figure 1. Reference-Gene-Based Human Transcriptome Deconvolution
Illustration of the approach employed by Butler et al. (2016) to identify endothelial-specific or -enriched genes from unfractionated human tissues using combined correlation coefficient analysis with three endothelial-cell-specific reference genes. Other cell types are included to indicate potential applications to deconvolute the human transcriptome for other cell-type-specific genes.

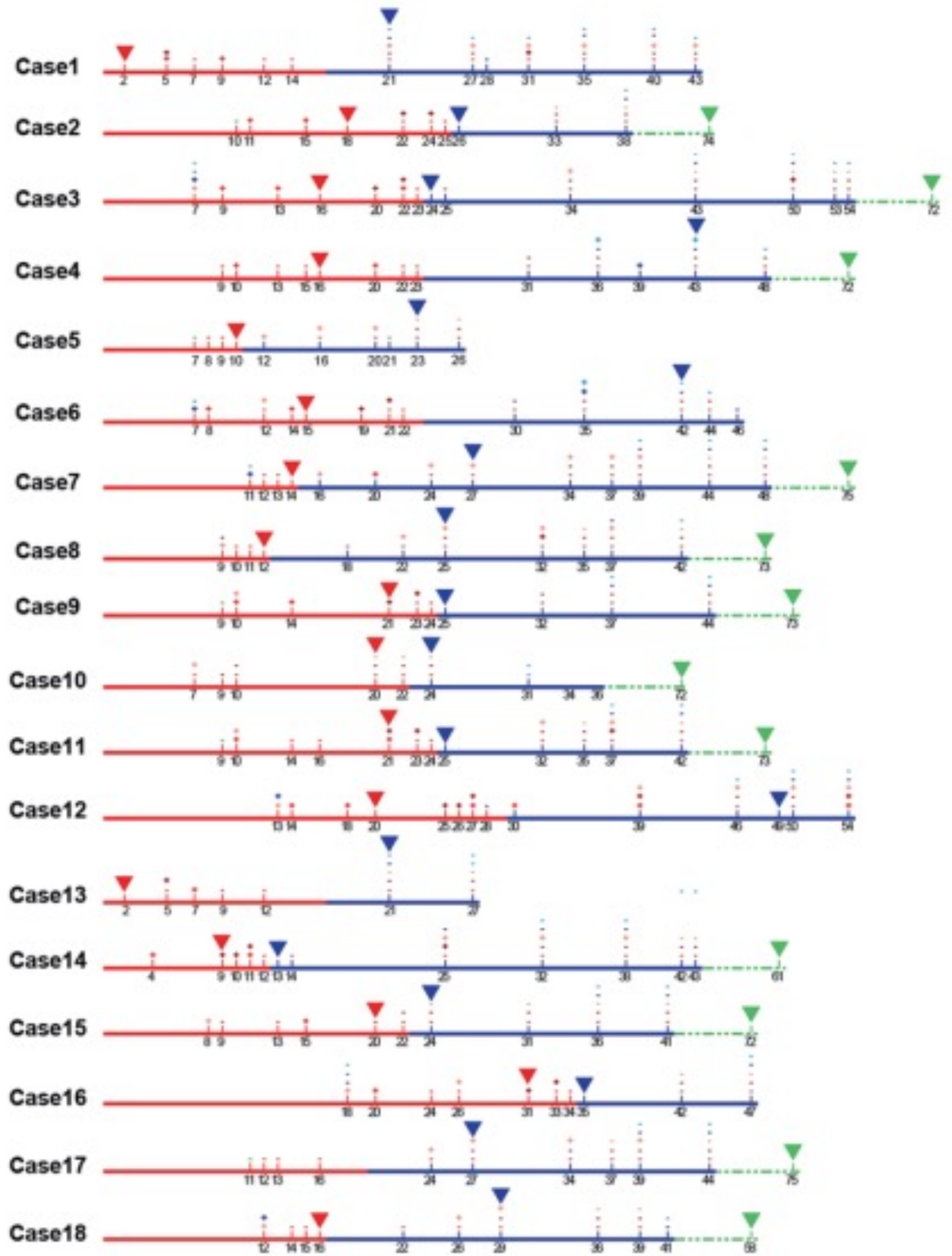
Fig. 1 Study design of whole transcriptome sequencing for PBMCs from patients with COVID-19 during recovery. **a** Patients are divided into mild, moderate and severe groups according to clinical symptoms of each one. **b** Timeline of the disease course of 18 patients infected with SARS-CoV-2. RT-PCR indicates the PCR test of SARS-CoV-2 nucleic acid. A positive RT-PCR (marked by "+") indicates that the SARS-CoV-2 nucleic acid was found in the throat swab, nasopharyngeal swab, sputum or other samples, which were distinguished by using different colors. "−", negative RT-PCR. The solid red lines stand for treatment stage, solid blue lines for convalescence stage and doted green lines for rehabilitation stage. The sampling days were marked below this line. The inverted triangle symbol is used to mark the time point for collecting PBMC samples at each stage. **c** A flowchart showing the process of library construction and RNA sequencing of PBMC samples from patients with COVID-19

ARTICLE **OPEN**

Longitudinal transcriptome analyses show robust T cell immunity during recovery from COVID-19

Hong-Yi Zheng[1], Min Xu[1], Cui-Xian Yang[2], Ren-Rong Tian[1], Mi Zhang[2], Jian-Jian Li[2], Xi-Cheng Wang[2], Zhao-Li Ding[3], Gui-Mei Li[3], Xiao-Lu Li[3], Yu-Qi He[3], Xing-Qi Dong[2], Yong-Gang Yao[1,4,5] and Yong-Tang Zheng[1,4]





+/- Positive/negative RT-PCR results from peripheral blood/urine/anal swabs/feces/ sputum/nasopharyngeal swabs/throat swabs

PBMC samples collected at the ▼treatment stage ▼convalescence stage ▼rehabilitation stage

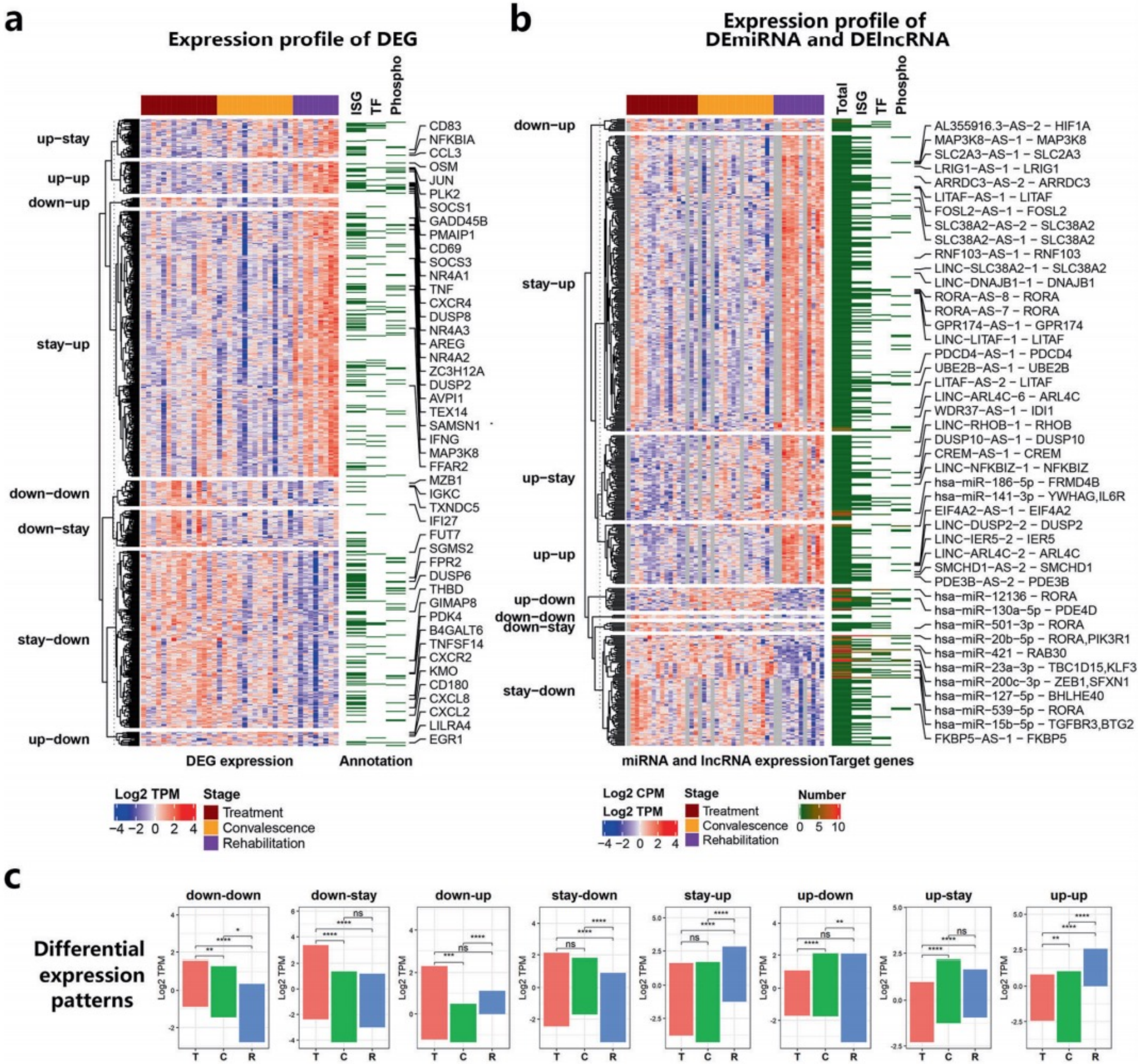
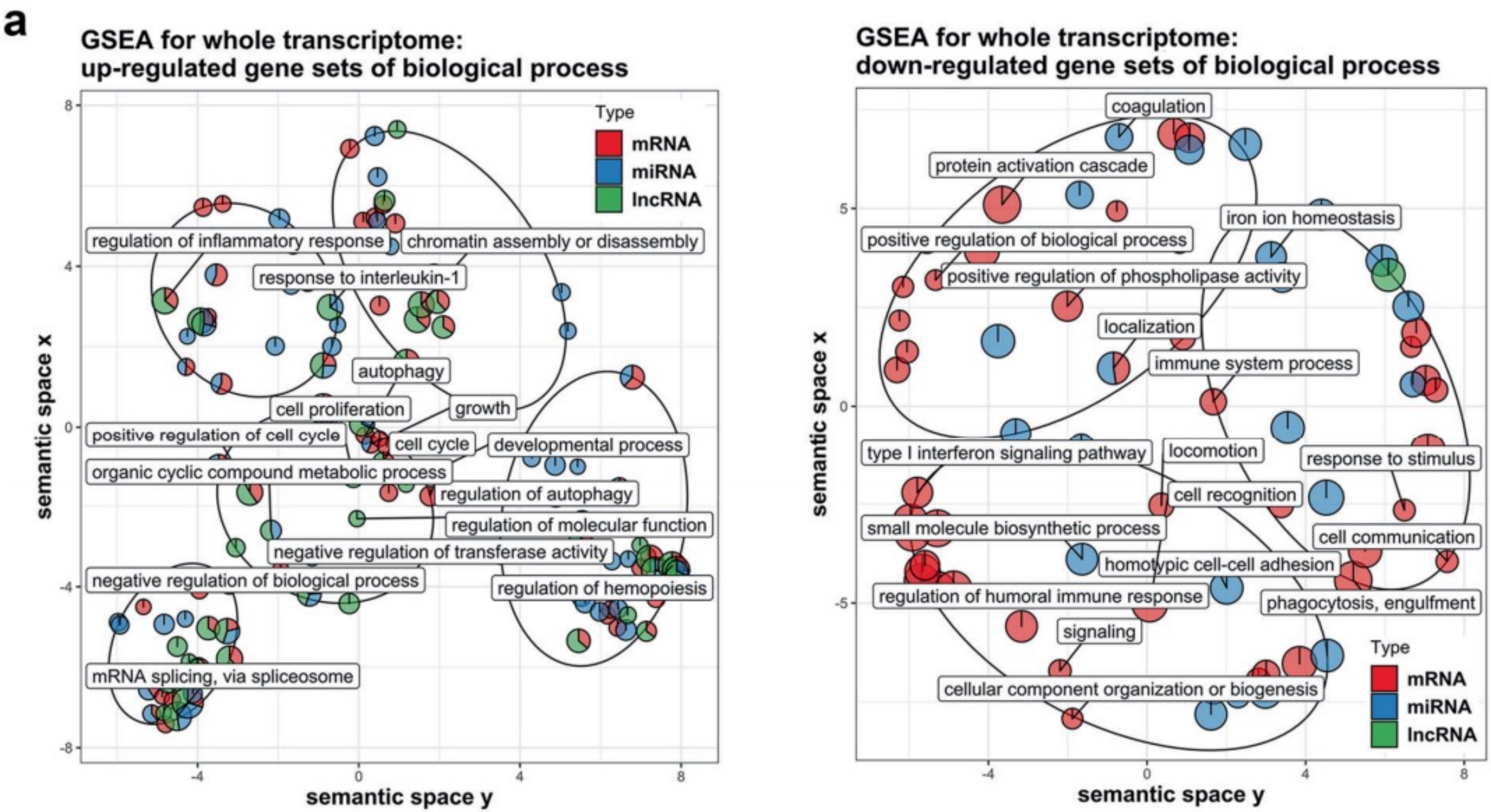


Fig. 2 Differentially expressed genes (DEG), miRNAs (DEmiRNA), and lncRNAs (DElncRNA) in PBMC samples from patients with COVID-19 during the recovery. **a** A heatmap showing the expression profiles of 643 DEGs. The blue-red gradient square maps the scaled $\log_2$ value of transcripts per kilobase million ($\log_2$ TPM) for each DEG in a sample. The grouping information on the left indicates genes with different differential expression patterns. The colored bars above the heatmap refer to the clinical stage of each sample. The gene function annotation on the right marks the interferon-stimulated genes (ISG), transcription factor (TF) and phosphorylation regulatory genes (Phospho) in DEGs. The hub genes with a more significant differential expression in the function networks constructed by all DEGs are marked with the respective gene names. **b** A heatmap showing the expression profiles of 67 DEmiRNAs and 405 DElncRNAs. The blue-red gradient square maps the scaled $\log_2$ TPM of each DElncRNA, or the scaled $\log_2$ value of counts per million ($\log_2$ CPM) of each DEmiRNA in a sample. The gene function annotation shows the number of target genes for each non-coding RNA. When the target gene is a DEG, we annotate the regulation module in the format of "non-coding RNA name - target gene name" on the right. **c** By comparing the *P* value and TPM/CPM at three different clinical stages, DEG, DEmiRNA or DElncRNA can be further clustered into 8 expression patterns. Each pattern reflects the characteristics of its RNAs being upregulated (up), downregulated (down) or stayed for no change (stay) at the convalescence (C) and rehabilitation (R) stages relative to the treatment (T) stage. Ns, *, **, ***, ****: $p > 0.05$, ≤ 0.05, ≤ 0.001, ≤ 0.0001, respectively


a
GSEA for whole transcriptome:
up-regulated gene sets of biological process
Type
mRNA
miRNA
lncRNA
regulation of inflammatory response
chromatin assembly or disassembly
response to interleukin-1
autophagy
cell proliferation
growth
positive regulation of cell cycle
cell cycle
developmental process
organic cyclic compound metabolic process
regulation of autophagy
regulation of molecular function
negative regulation of transferase activity
negative regulation of biological process
regulation of hemopoiesis
mRNA splicing, via spliceosome
semantic space x
semantic space y
GSEA for whole transcriptome:
down-regulated gene sets of biological process
coagulation
protein activation cascade
iron ion homeostasis
positive regulation of biological process
positive regulation of phospholipase activity
localization
immune system process
type I interferon signaling pathway
locomotion
response to stimulus
cell recognition
small molecule biosynthetic process
cell communication
homotypic cell-cell adhesion
regulation of humoral immune response
phagocytosis, engulfment
signaling
cellular component organization or biogenesis
Type
mRNA
miRNA
lncRNA
semantic space x
semantic space y

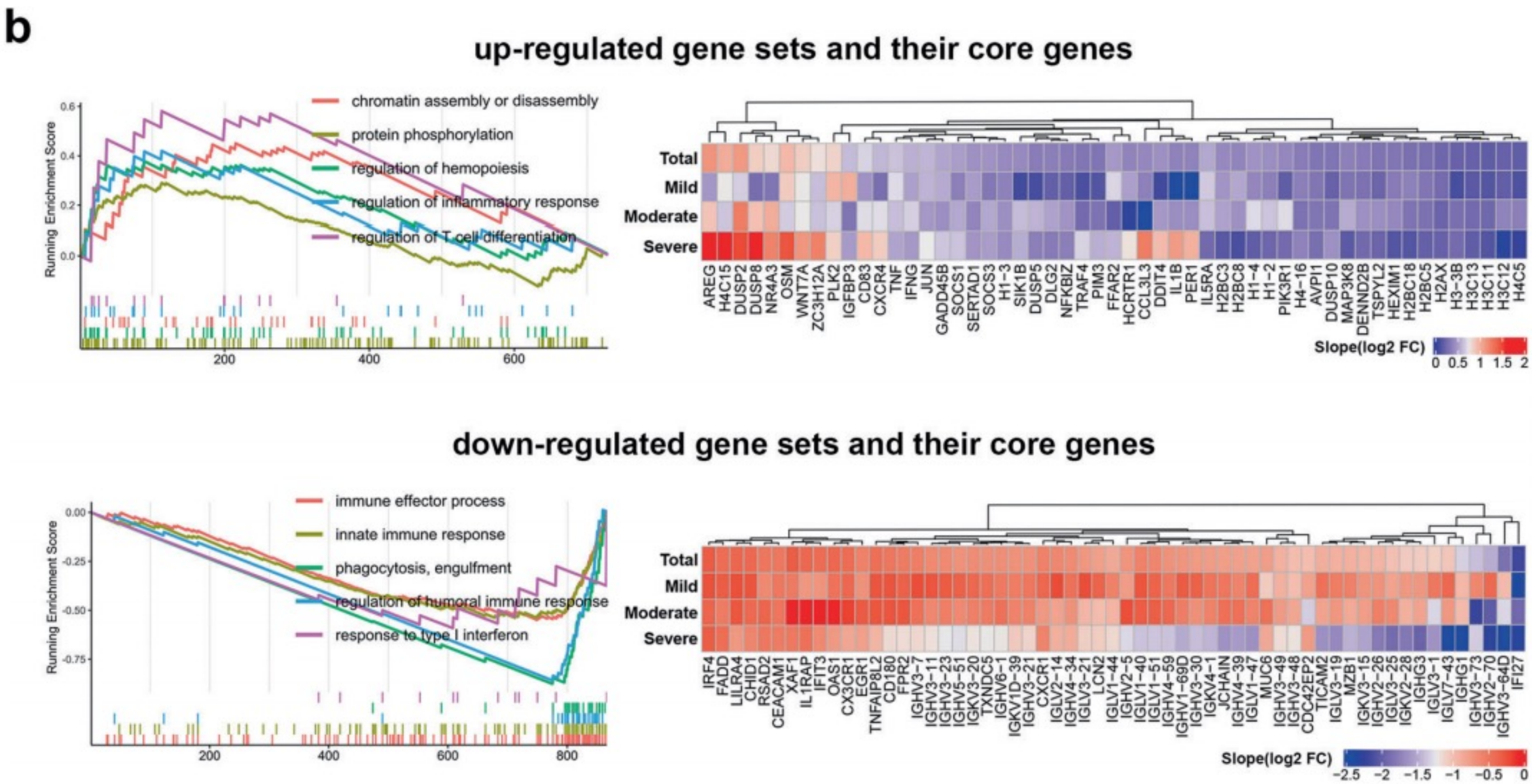


Fig. 3 Gene set enrichment analysis (GSEA) showing an immune system remolding at the whole transcriptome level. a The bubble-pie graphs displaying upregulated and downregulated gene ontology (GO) gene sets enriched by the differentially expressed mRNAs, miRNAs, and lncRNAs. The color refers to the indicated type of transcripts enriched for a gene set. The bubble size maps the sum enrichment scores of all RNA types in a gene set. The X and Y values of a gene set are calculated by REVIGO's GO semantic space algorithm.[69] The ellipses refer to clusters formed by gene sets with a high semantic similarity. **b** GSEA plot showing the distribution of the upregulated or downregulated gene sets and the enrichment scores based on DEGs for all samples. The heat map shows the core genes in these gene sets of different clinical stages for all samples, mild, moderate and severe groups, respectively. The blue-red gradient square maps the slope of the linear regression equation of Log2 fold change of DEG

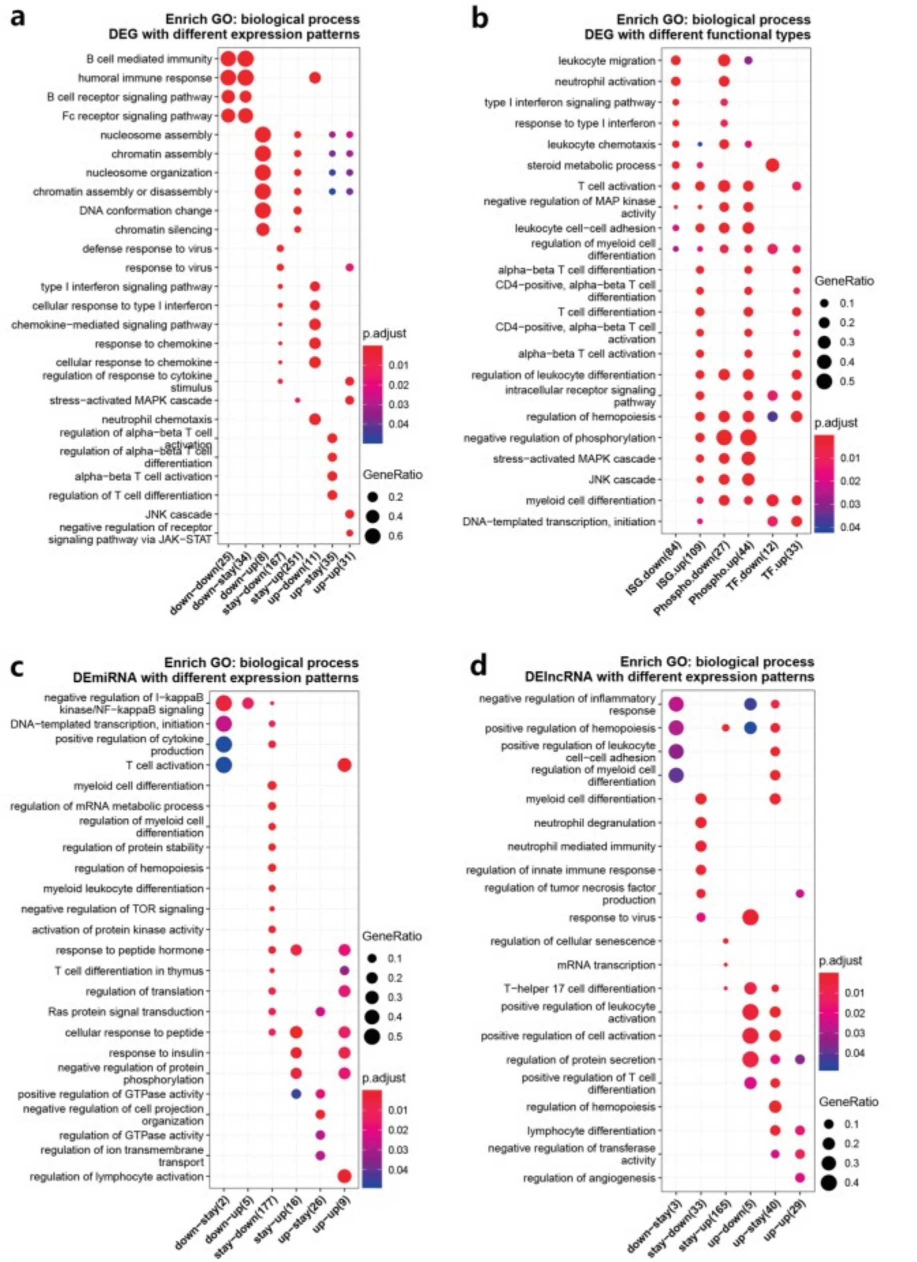
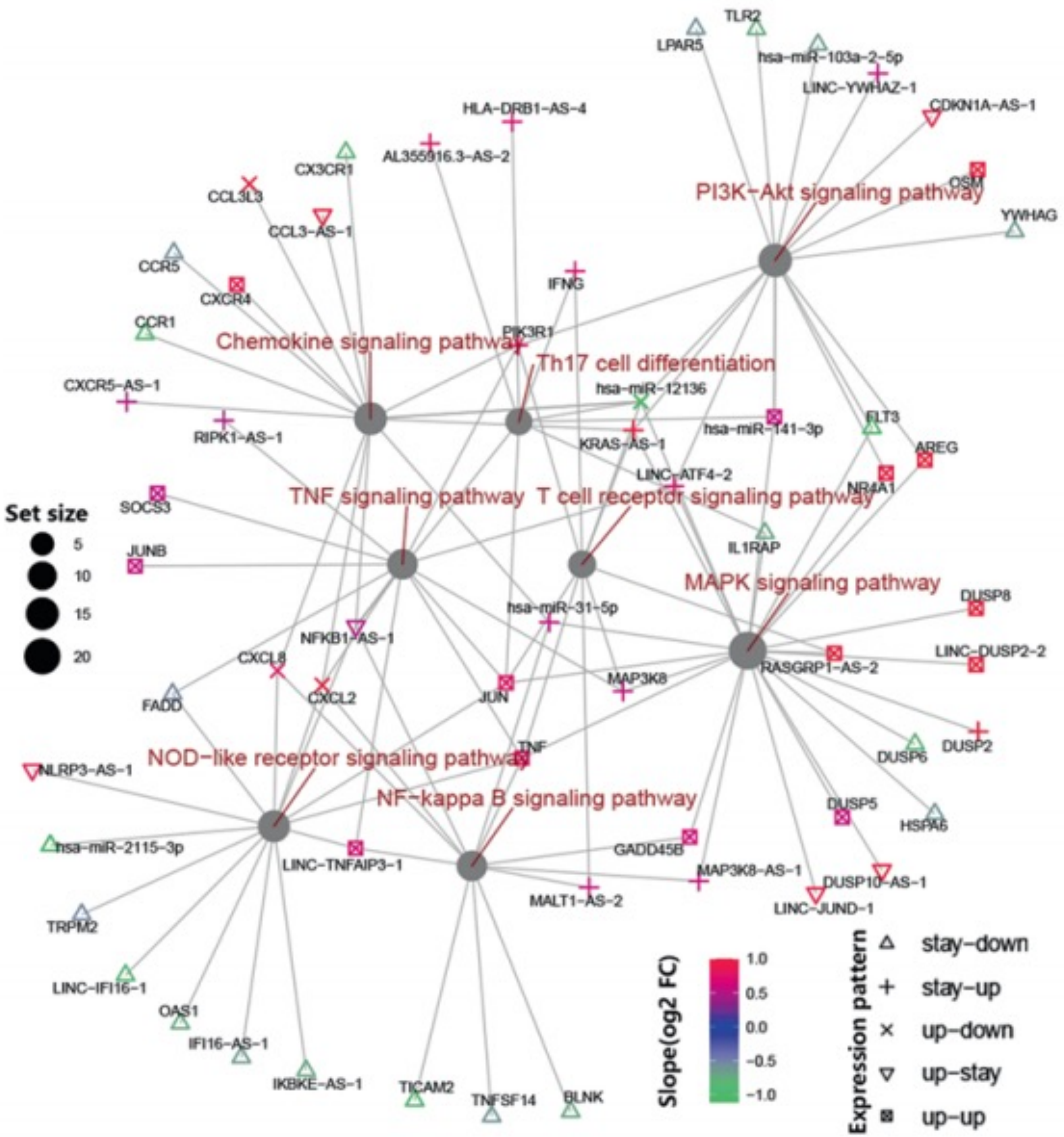


Fig. 4 Gene ontology (GO) enrichment analysis identified T cell immunity as the main regulatory target during the recovery of COVID-19. **a** GO enrichment results of DEGs in 8 differential expression clusters defined in Fig. 2b. The numbers of significantly enriched genes in each cluster were included in parentheses. **b** GO enrichment results of interferon-stimulated genes (ISG), transcription factor genes (TF), and phosphorylation regulatory genes (Phospho) in DEGs. **c**, **d** GO enrichment results of the genes targeted by DEmiRNAs (**c**) and DElncRNAs (**d**) in 6 differential expression clusters defined in Fig. 2c. The bubble size indicates the gene enrichment ratio (Generatio) of a biological process GO term, with color maps the FDR value (p.adjust) of the enrichment analysis

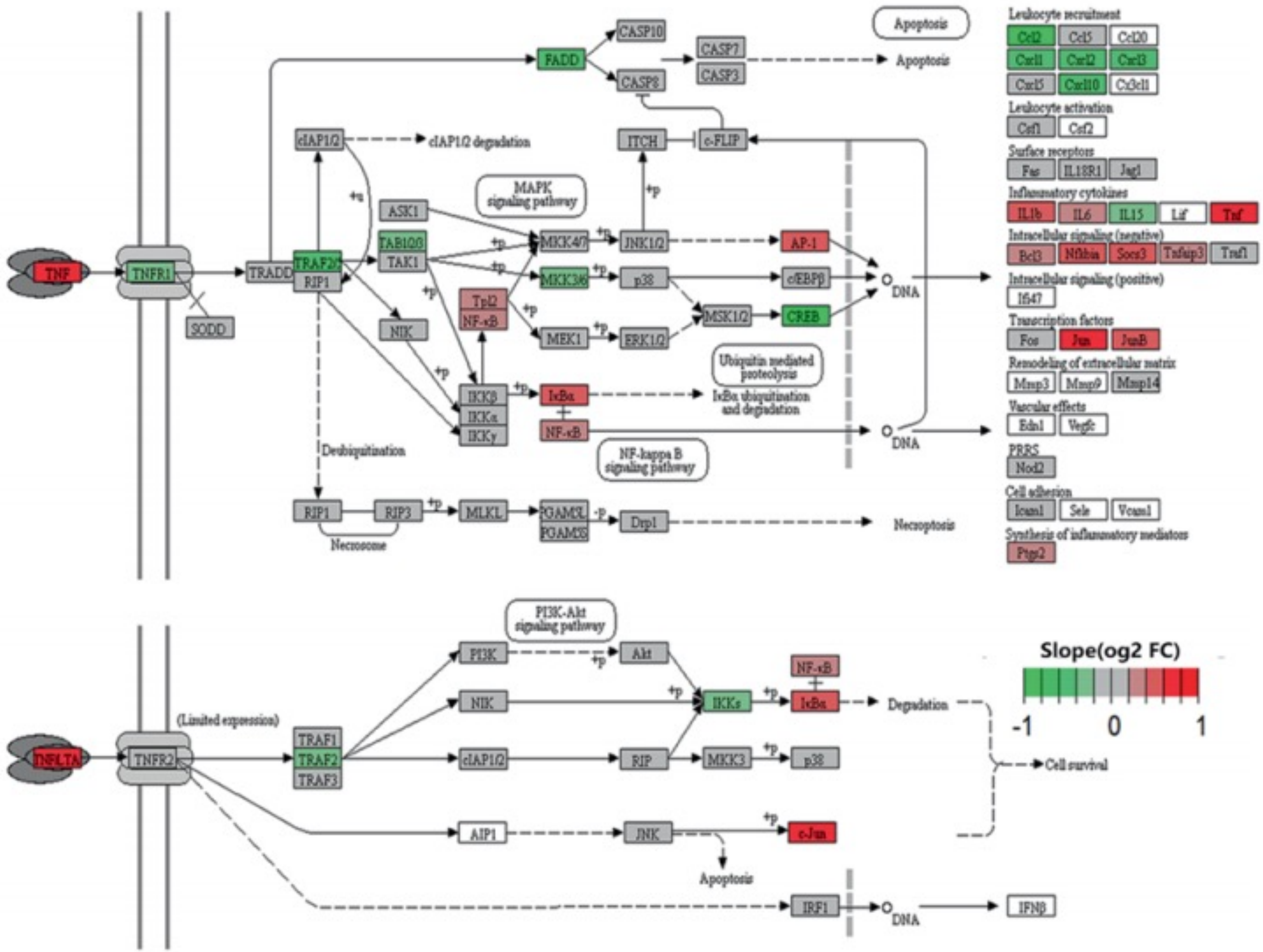
a

KEGG network of DEG, DEmiRNA and DElncRNA



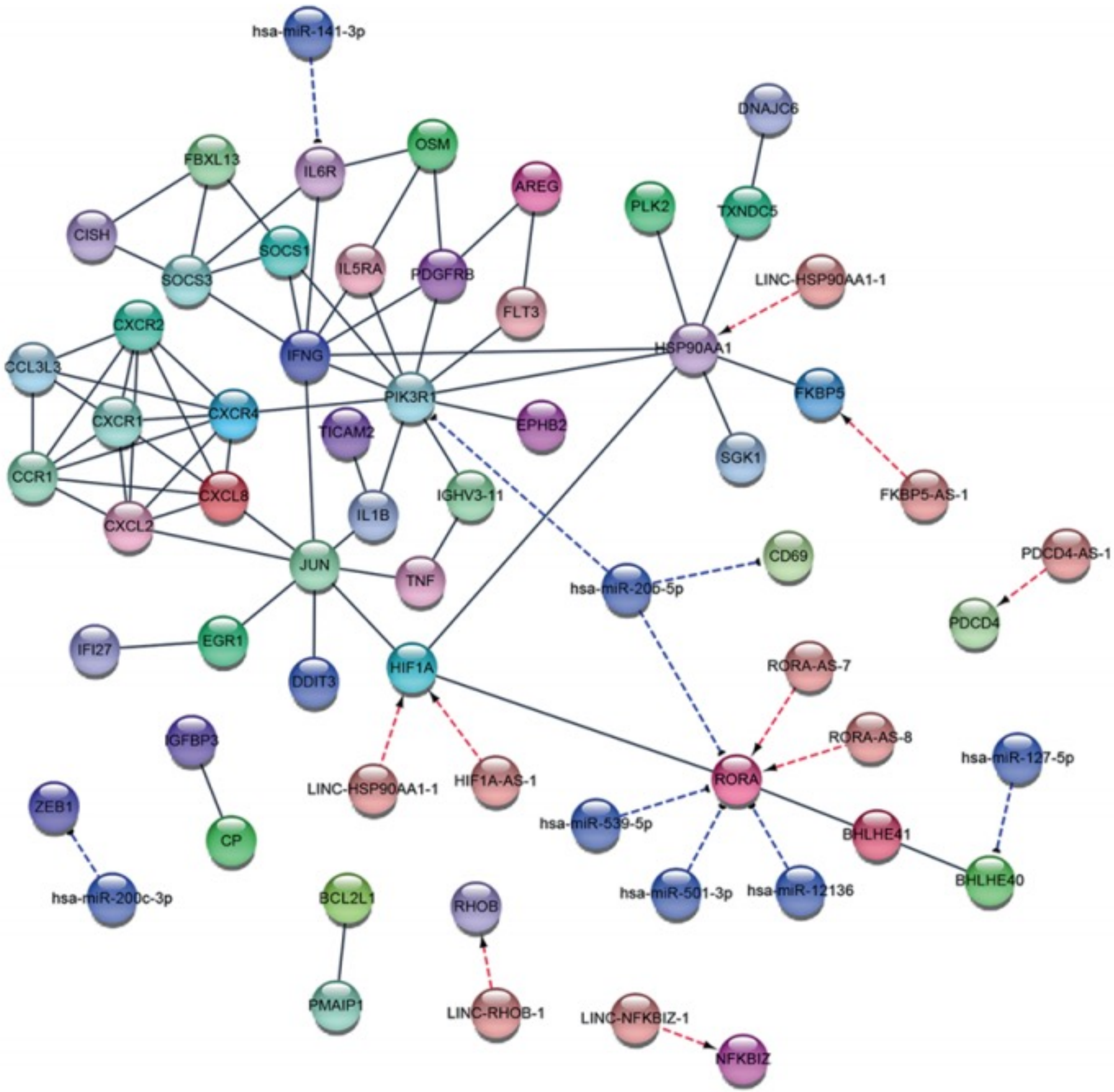
b

KEGG graph of TNF signaling pathway



c

PPI network for key DEG, DEmiRNA and DElncRNA



d

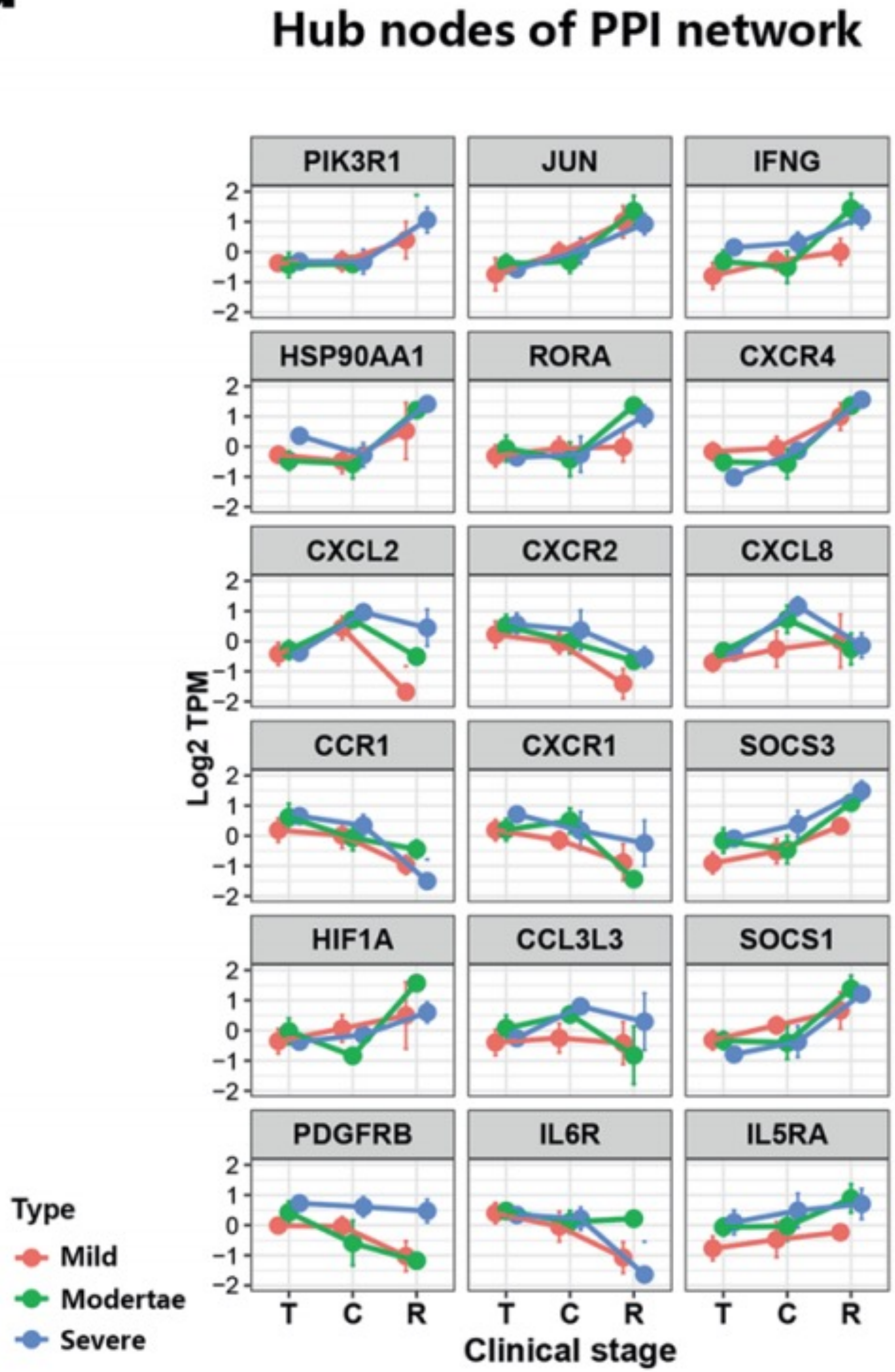
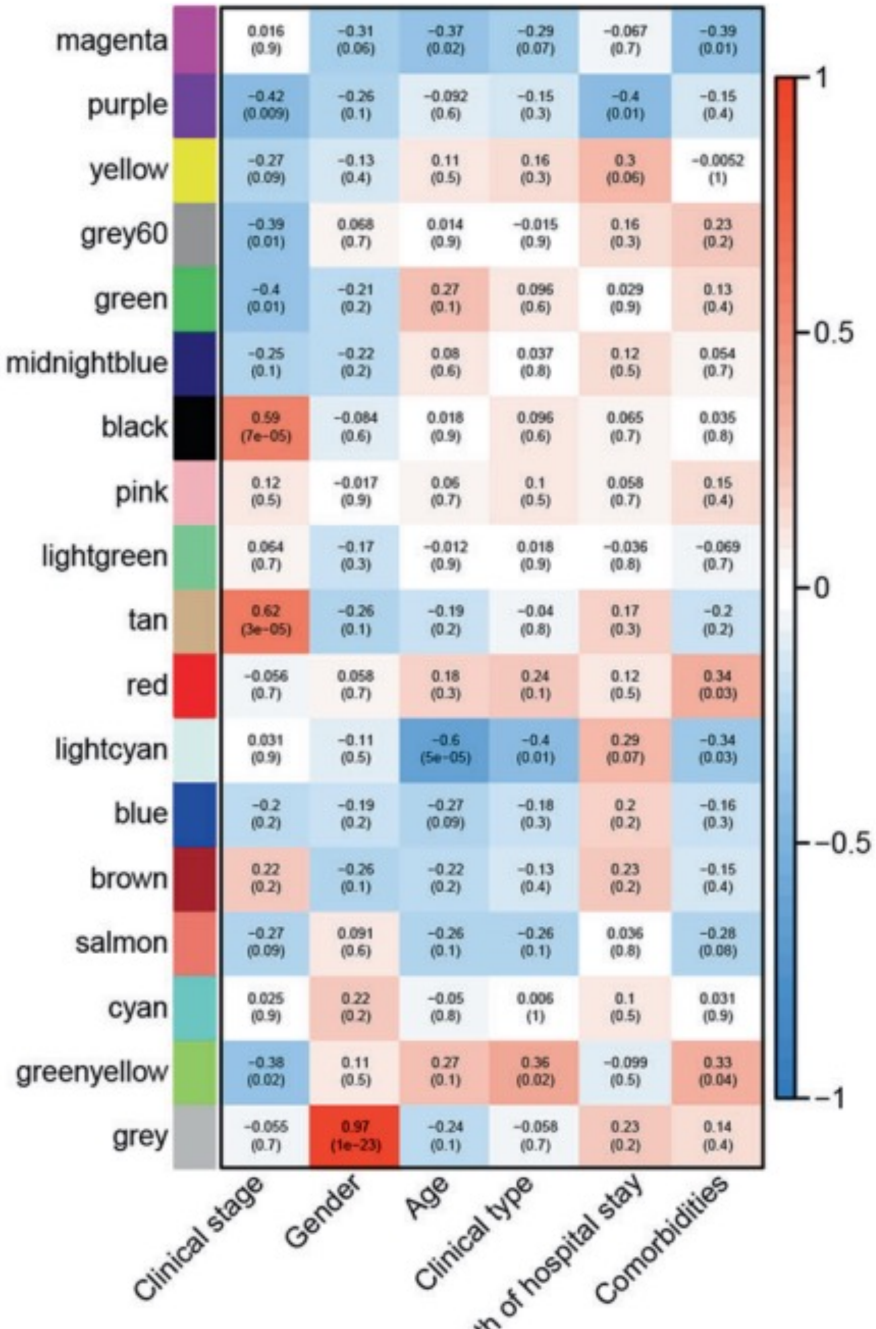


Fig. 5 Gene function network revealed a core role of the AP-1 linked signaling pathways during the recovery of COVID-19. **a** A network showing the relationship between genes and enriched KEGG pathway. The genes displayed are selected from the DEGs and the genes regulated by DEmiRNAs or DElncRNAs with a more significant differential expression and a higher connectivity. The color maps the linear regression slope of Log_2 fold change of a gene. The size of the circle is proportional to the number of genes enriched in a KEGG pathway. **b** A KEGG graph showing the interactions of genes involved in the TNF signaling pathway. The color indicates linear regression slope of Log_2 fold change of a gene. **c** A protein–protein interaction network (PPI) showing the interactions between genes and non-coding RNA. The genes displayed are selected from the top hub genes with the most significant differential expression. **d** A line graph showing the expression profiles of the main hub genes during COVID-19 recovery. The color indicates samples with different clinical types. T, treatment stage; C, convalescence stage; R, rehabilitation stage. The data are shown as mean ± SD

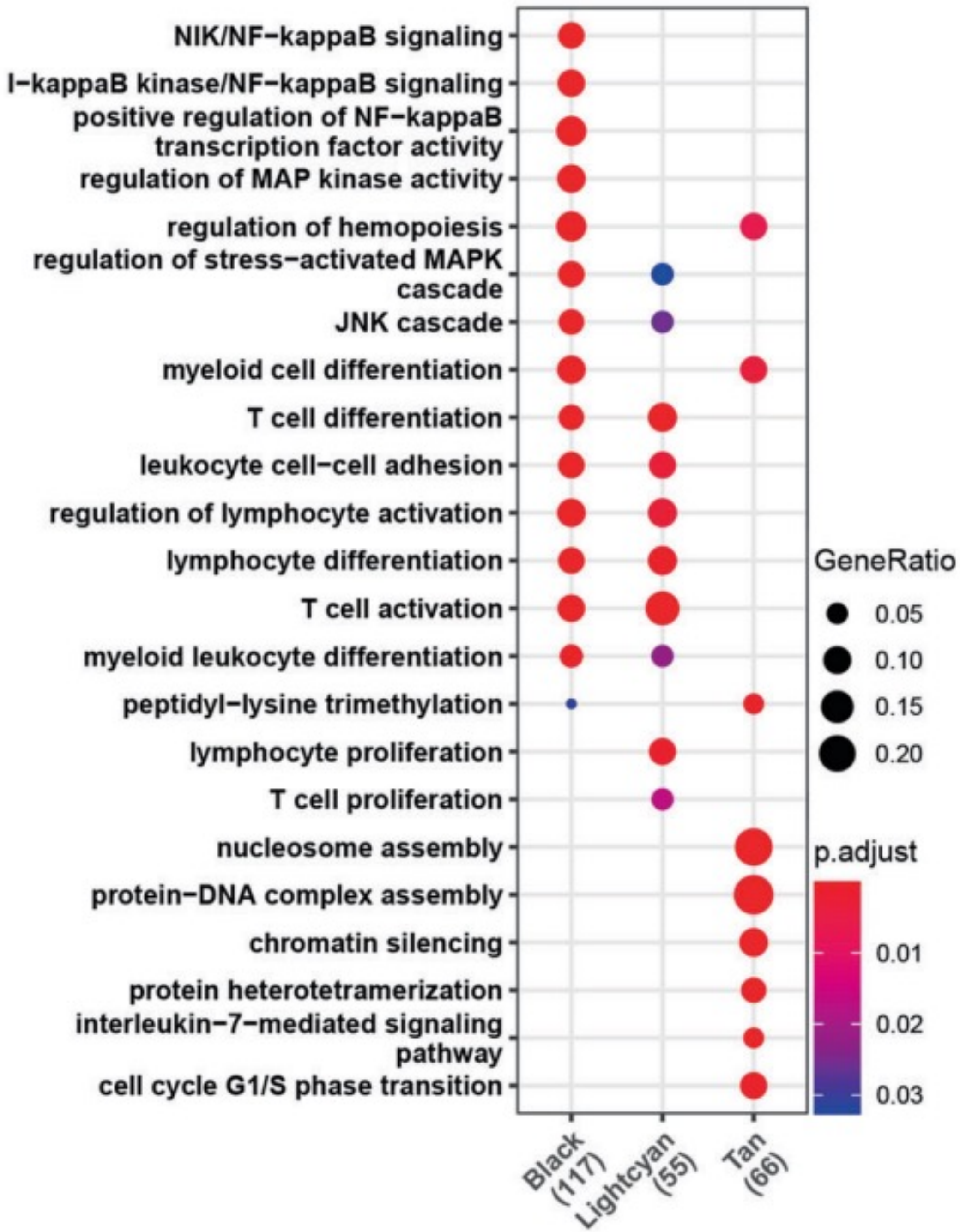
b

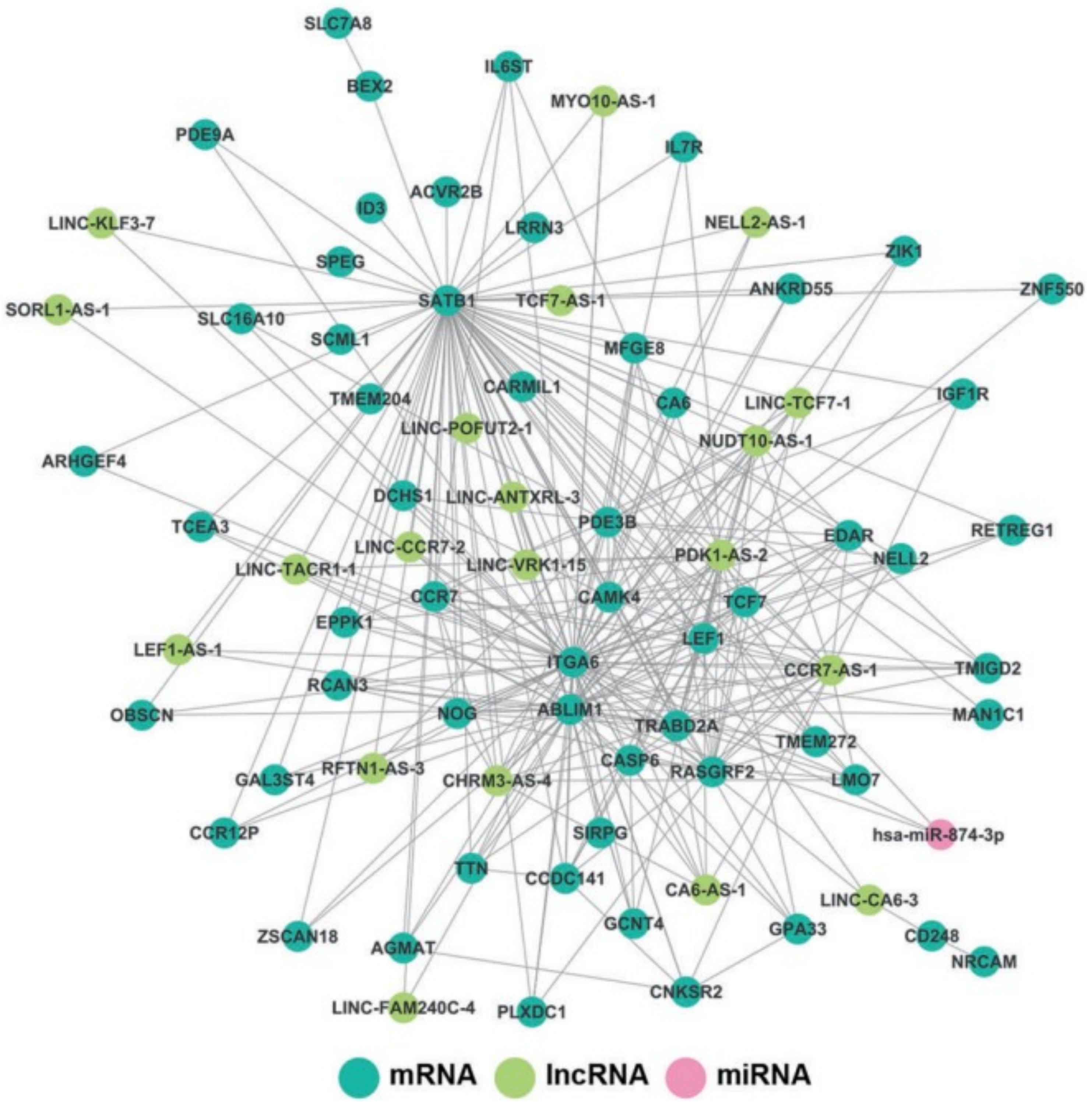


magenta
purple
yellow
grey60
green
midnightblue
black
pink
lightgreen
tan
red
lightcyan
blue
brown
salmon
cyan
greenyellow
grey
0.016 (0.9) −0.31 (0.06) −0.37 (0.02) −0.29 (0.07) −0.067 (0.7) −0.39 (0.01)
−0.42 (0.009) −0.26 (0.1) −0.092 (0.6) −0.15 (0.3) −0.4 (0.01) −0.15 (0.4)
−0.27 (0.09) −0.13 (0.4) 0.11 (0.5) 0.16 (0.3) 0.3 (0.06) −0.0052 (1)
−0.39 (0.01) 0.068 (0.7) 0.014 (0.9) −0.015 (0.9) 0.16 (0.3) 0.23 (0.2)
−0.4 (0.01) −0.21 (0.2) 0.27 (0.1) 0.096 (0.6) 0.029 (0.9) 0.13 (0.4)
−0.25 (0.1) −0.22 (0.2) 0.08 (0.6) 0.037 (0.8) 0.12 (0.5) 0.054 (0.7)
0.59 (7e−05) −0.084 (0.6) 0.018 (0.9) 0.096 (0.6) 0.065 (0.7) 0.035 (0.8)
0.12 (0.5) −0.017 (0.9) 0.06 (0.7) 0.1 (0.5) 0.058 (0.7) 0.15 (0.4)
0.064 (0.7) −0.17 (0.3) −0.012 (0.9) 0.018 (0.9) −0.036 (0.8) −0.069 (0.7)
0.62 (3e−05) −0.26 (0.1) −0.19 (0.2) −0.04 (0.8) 0.17 (0.3) −0.2 (0.2)
−0.056 (0.7) 0.058 (0.7) 0.18 (0.3) 0.24 (0.1) 0.12 (0.5) 0.34 (0.03)
0.031 (0.9) −0.11 (0.5) −0.6 (5e−05) −0.4 (0.01) 0.29 (0.07) −0.34 (0.03)
−0.2 (0.2) −0.19 (0.2) −0.27 (0.09) −0.18 (0.3) 0.2 (0.2) −0.16 (0.3)
0.22 (0.2) −0.26 (0.1) −0.22 (0.2) −0.13 (0.4) 0.23 (0.2) −0.15 (0.4)
−0.27 (0.09) 0.091 (0.6) −0.26 (0.1) −0.26 (0.1) 0.036 (0.8) −0.28 (0.08)
0.025 (0.9) 0.22 (0.2) −0.05 (0.8) 0.006 (1) 0.1 (0.5) 0.031 (0.9)
−0.38 (0.02) 0.11 (0.5) 0.27 (0.1) 0.36 (0.02) −0.099 (0.5) 0.33 (0.04)
−0.055 (0.7) 0.97 (1e−23) −0.24 (0.1) −0.058 (0.7) 0.23 (0.2) 0.14 (0.4)
Clinical stage
Gender
Age
Clinical type
Length of hospital stay
Comorbidities
1
0.5
0
−0.5
−1

SLC7A8

c


NIK/NF−kappaB signaling
I−kappaB kinase/NF−kappaB signaling
positive regulation of NF−kappaB transcription factor activity
regulation of MAP kinase activity
regulation of hemopoiesis
regulation of stress−activated MAPK cascade
JNK cascade
myeloid cell differentiation
T cell differentiation
leukocyte cell−cell adhesion
regulation of lymphocyte activation
lymphocyte differentiation
T cell activation
myeloid leukocyte differentiation
peptidyl−lysine trimethylation
lymphocyte proliferation
T cell proliferation
nucleosome assembly
protein−DNA complex assembly
chromatin silencing
protein heterotetramerization
interleukin−7−mediated signaling pathway
cell cycle G1/S phase transition
Black (117)
Lightcyan (55)
Tan (66)
GeneRatio
0.05
0.10
0.15
0.20
p.adjust
0.01
0.02
0.03


SLC7A8
IL6ST
BEX2
MYO10-AS-1
PDE9A
IL7R
ACVR2B
ID3
LINC-KLF3-7
LRRN3
NELL2-AS-1
ZIK1
SPEG
ANKRD55
ZNF550
SORL1-AS-1
SATB1
TCF7-AS-1
SLC16A10
SCML1
MFGE8
CARMIL1
IGF1R
TMEM204
CA6
LINC-TCF7-1
LINC-POFUT2-1
NUDT10-AS-1
ARHGEF4
DCHS1
LINC-ANTXRL-3
PDE3B
TCEA3
RETREG1
EDAR
LINC-CCR7-2
PDK1-AS-2
NELL2
LINC-TACR1-1
LINC-VRK1-15
CCR7
CAMK4
TCF7
EPPK1
LEF1
LEF1-AS-1
ITGA6
CCR7-AS-1
TMIGD2
RCAN3
OBSCN
NOG
ABLIM1
MAN1C1
TRABD2A
TMEM272
CASP6
GAL3ST4
RFTN1-AS-3
CHRM3-AS-4
RASGRF2
LMO7
CCR12P
SIRPG
hsa-miR-874-3p
TTN
CCDC141
CA6-AS-1
LINC-CA6-3
GCNT4
GPA33
CD248
ZSCAN18
AGMAT
NRCAM
CNKSR2
LINC-FAM240C-4
PLXDC1
mRNA
lncRNA
miRNA

ARTICLE

doi:10.1038/nature09715

The developmental transcriptome of *Drosophila melanogaster*

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Drosophila melanogaster is one of the most well studied genetic model organisms; nonetheless, its genome still contains unannotated coding and non-coding genes, transcripts, exons and RNA editing sites. Full discovery and annotation are pre-requisites for understanding how the regulation of transcription, splicing and RNA editing directs the development of this complex organism. Here we used RNA-Seq, tiling microarrays and cDNA sequencing to explore the transcriptome in 30 distinct developmental stages. We identified 111,195 new elements, including thousands of genes, coding and non-coding transcripts, exons, splicing and editing events, and inferred protein isoforms that previously eluded discovery using established experimental, prediction and conservation-based approaches. These data substantially expand the number of known transcribed elements in the _Drosophila_ genome and provide a high-resolution view of transcriptome dynamics throughout development.

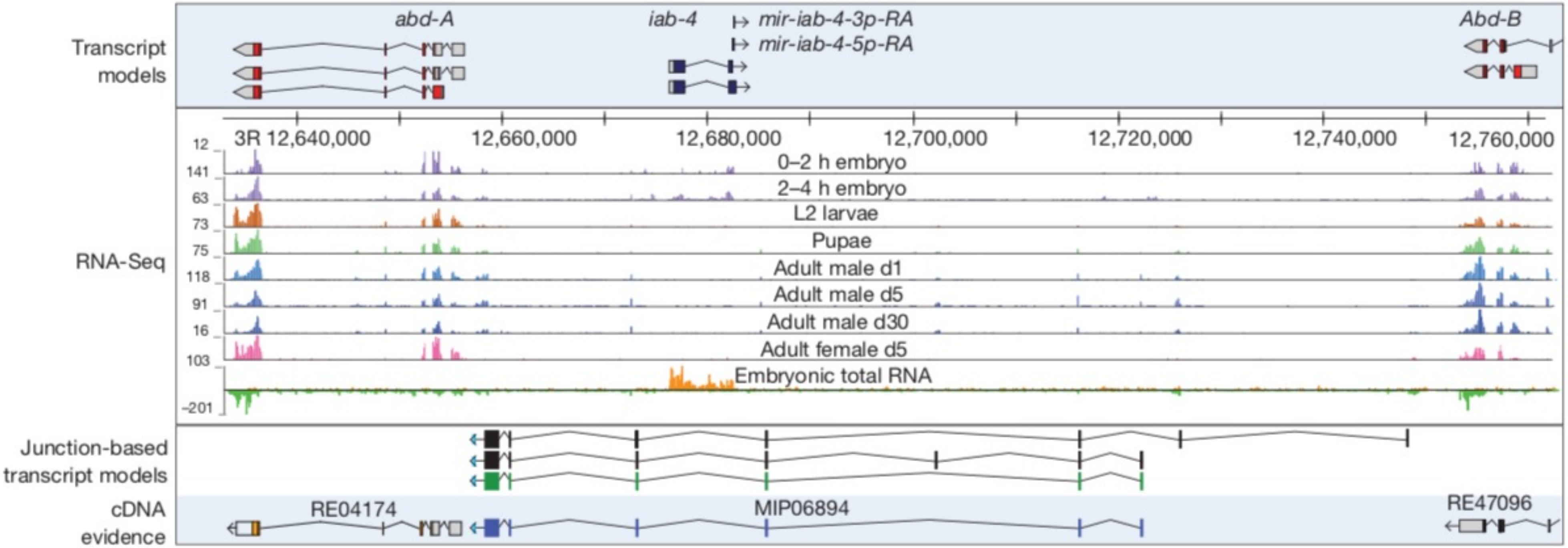


Figure 1 | Discovery of new RNAs in the Bithorax complex. Genomic organization and experimental evidence for new transcripts located between the HOX genes, *abd-A* and *Abd-B*, based on short poly(A)$^{+}$ RNA and total RNA-Seq expression profiles. The numbers to the left of each track indicate the maximal number of reads for that sample. Three manually curated junction-based transcript models are shown; the green transcript model was fully validated by a cDNA, MIP06894.

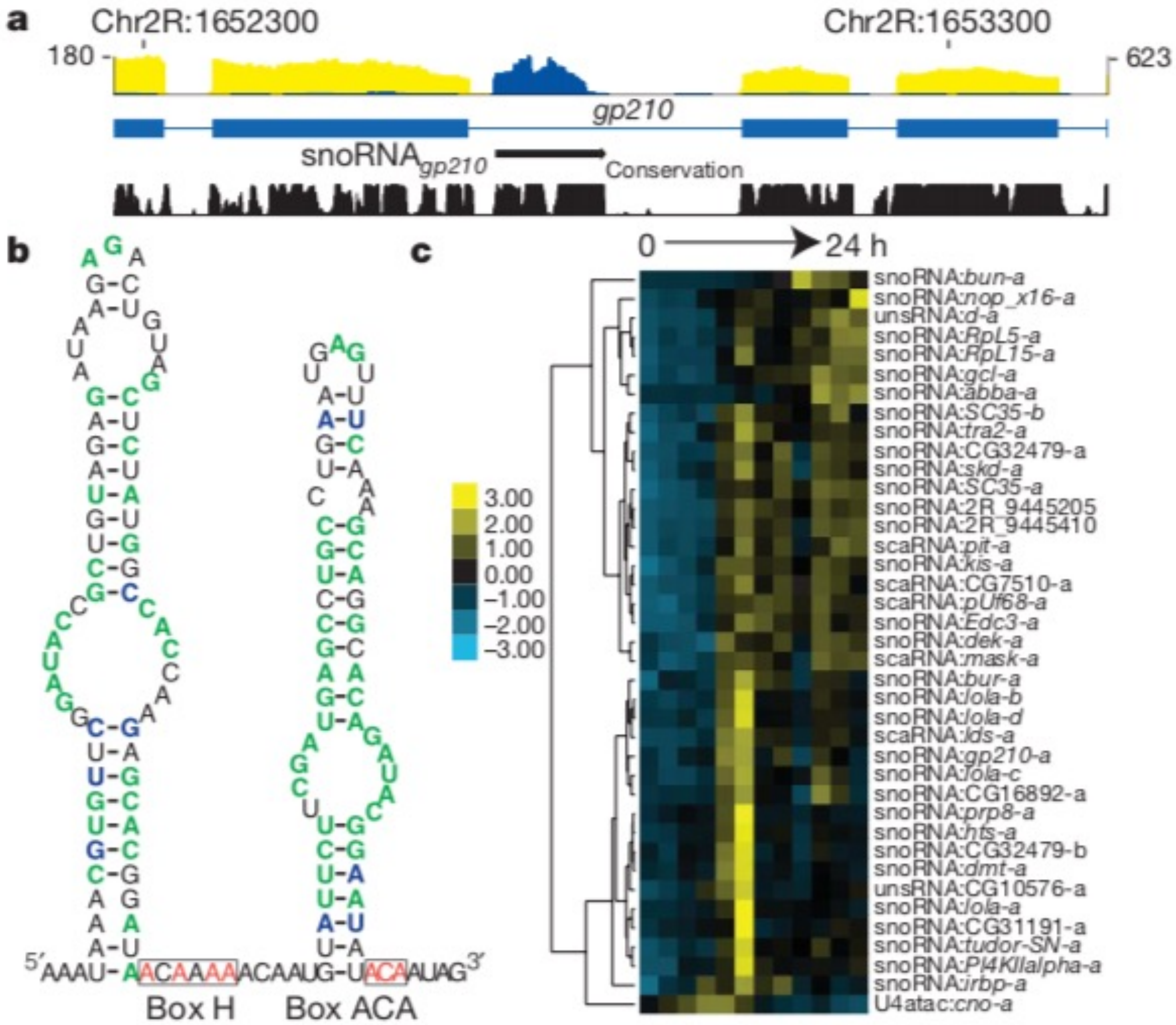


Figure 2 | Discovery of small non-coding RNAs. **a**, Poly(A)$^+$ (yellow) and total RNA (blue) data from 10–12-h embryos are shown for the *gp210* gene which hosts a representative new snoRNA. The maximal number of reads in the poly(A)$^+$ and total RNA-Seq data are shown on the left and right of the track, respectively. **b**, The predicted RNA secondary structure of snoRNA$_{gp210}$ is characteristic of a H/ACA-box snoRNA. Nucleotides that are 100% conserved in sequence or base-pairing are indicated in green and blue, respectively. **c**, Embryonic expression of the new small RNAs. The scale bar indicates FPKM *Z*-scores. unsRNA, unclassified small RNA.

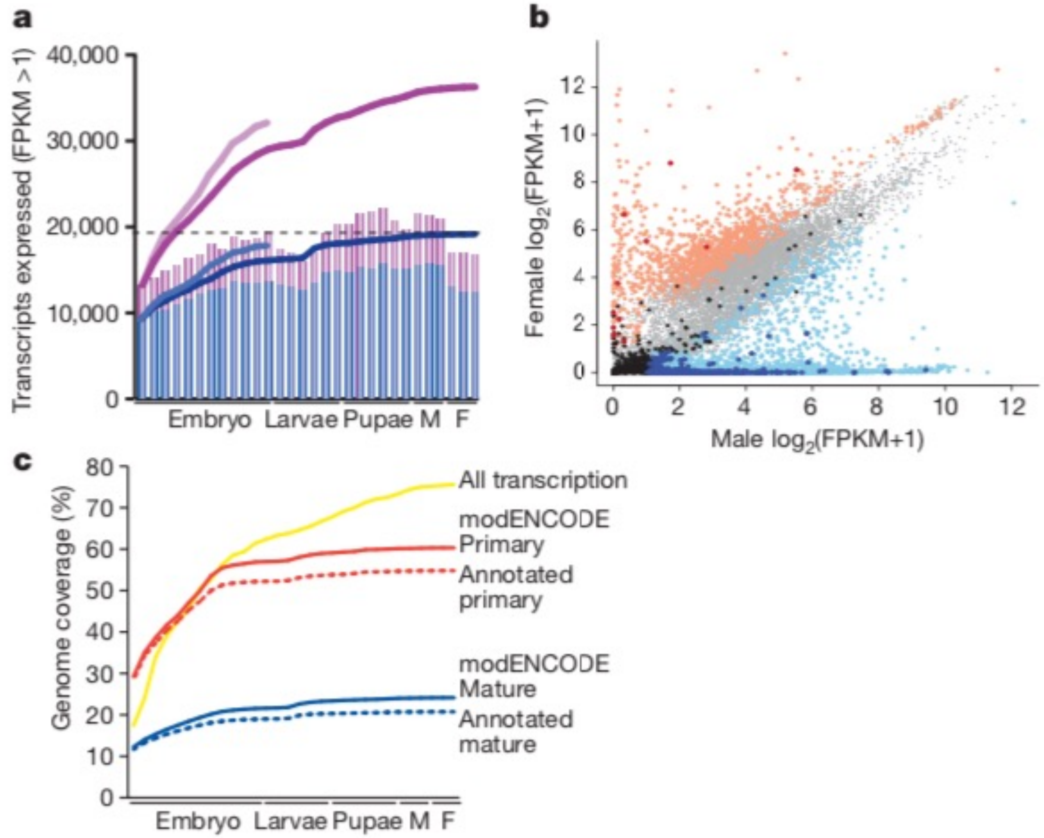


Figure 3 | Dynamics of gene expression. **a**, Transcripts expressed (FPKM >1) in the short poly(A)$^{+}$ RNA-Seq data: FlyBase 5.12, blue; modENCODE, purple. The bar graphs indicate the number of transcripts expressed in each sample (Supplementary Table 1); the lines indicate the cumulative number of expressed transcripts. The lighter blue and purple lines indicate the cumulative number of transcripts expressed in the embryonic total RNA-Seq samples. The horizontal dotted lines indicate the number of expressed previously annotated transcripts. F, female; M, male. **b**, Scatter plot of sex-biased gene expression. Light red, female-biased annotated ($n = 960$); dark red, female-biased NTRs ($n = 12$); light blue, male-biased annotated ($n = 2,401$); dark blue, male-biased NTRs ($n = 431$); light grey, unbiased annotated ($n = 8,217$); black, unbiased NTRs ($n = 136$). **c**, Genome coverage. For each developmental sample, the short poly(A)$^{+}$ reads were used to estimate the percentage of the genome covered using a cutoff of two reads. The mature and primary transcripts were inferred for the previously FlyBase 5.12 (dotted lines) and modENCODE (solid lines) gene models.

Table 1 | Classification of alternative splicing events

Splicing event	Diagram	FlyBase r5.12	modENCODE	New events	Short poly(A)$^{+}$ RNA-Seq	Significantly changing
Cassette exons		793	2,717	2,014	2,369	1,539
Alternative 5′ splice sites		843	5,192	4,599	4,583	3,142
Alternative 3′ splice sites		879	6,253	5,505	5,579	3,242
Mutually exclusive exons		229	251	123	228	226
Coordinate cassette exons		301	1,227	979	992	467
Alternative first exons		1,767	4,936	3,442	4,473	3,996
Alternative last exons		227	604	432	553	471
Retained/unprocessed introns		1,434	2,679 (5,667)	1,275 (4,263)	2,439 (35,641)	868 (8,998)
Total		6,437	23,859 (26,847)	18,369 (21,478)	21,216 (54,418)	13,951 (22,081)

The number of retained/unprocessed introns in parentheses indicates the total number identified, whereas the number not in parentheses indicates the subset of identified events that have been validated by cDNA sequences or FlyBase 5.12 annotations.

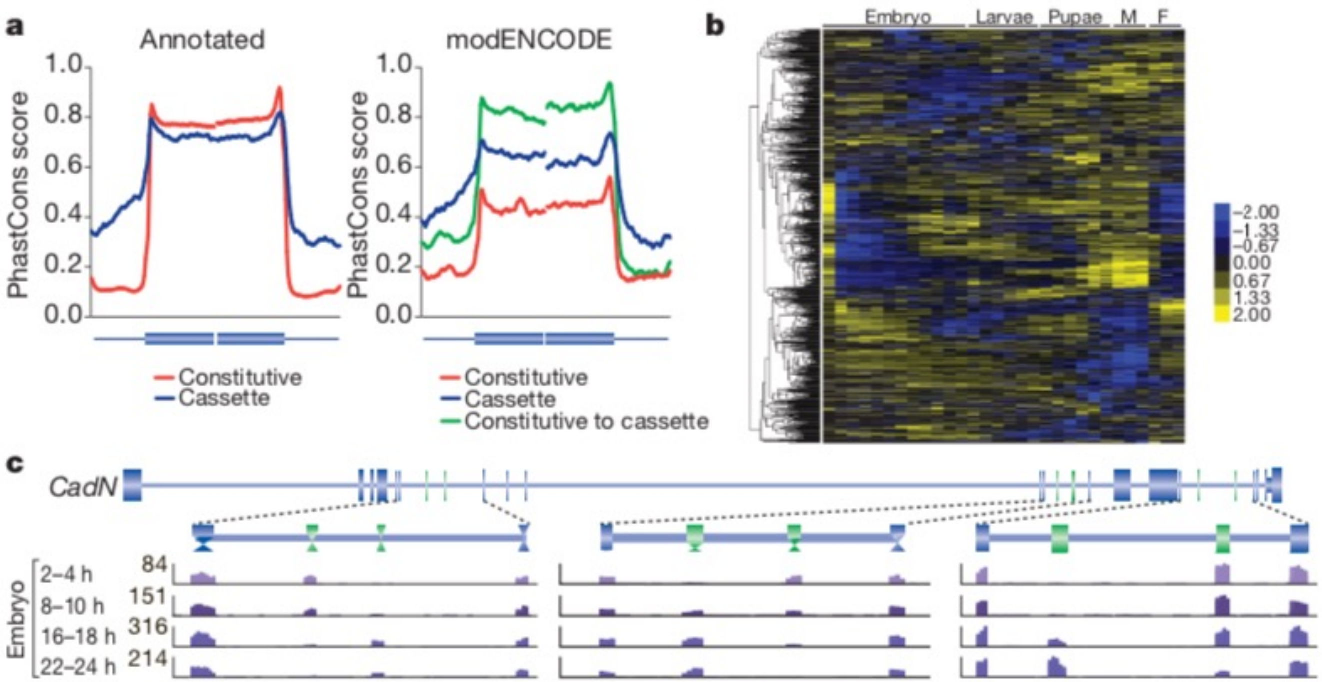


Figure 4 | Developmentally regulated splicing events. a, Conservation of internal constitutive and cassette exons >50 nucleotides that were annotated or new discoveries. (Annotated constitutive, $n = 26{,}127$; annotated cassette, $n = 438$; modENCODE cassette, $n = 173$; modENCODE constitutive, $n = 306$; FlyBase 5.12 constitutive to modENCODE cassette, $n = 304$.) **b**, Clusters of regulated cassette exon events during development. The scale bar indicates Z-scores of Ψ. **c**, Regulated alternative splicing in *CadN* during embryogenesis. The maximal number of reads in the poly(A)$^+$ RNA-Seq data are indicated for each track.

ARTICLE

OPEN
doi:10.1038/nature12962

Diversity and dynamics of the *Drosophila* transcriptome

James B. Brown[1,2]*, Nathan Boley[1]*, Robert Eisman[3]*, Gemma E. May[4]*, Marcus H. Stoiber[1]*, Michael O. Duff[4], Ben W. Booth[2], Jiayu Wen[5], Soo Park[2], Ana Maria Suzuki[6,7], Kenneth H. Wan[2], Charles Yu[2], Dayu Zhang[8], Joseph W. Carlson[2], Lucy Cherbas[3], Brian D. Eads[3], David Miller[3], Keithanne Mockaitis[3], Johnny Roberts[8], Carrie A. Davis[9], Erwin Frise[2], Ann S. Hammonds[2], Sara Olson[4], Sol Shenker[5], David Sturgill[10], Anastasia A. Samsonova[11,12], Richard Weiszmann[2], Garret Robinson[1], Juan Hernandez[1], Justen Andrews[3], Peter J. Bickel[1], Piero Carninci[6,7], Peter Cherbas[3,8], Thomas R. Gingeras[9], Roger A. Hoskins[2], Thomas C. Kaufman[3], Eric C. Lai[5], Brian Oliver[10], Norbert Perrimon[11,12], Brenton R. Graveley[4] & Susan E. Celniker[2]

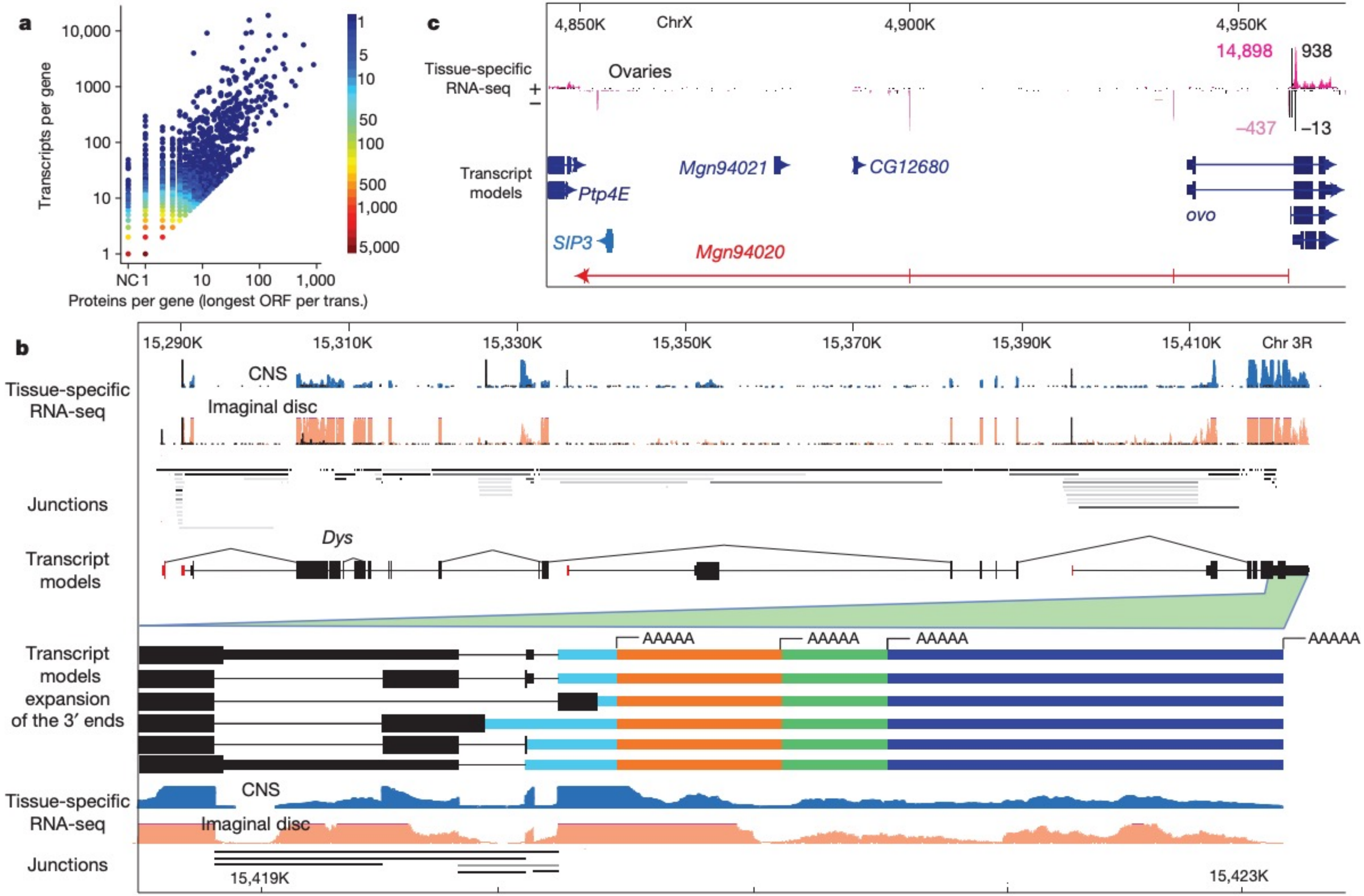
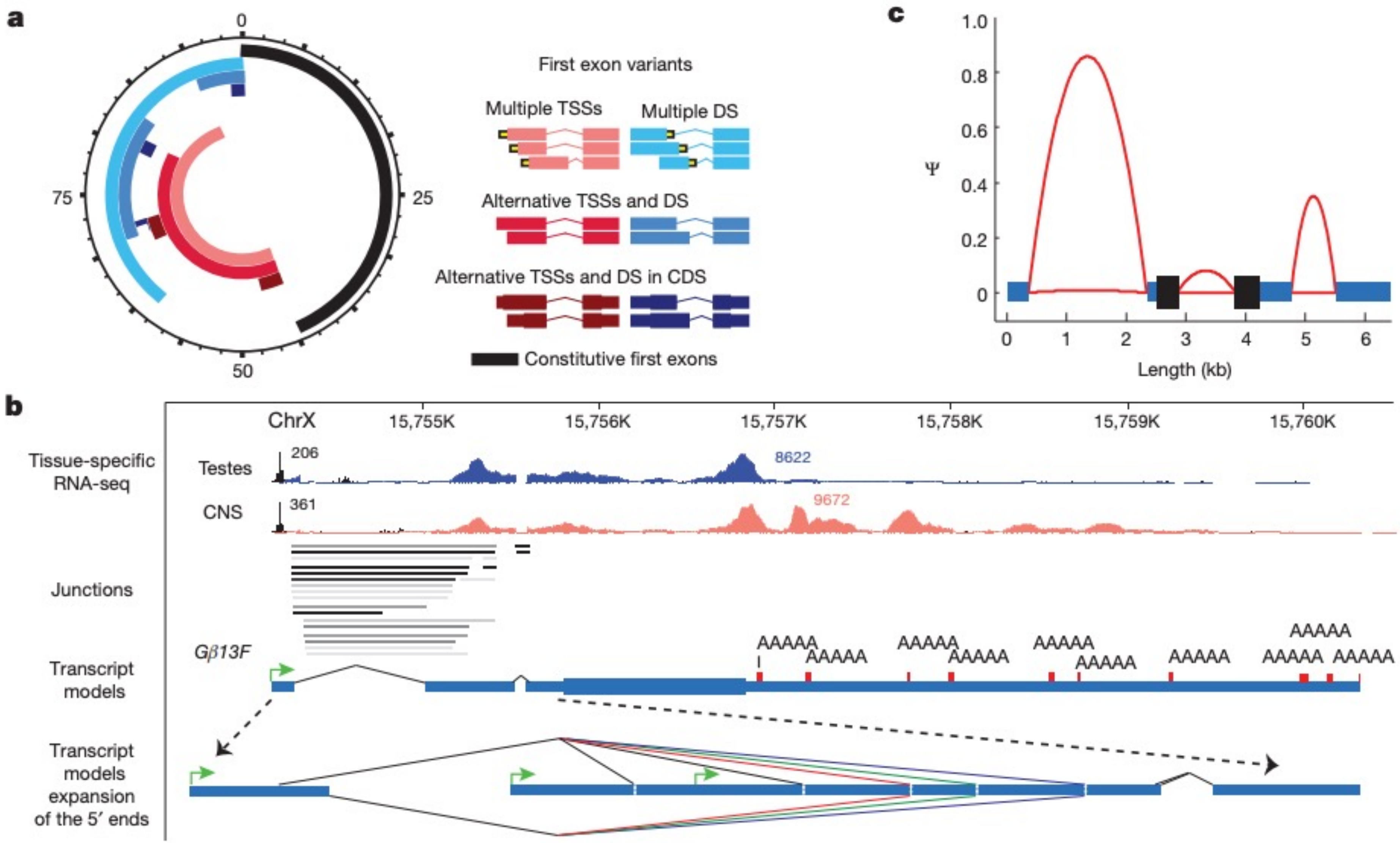


Figure 1 | Overview of the annotation of the *Drosophila melanogaster* transcriptome. **a**, Scatterplot showing the per gene correlation between number of proteins and number of transcripts. The genes *Dscam* and *para* are omitted as extreme outliers both encoding >10,000 unique proteins. **b**, *Dystrophin* (*Dys*) produces 72 transcripts and encodes 32 proteins. Highlighted is alternative splicing and polyadenylation at the 3′ end. CAGE (black), RNA-seq (tan, blue), splice junctions (shaded grey as a function of usage). **c**, An internal promoter of *ovo* is bidirectional in ovaries and produces a lncRNA (430 bp, red) bridging two gene deserts. CAGE (black), RNA-seq (pink), counts are read-depth (minus-strand given as negative).

Splicing complexity across the gene body



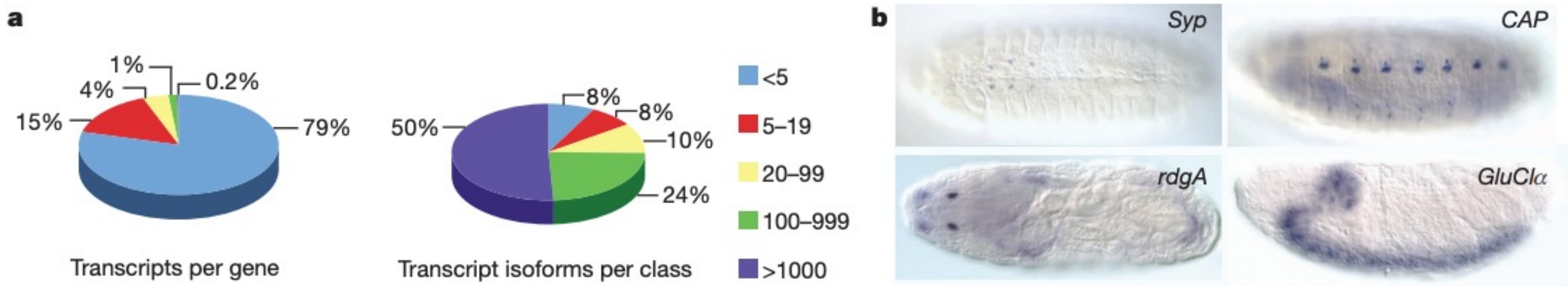


Figure 3 | Complex splicing patterns are mainly limited to neural tissues. **a**, A small minority of genes (47, 0.2%) encode most transcripts. **b**, *In situ* RNA staining of constitutive exons of four genes with highly complex splicing patterns in the embryo. *Syncrip* (*Syp*), *CAP*, *retinal degeneration A* (*rdgA*) and *GluClα* show specific late embryonic neural expression in the ventral midline neurons; dorsal/lateral and ventral sensory complexes; Bolwig's organ or larval eye; and central nervous system, respectively.

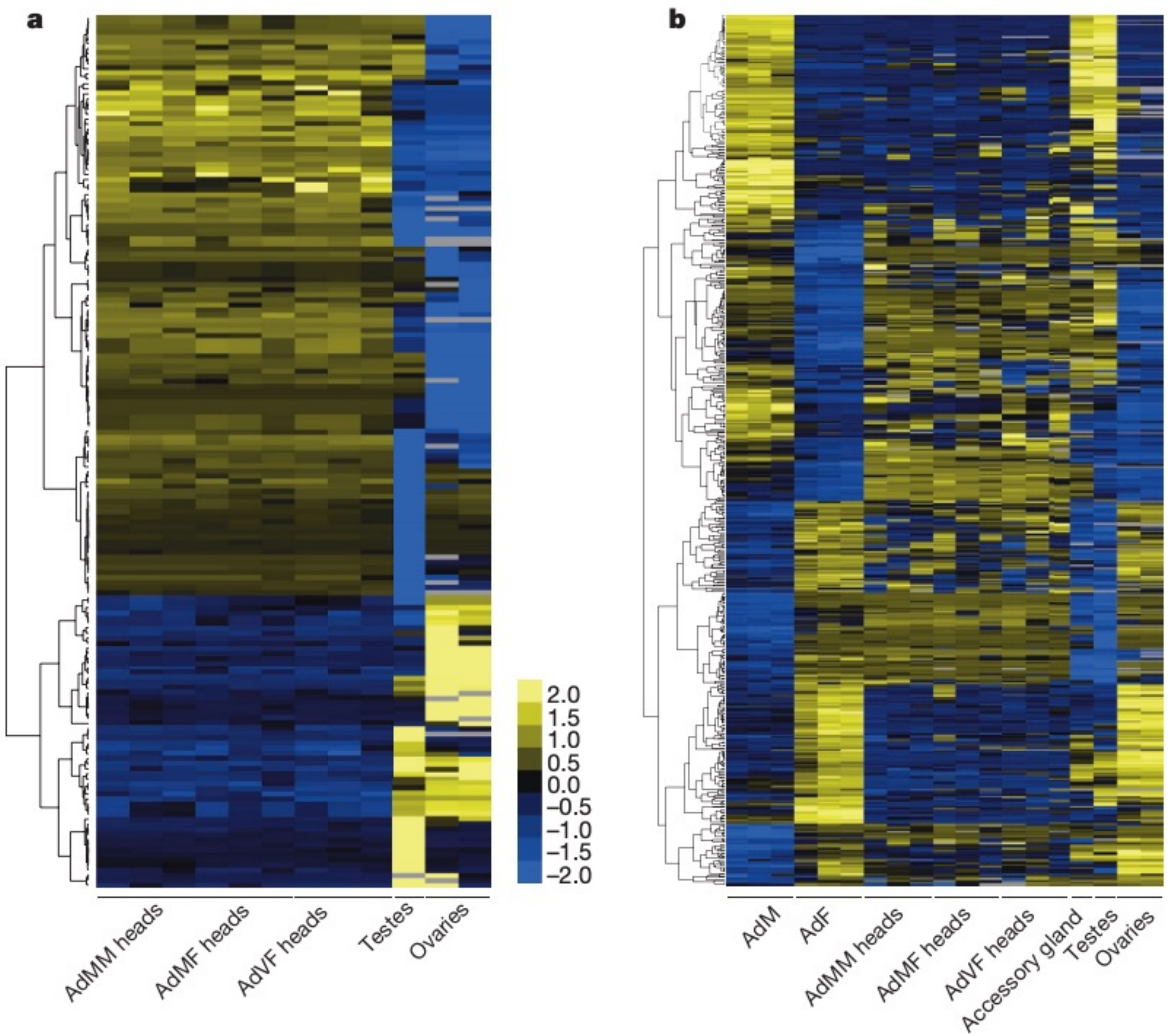


Figure 4 | Sex-specific splicing is mainly tissue-specific splicing. a, Clusters of tissue-specific splicing events. The scale bar indicates *z*-scores of Ψ. Adult mated male (AdMM), adult mated female (AdMF) and adult virgin female (AdVF) heads are from 1-, 4- and 20-day-old animals, respectively. Testes are from 4-day-old adult males, and ovaries are from mated and virgin 4-day-old adult females. **b,** Sex-specific splicing events in whole animals are primarily testes- or ovary-specific splicing events. Adult male (AdM) and adult female (AdF) animals are 1, 5 and 30 days old. Accessory glands were dissected from 4-day-old adult males. The RNA-seq columns from heads, testes and ovaries are as described in **a**.

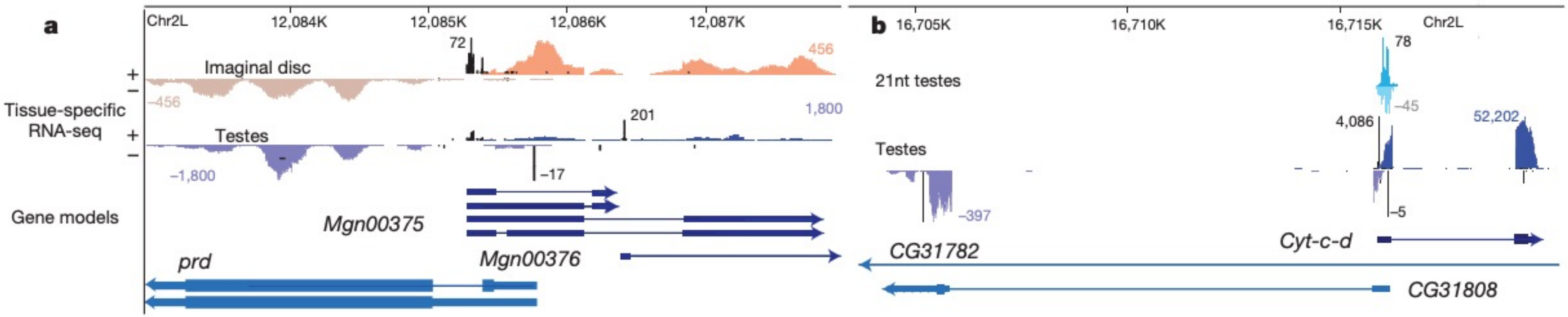


Figure 5 | Examples of antisense transcription. a, 5′/5′ overlapping bidirectional antisense transcription at the *prd* locus. Short RNA sequencing does not reveal substantial siRNA (that is, 21-nt-dominant small RNA) signal in this region (data not shown). **b,** A 5′/5′ antisense region that produces substantial small RNA signal on both strands. nt, nucleotide.

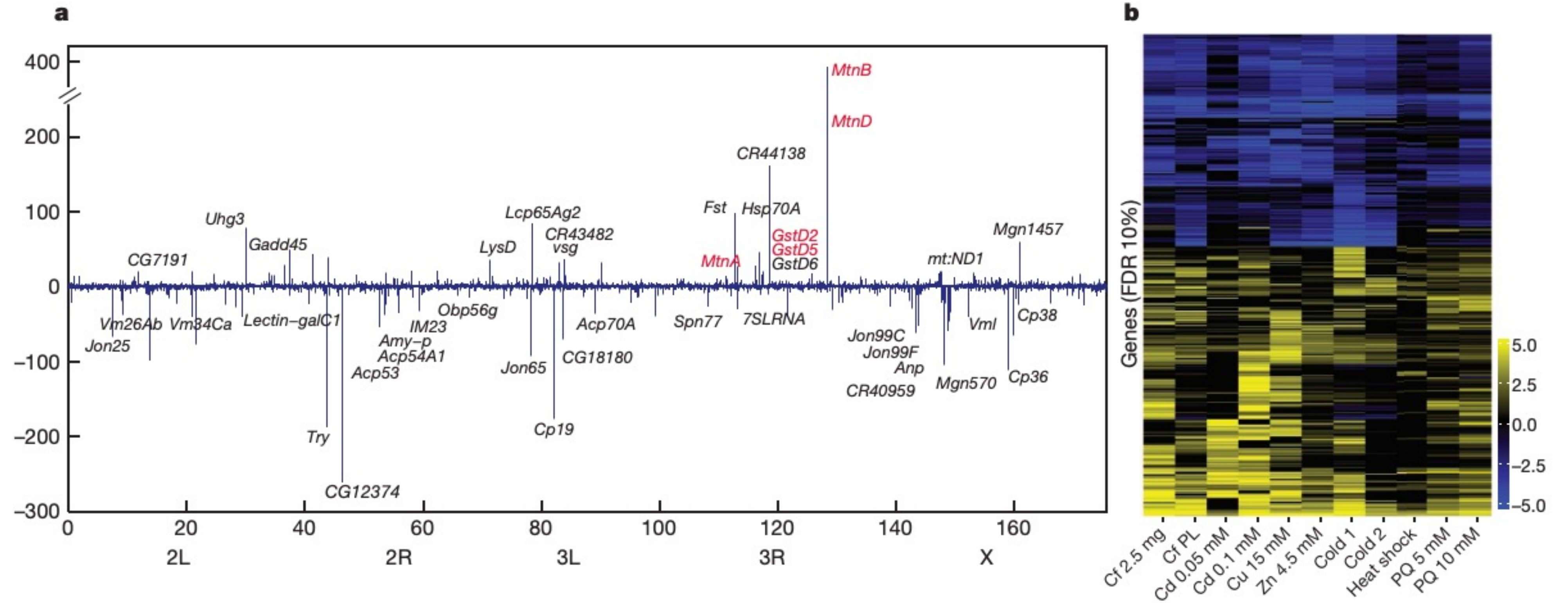
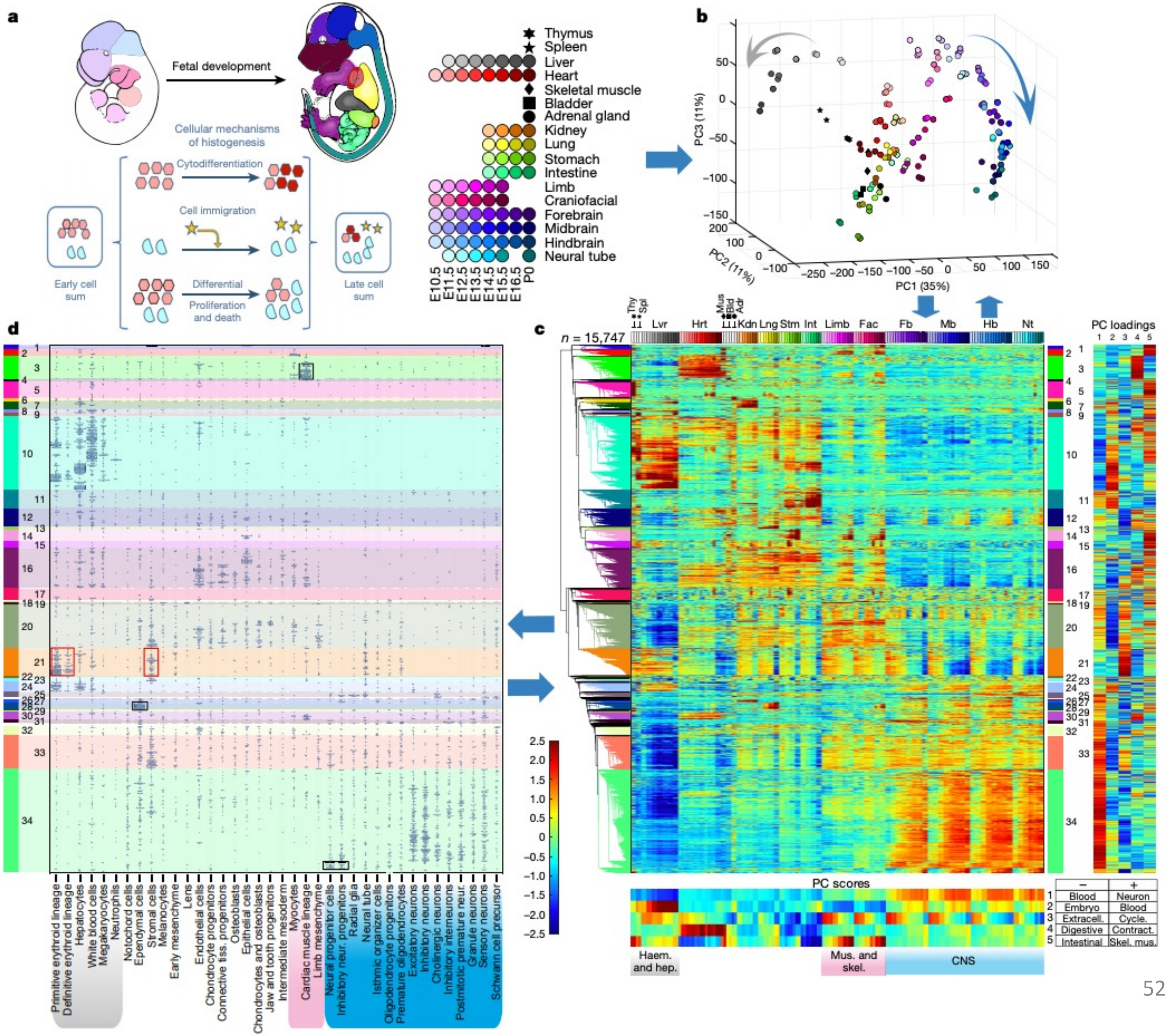


Figure 6 | Effects of environmental perturbations on the *Drosophila* transcriptome. Adults were treated with caffeine (Cf), Cd, Cu, Zn, cold, heat or paraquat (PQ). **a**, A genome-wide map of genes that are up- or downregulated as a function of Cd treatment. Labelled genes are those that showed a 20-fold (<10% FDR) change in response (linear scale). Genes highlighted in red are those identified in larvae[50]. Some genes are omitted for readability, the complete figure and list of omitted genes are given in Supplementary Fig. 8a. **b**, Heat map showing the fold change of genes with a FDR < 10% (differential expression) in at least one sample ($\log_2$ scale). PL, pre-lethal.

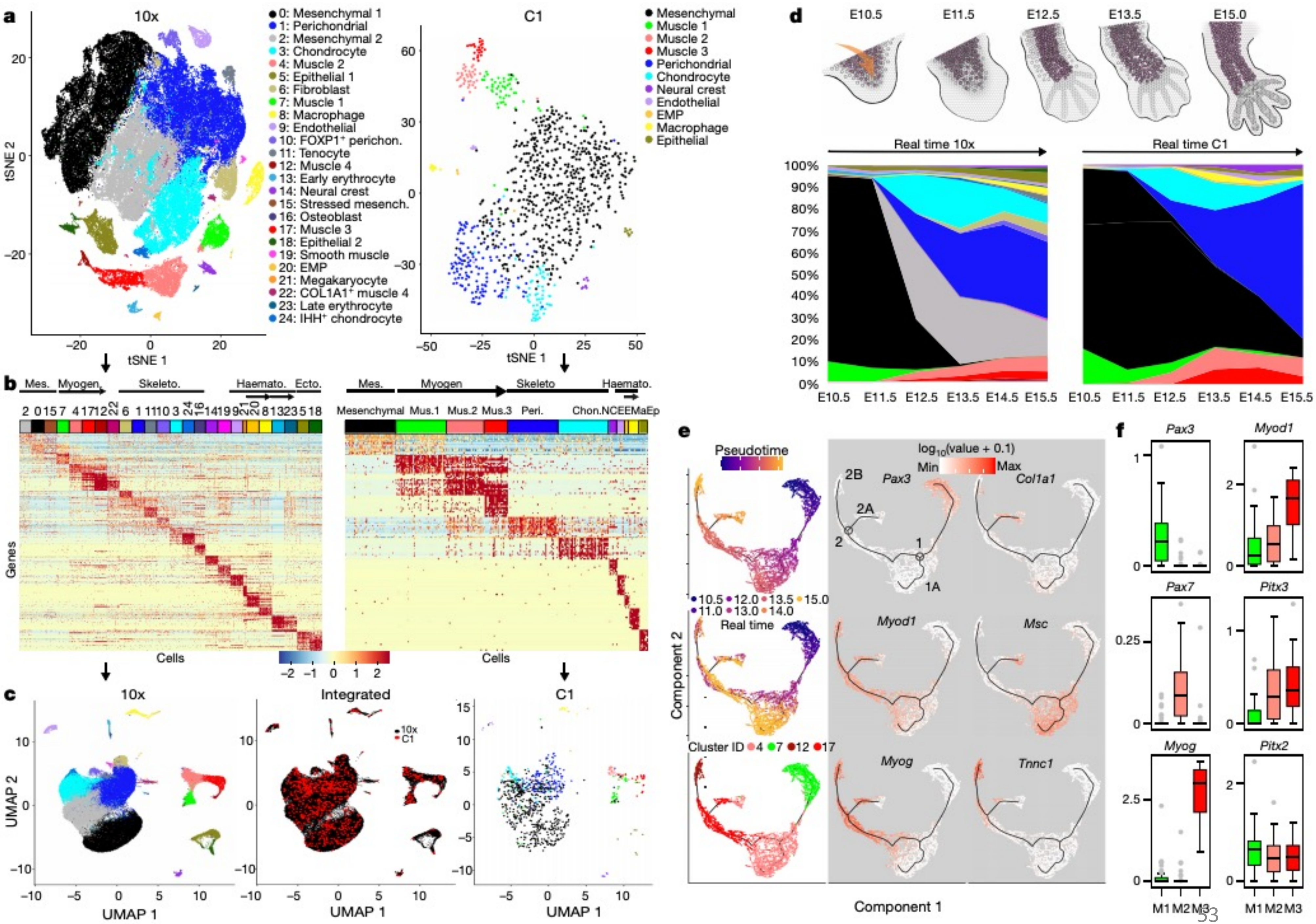
The changing mouse embryo transcriptome at whole tissue and single-cell resolution

Peng He, Brian A. Williams, Diane Trout, Georgi K. Marinov, Henry Amrhein, Libera Berghella, Say-Tar Goh, Ingrid Plajzer-Frick, Veena Afzal, Len A. Pennacchio, Diane E. Dickel, Axel Visel, Bing Ren, Ross C. Hardison, Yu Zhang & Barbara J. Wold

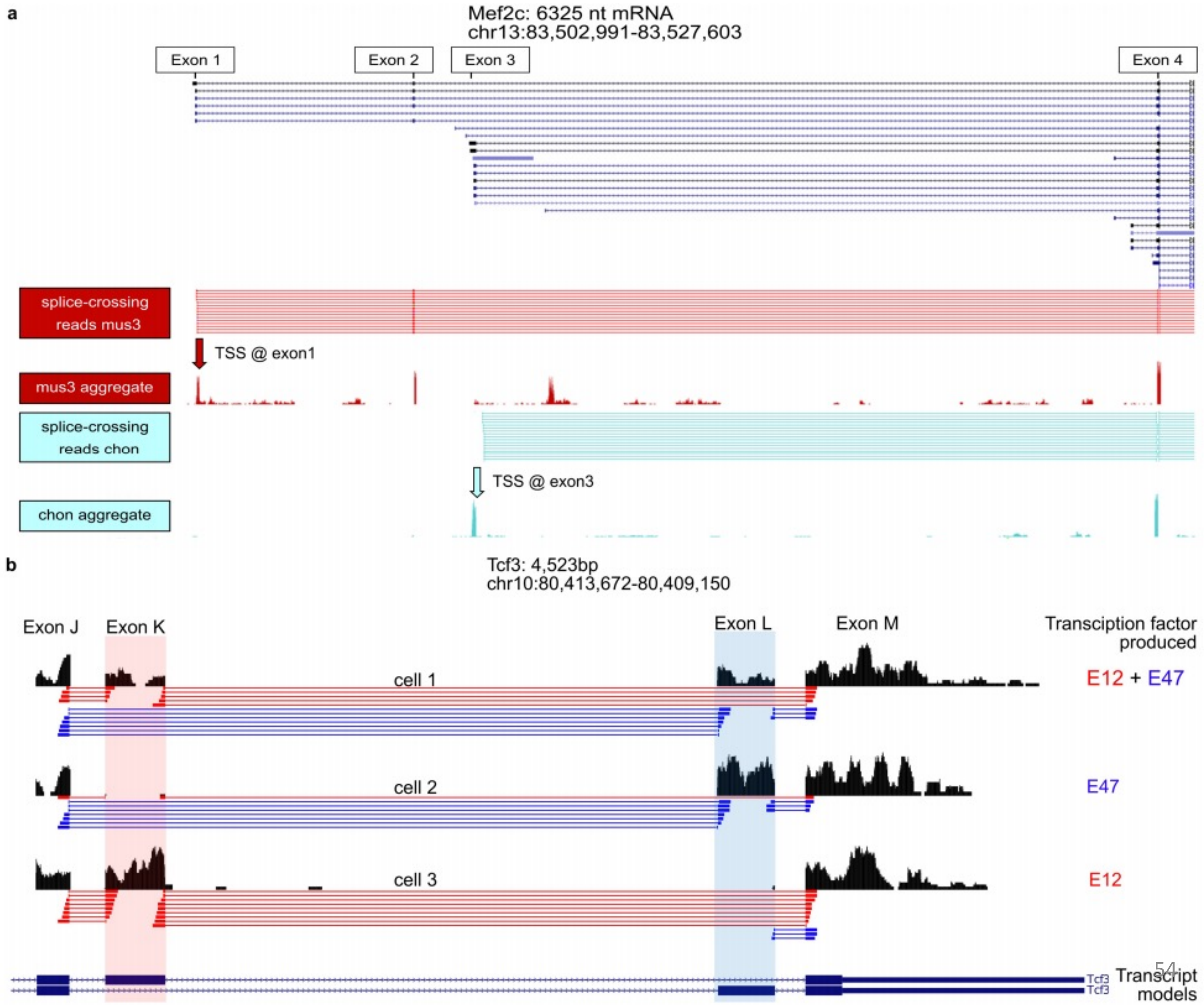
Nature **583**, 760–767(2020) | Cite this article



a
Fetal development
Cellular mechanisms of histogenesis
Cytodifferentiation
Cell immigration
Differential Proliferation and death
Early cell sum
Late cell sum
Thymus
Spleen
Liver
Heart
Skeletal muscle
Bladder
Adrenal gland
Kidney
Lung
Stomach
Intestine
Limb
Craniofacial
Forebrain
Midbrain
Hindbrain
Neural tube
E10.5 E11.5 E12.5 E13.5 E14.5 E15.5 E16.5 P0
b
PC3 (11%)
PC2 (11%)
PC1 (35%)
c
n = 15,747
Thy Spl Lvr Hrt Mus Bld Adr Kdn Lng Stm Int Limb Fac Fb Mb Hb Nt
PC loadings
PC scores
Haem. and hep.
Mus. and skel.
CNS
− +
1 Blood Neuron
2 Embryo Blood
3 Extracell. Cycle.
4 Digestive Contract.
5 Intestinal Skel. mus.
d
Primitive erythroid lineage
Definitive erythroid lineage
Hepatocytes
White blood cells
Megakaryocytes
Neutrophils
Notochord cells
Ependymal cells
Stromal cells
Melanocytes
Early mesenchyme
Lens
Endothelial cells
Chondrocyte progenitors
Connective tiss. progenitors
Osteoblasts
Epithelial cells
Chondrocytes and osteoblasts
Jaw and tooth progenitors
Intermediate mesoderm
Myocytes
Cardiac muscle lineage
Limb mesenchyme
Neural progenitor cells
Inhibitory neur. progenitors
Radial glia
Neural tube
Isthmic organizer cells
Oligodendrocyte progenitors
Premature oligodendrocytes
Excitatory neurons
Inhibitory neurons
Cholinergic neurons
Inhibitory interneurons
Postmitotic premature neur.
Granule neurons
Sensory neurons
Schwann cell precursor

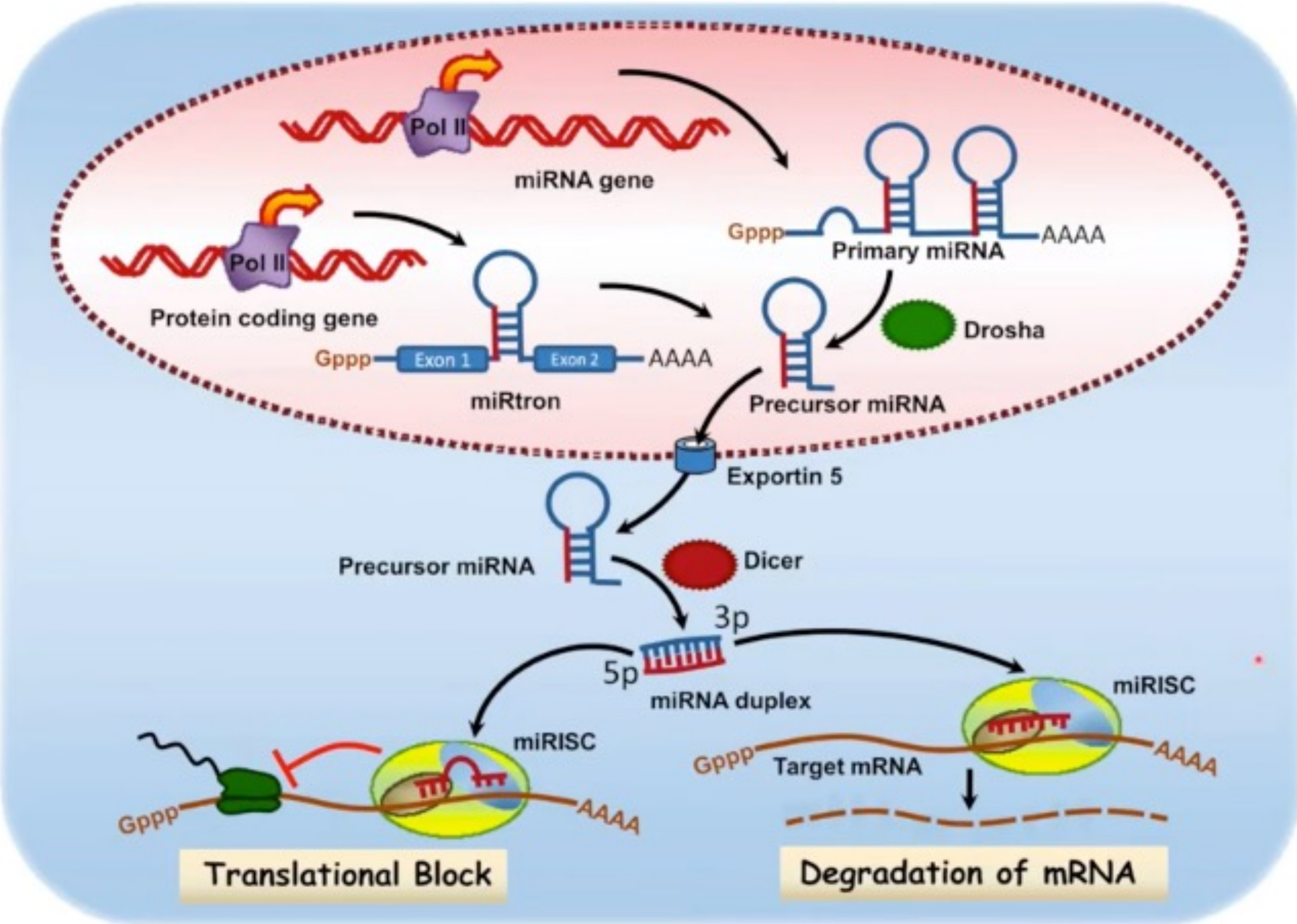


a
10x
0: Mesenchymal 1
1: Perichondrial
2: Mesenchymal 2
3: Chondrocyte
4: Muscle 2
5: Epithelial 1
6: Fibroblast
7: Muscle 1
8: Macrophage
9: Endothelial
10: FOXP1+ perichon.
11: Tenocyte
12: Muscle 4
13: Early erythrocyte
14: Neural crest
15: Stressed mesench.
16: Osteoblast
17: Muscle 3
18: Epithelial 2
19: Smooth muscle
20: EMP
21: Megakaryocyte
22: COL1A1+ muscle 4
23: Late erythrocyte
24: IHH+ chondrocyte
tSNE 1
tSNE 2
C1
Mesenchymal
Muscle 1
Muscle 2
Muscle 3
Perichondrial
Chondrocyte
Neural crest
Endothelial
EMP
Macrophage
Epithelial
d
E10.5
E11.5
E12.5
E13.5
E15.0
Real time 10x
Real time C1
b
Mes.
Myogen.
Skeleto.
Haemato.
Ecto.
Mesenchymal
Mus.1
Mus.2
Mus.3
Peri.
Chon.
Genes
Cells
c
10x
Integrated
10x
C1
UMAP 1
UMAP 2
e
Pseudotime
Real time
Cluster ID
log10(value + 0.1)
Min
Max
Pax3
Col1a1
Myod1
Msc
Myog
Tnnc1
Component 1
Component 2
f
Pax3
Myod1
Pax7
Pitx3
Myog
Pitx2
M1
M2
M3



a
Mef2c: 6325 nt mRNA
chr13:83,502,991-83,527,603
Exon 1
Exon 2
Exon 3
Exon 4
splice-crossing reads mus3
TSS @ exon1
mus3 aggregate
splice-crossing reads chon
TSS @ exon3
chon aggregate
b
Tcf3: 4,523bp
chr10:80,413,672-80,409,150
Exon J
Exon K
Exon L
Exon M
Transciption factor produced
cell 1
E12 + E47
cell 2
E47
cell 3
E12
Tcf3
Tcf3
Transcript models

MicroRNAs as post-transcriptional regulators of mRNA



(Comer, 2015)

Today: from very short to very long

RNA Length

microRNAs

21-28 nt long

UGAGGUAGUAGGUUGUAUAGUU
let7a-1

ENCODE3/4: sequence microRNAs

First Half

messenger RNAs

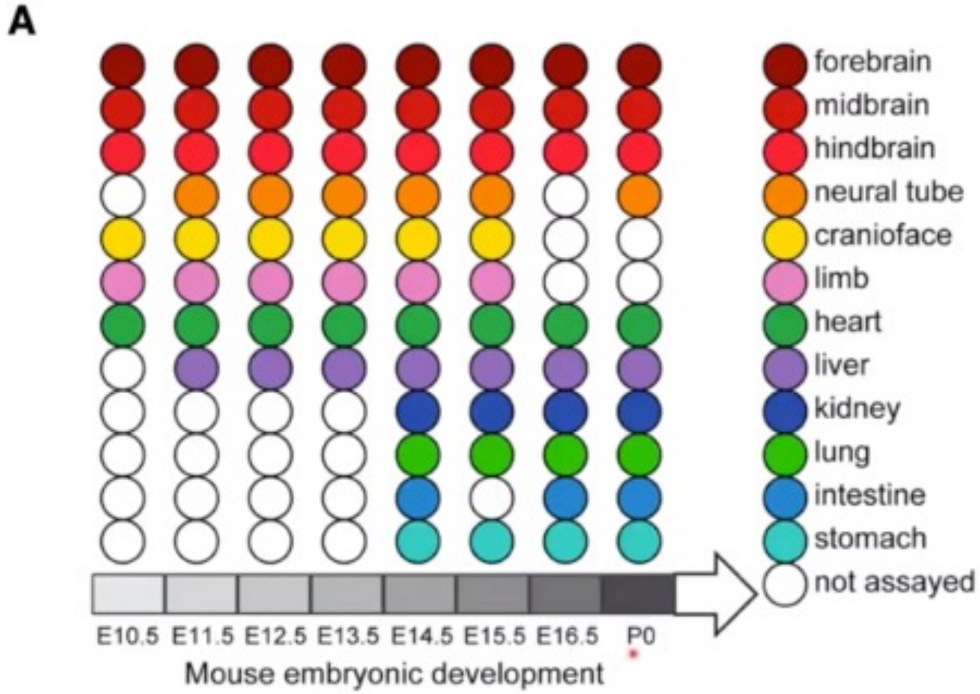
300-110,000 nt long

Most annotated mRNAs less than 5kb

ENCODE4: full-length transcript isoforms

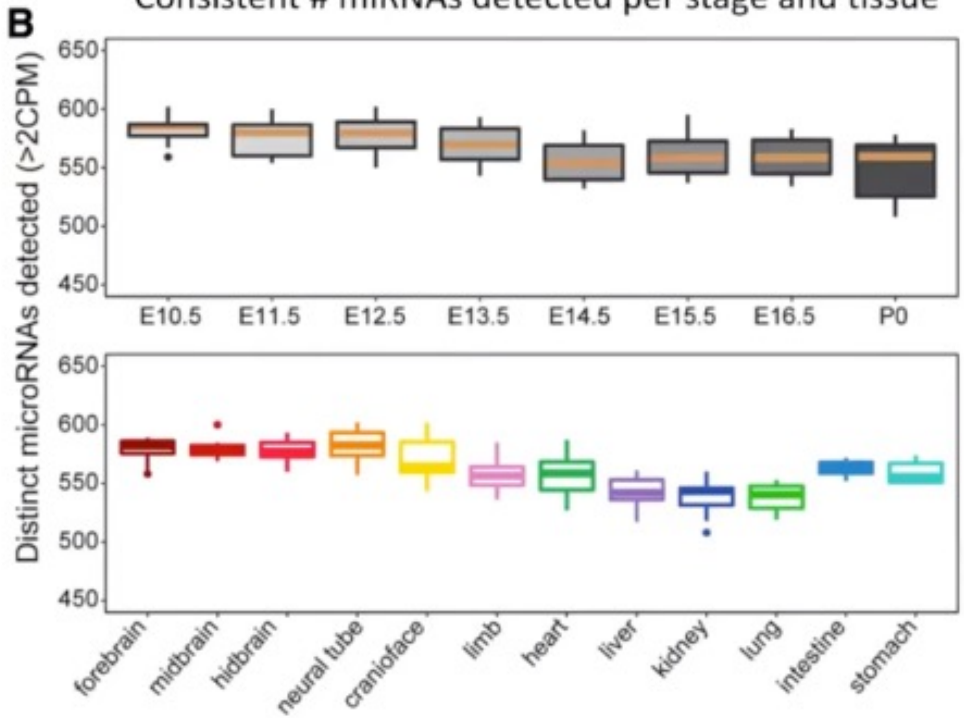
Second Half

Overview of mouse ENCODE miRNA data sets



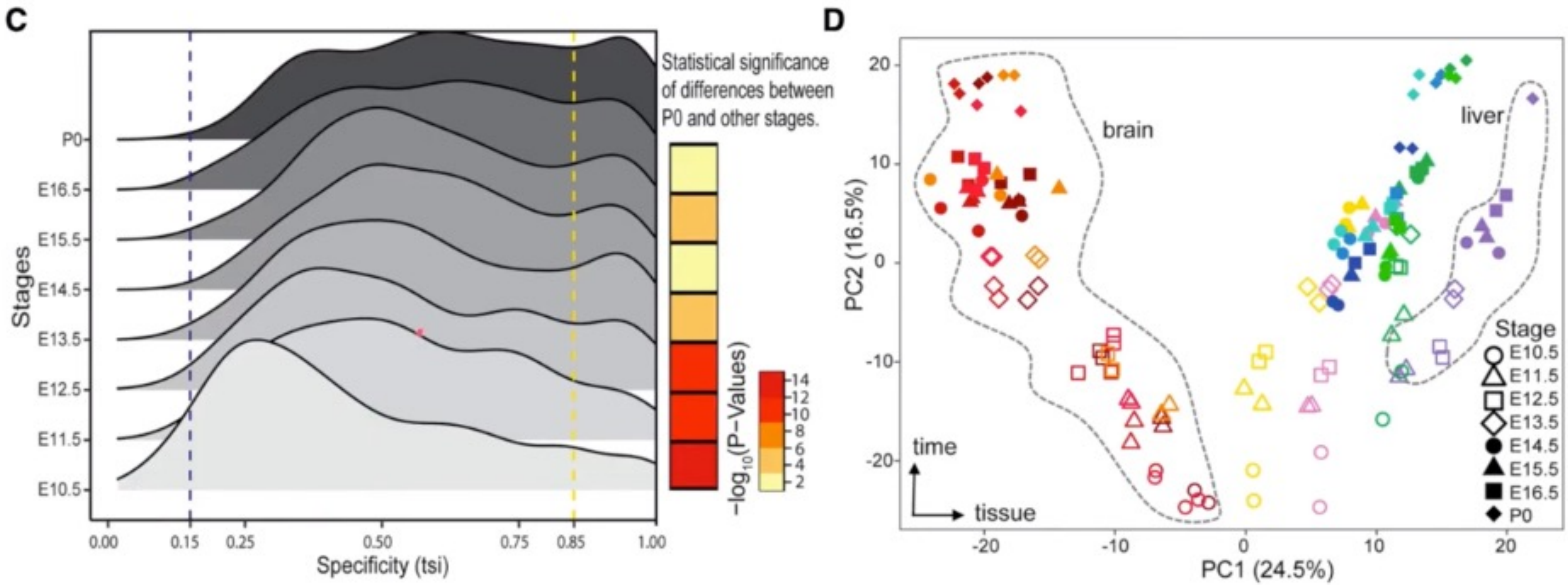
Matches other ENCODE3 mouse samples

Consistent # miRNAs detected per stage and tissue



Overall, detected 785 distinct miRNA > 2 CPM

MicroRNA expression profiles become more tissue specific during development

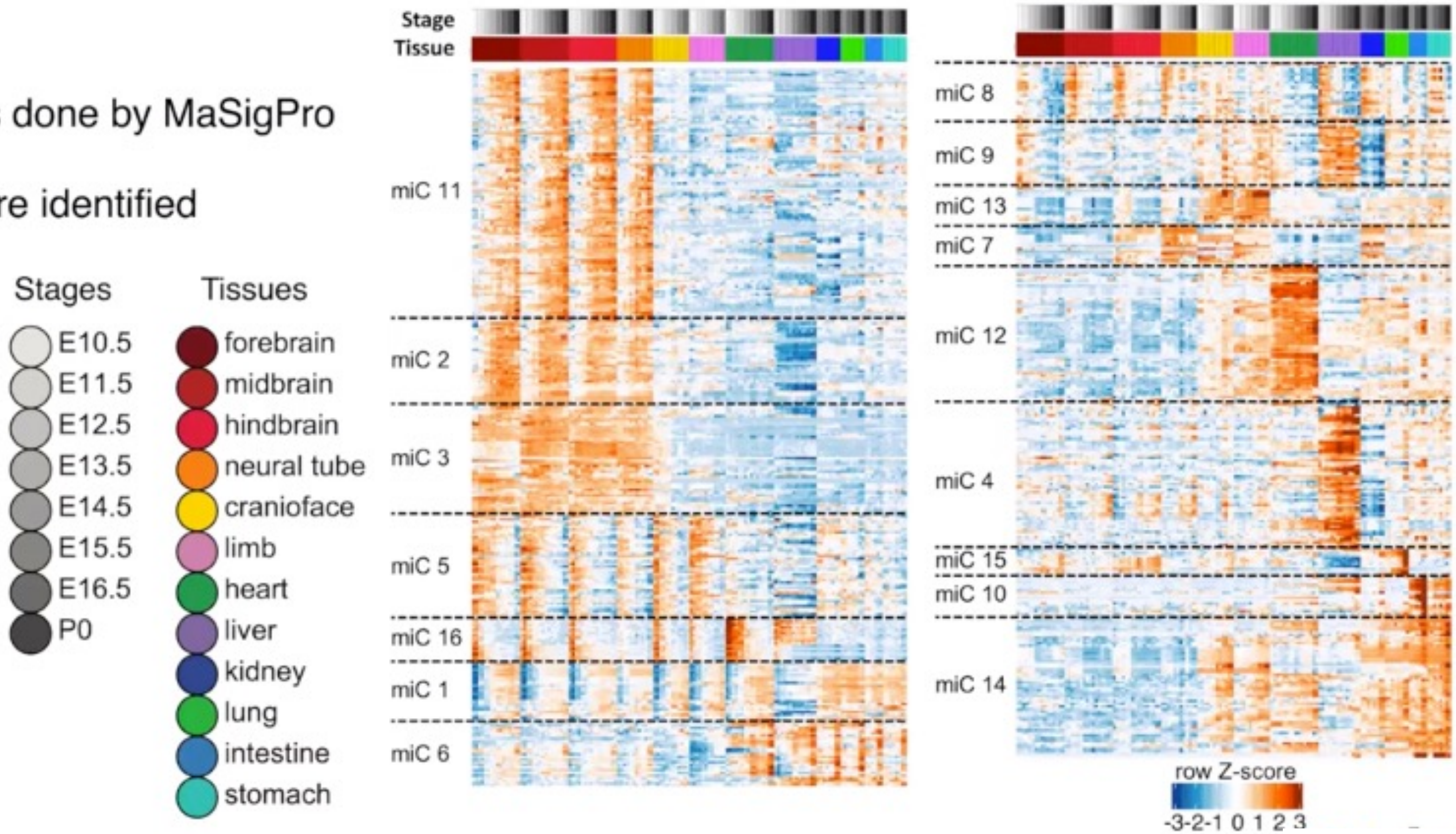


TSI = Tissue Specificity Index

(Rahmanian and Murad et al, 2019)

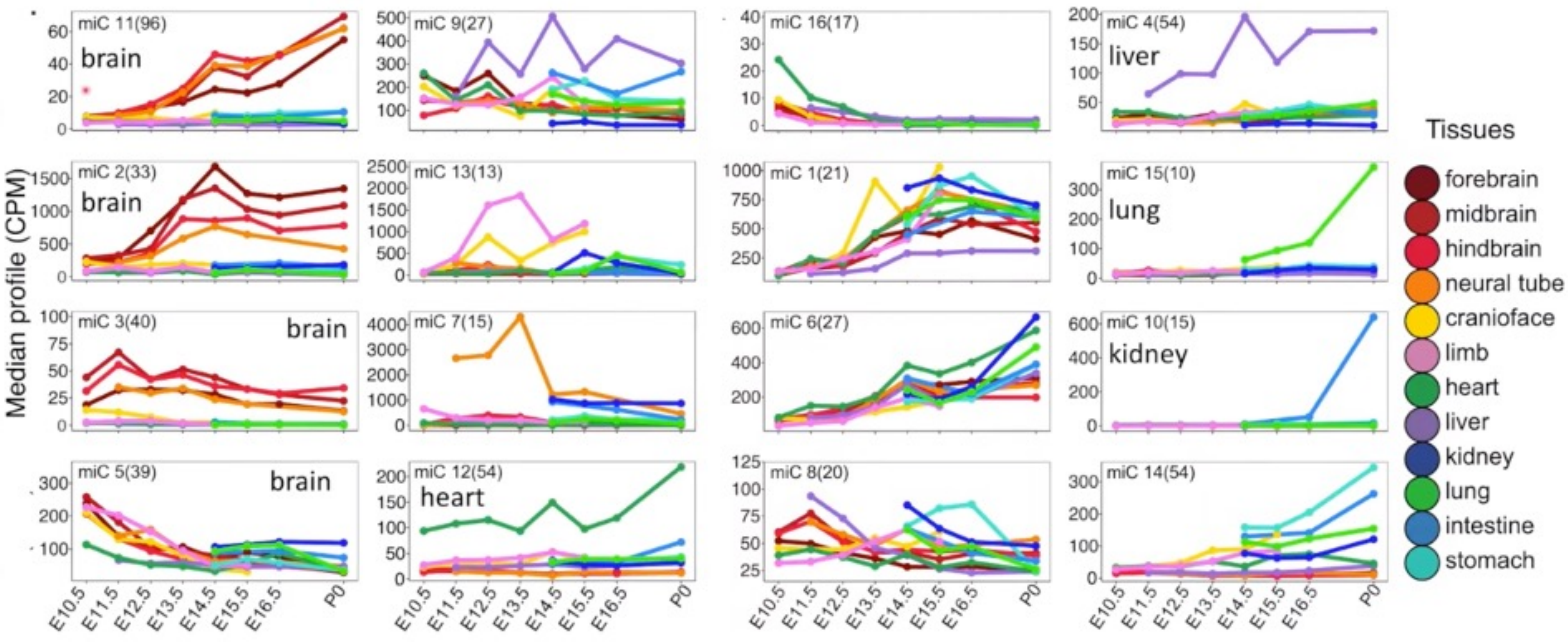
Clustering of MicroRNAs expression captures ttissue specific expression patterns

- Clustering was done by MaSigPro
- 16 clusters were identified



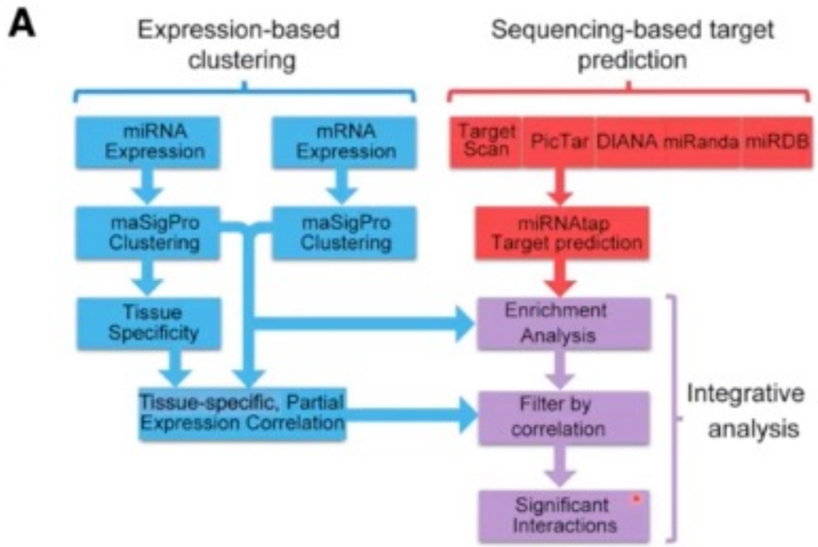
(Rahmanian and Murad et al, 2019)

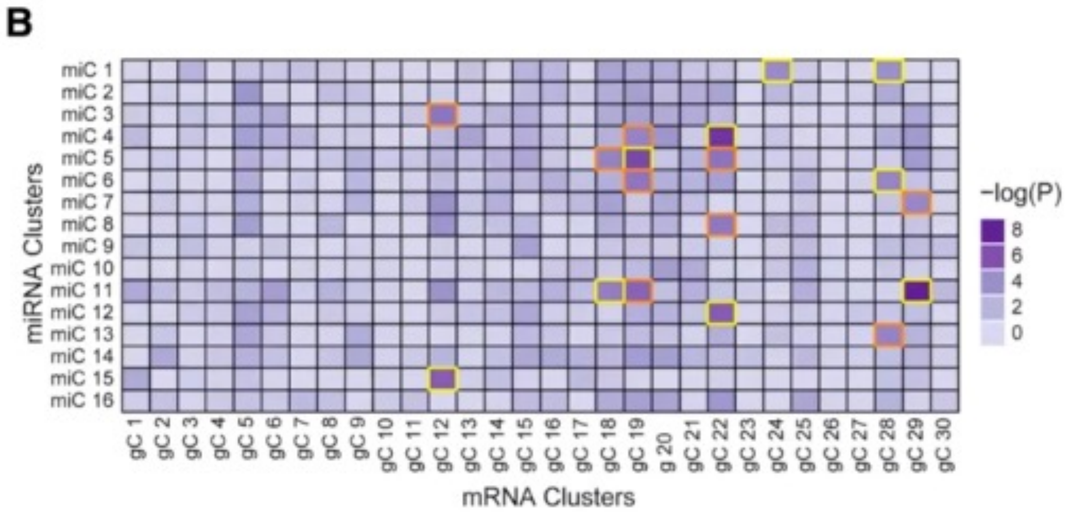
Each miRNA cluster exhibits a distinct tissue-specific expression



(Rahmanian and Murad et al, 2019)

Identifying microRNA expression clusters that are repressing their mRNA targets



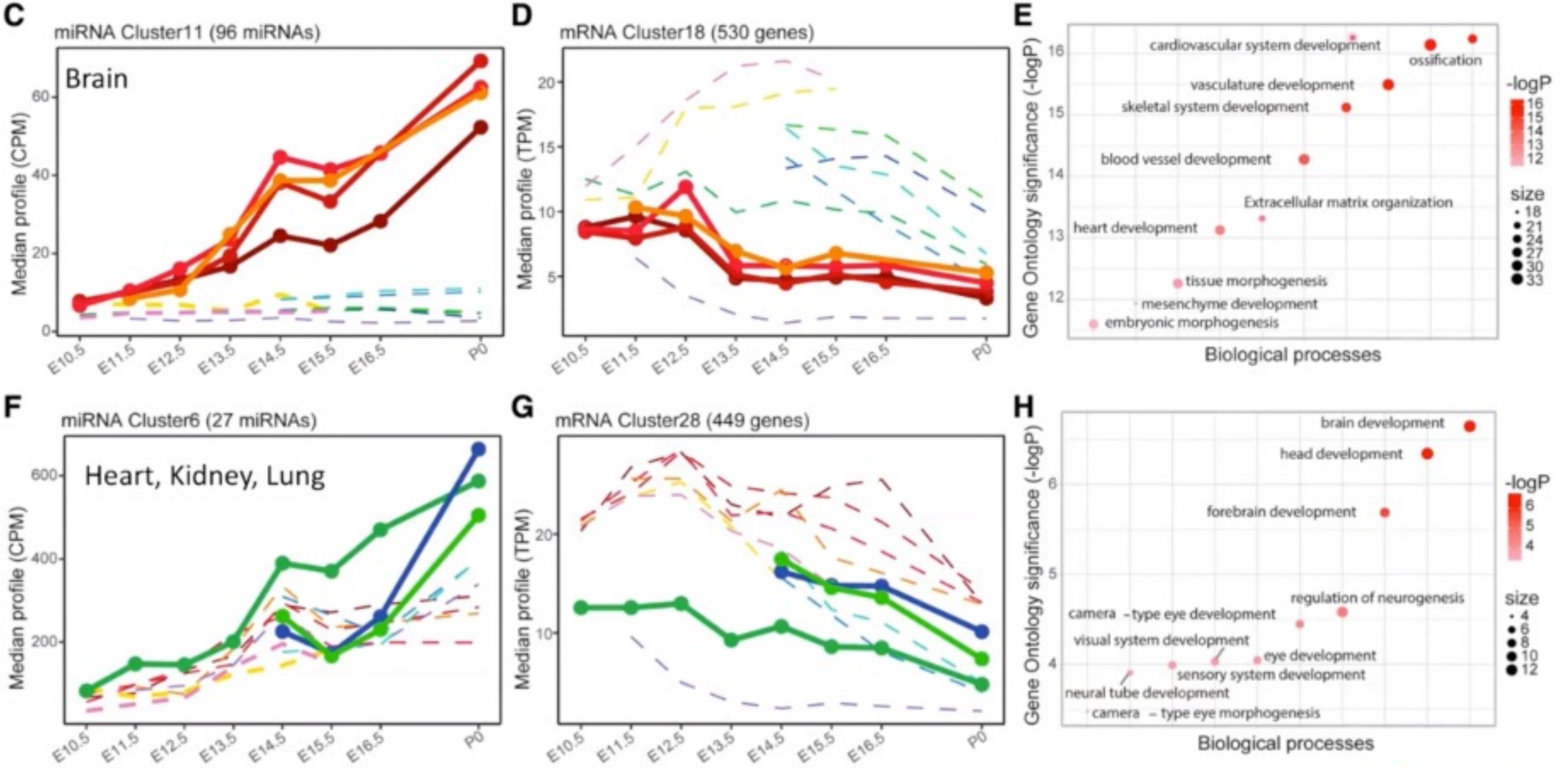


18 are significant by χ^2

negative correlation with mRNA targets
positive correlation with mRNA targets

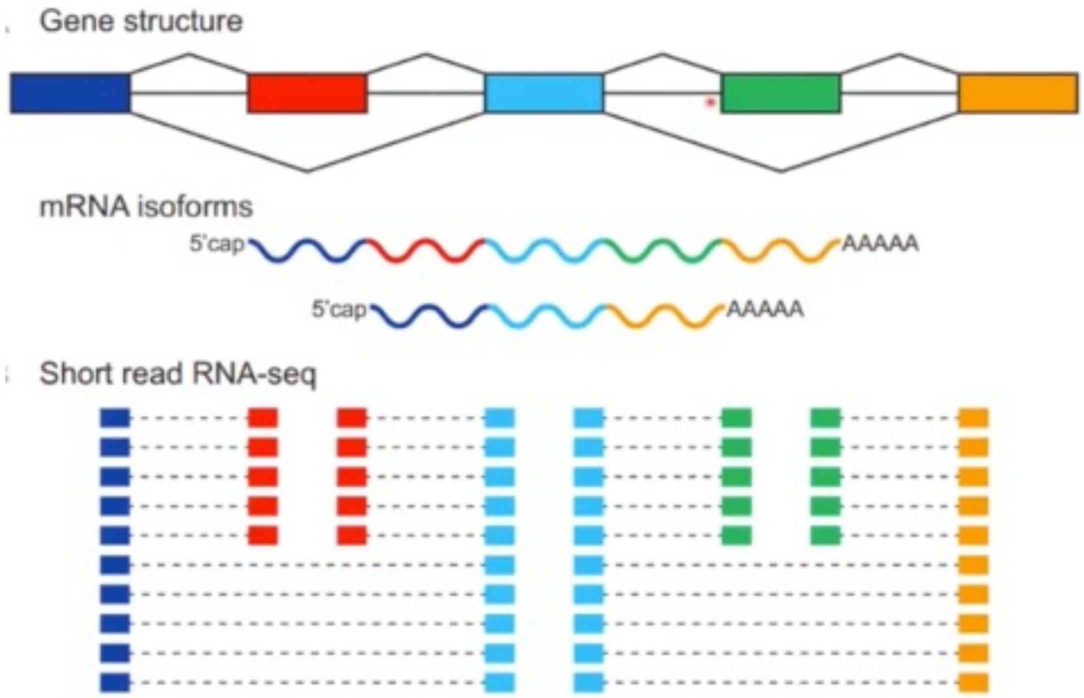
9 have negative correlation to their target mRNA

Examples of significant anti-correlated miRNA-mRNA clusters



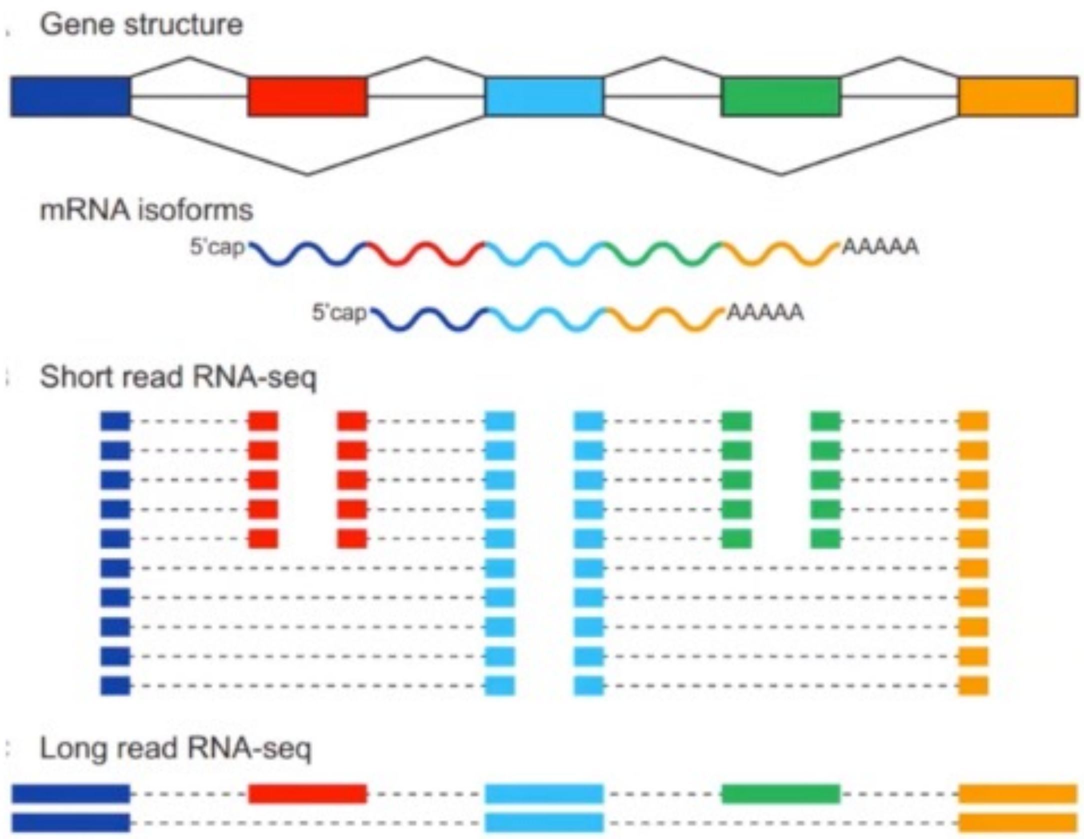
(Rahmanian and Murad et al, 2019)

Short-read sequencing falls short of identifying mRNA isoforms



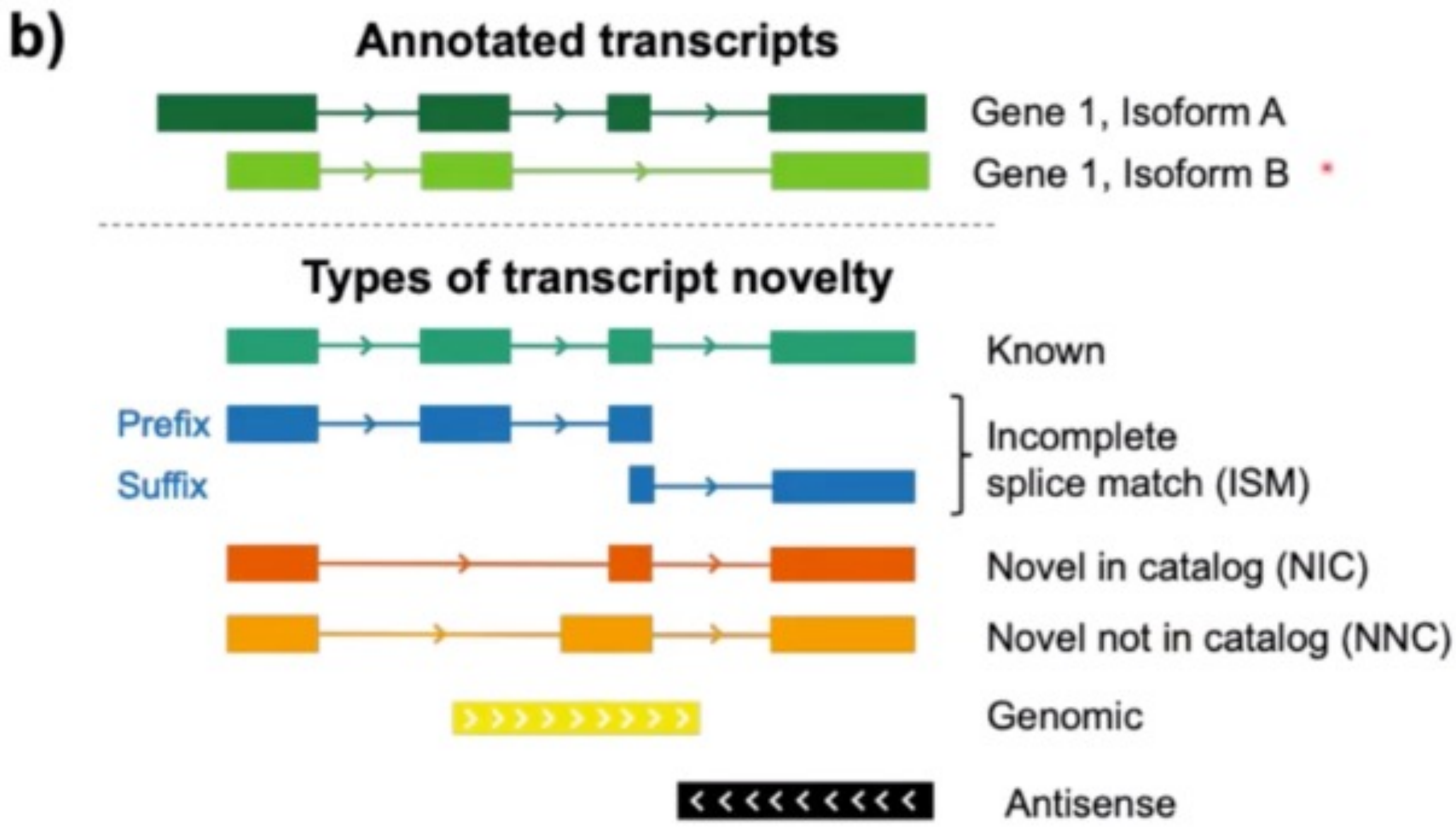
(Park et al., 2017)

Short-read sequencing falls short of identifying mRNA isoforms



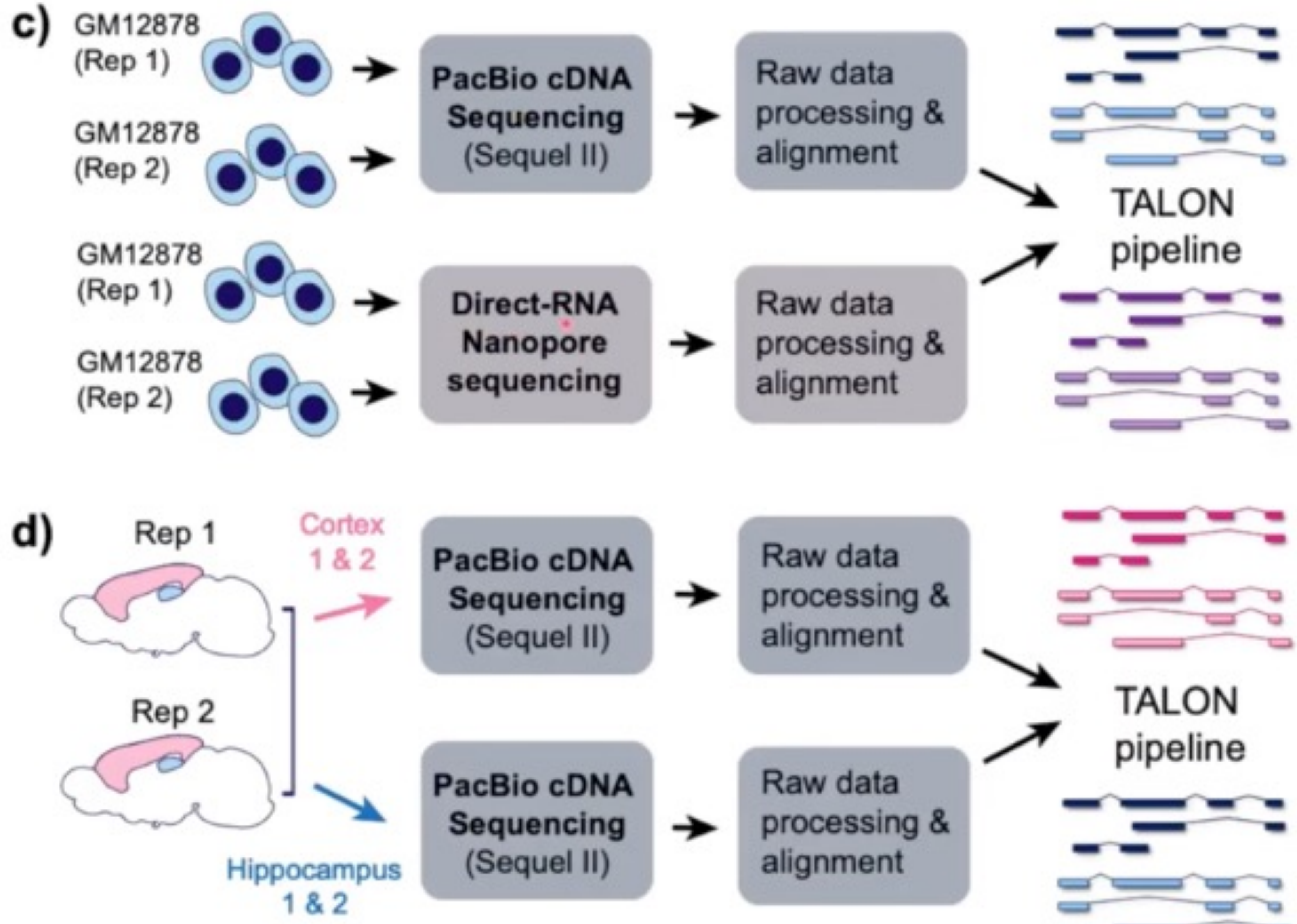
(Park et al., 2017)

Isoform novelty categories



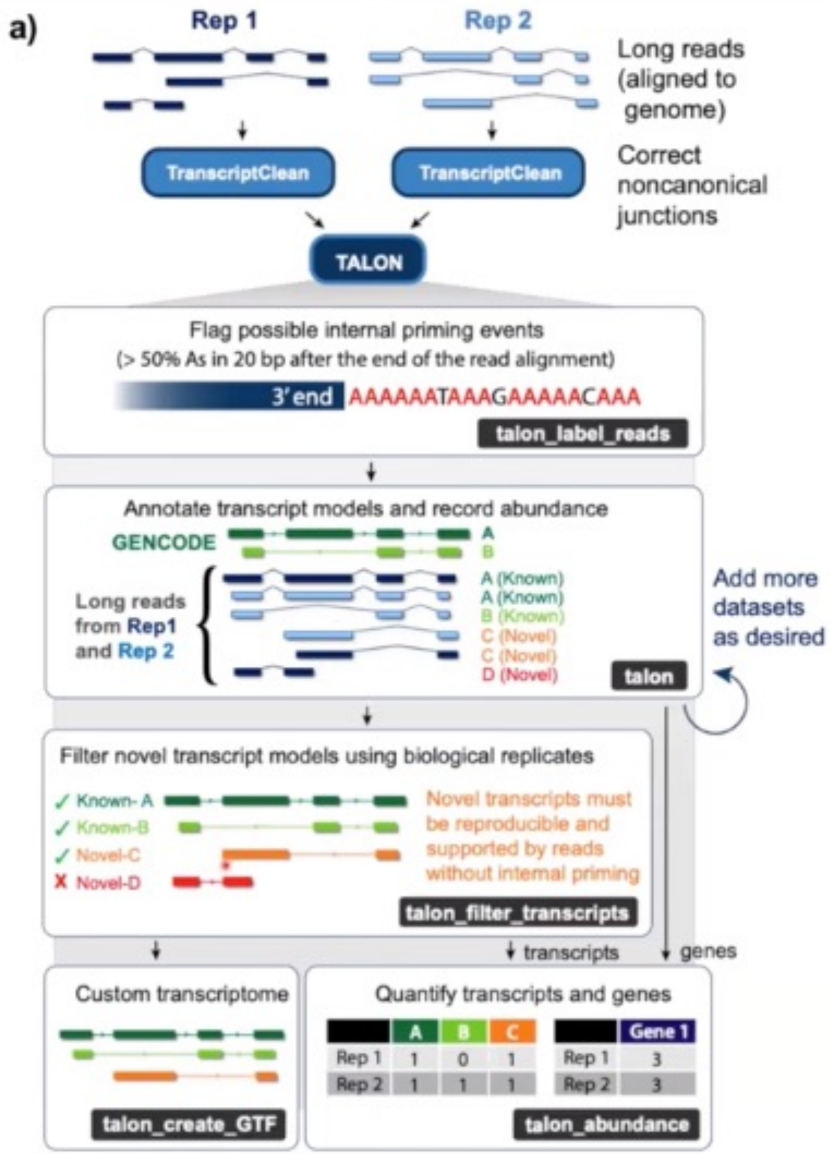
Using SQANTI novelty categories

Comparing sequencing technologies and tissues



(Wyman and Balderrama-Gutierrez et al, BioRXiv, 2020)

TALON: a pipeline for long-read transcriptomes



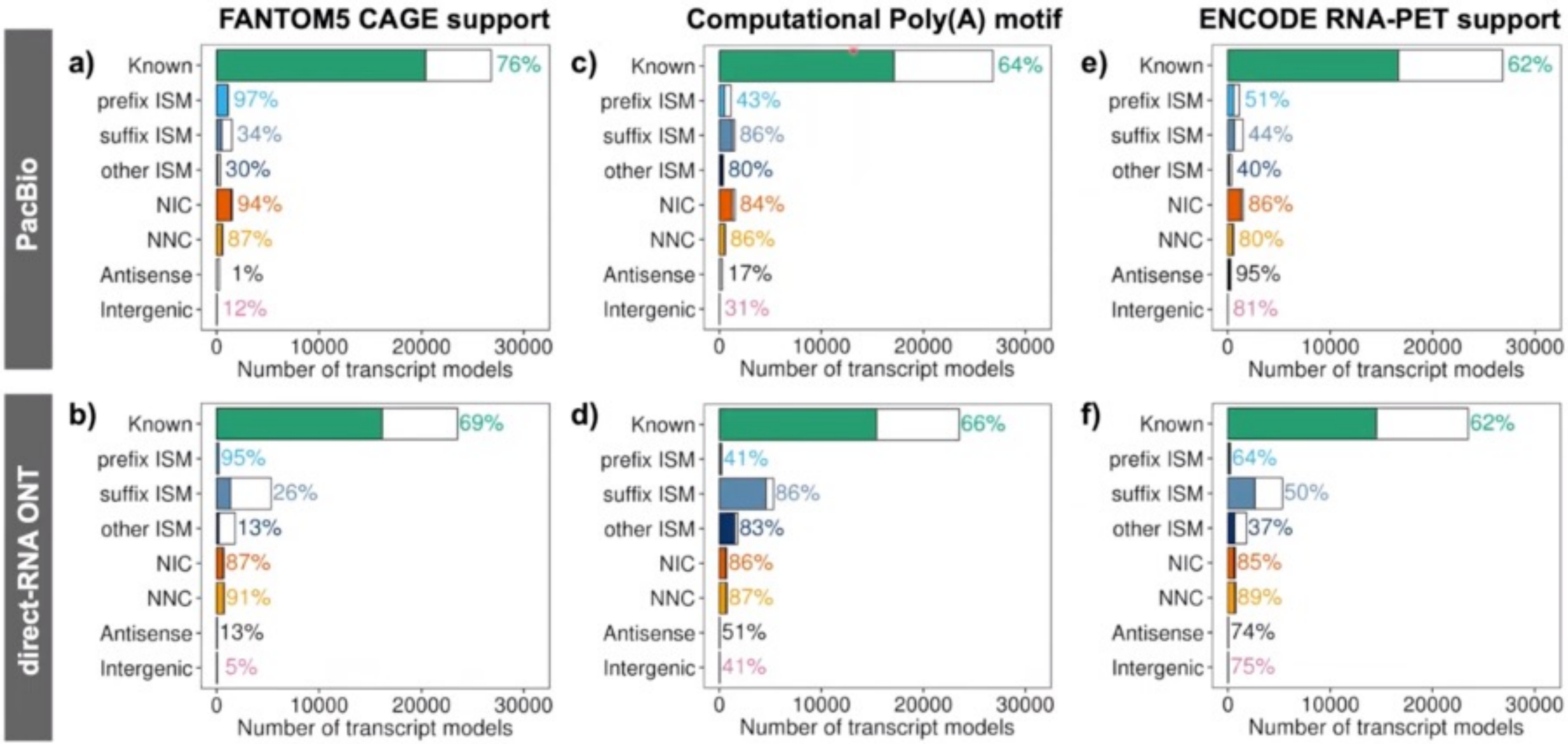
https://github.com/mortazavilab/TranscriptClean

https://github.com/mortazavilab/TALON

(Wyman and Balderrama-Gutierrez et al, BioRXiv, 2020)

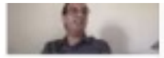
UCI University of California, Irvine

Pacbio has higher rates of 5’ and 3’ completeness

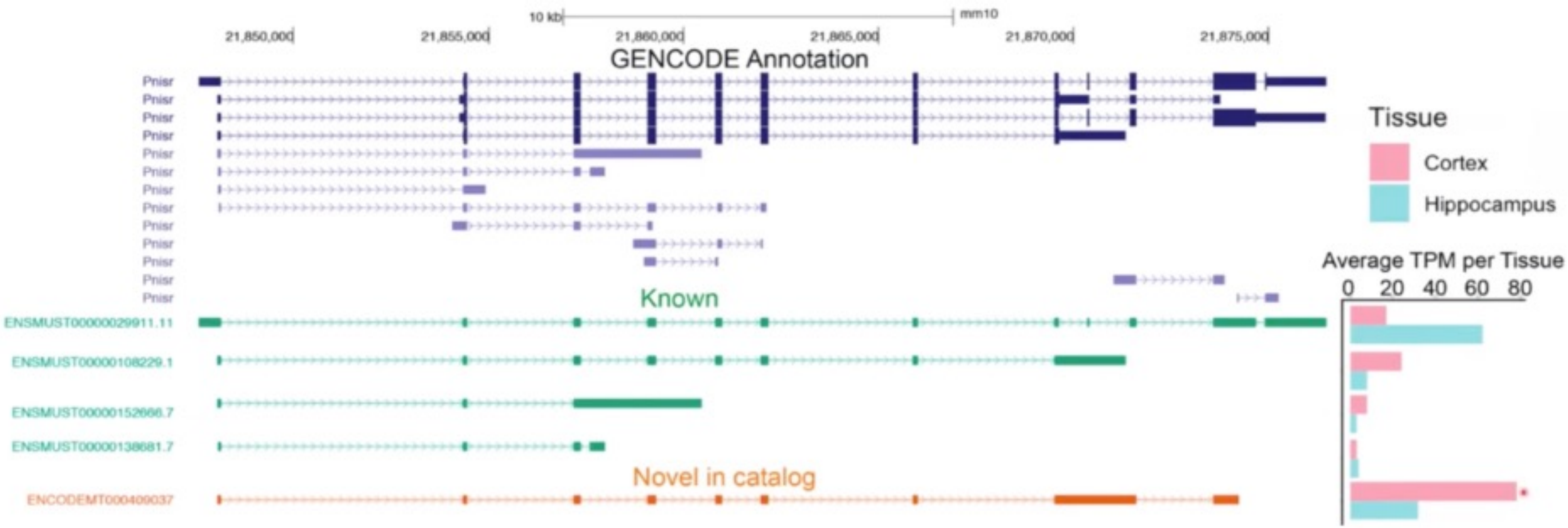


(Wyman and Balderrama-Gutierrez et al, BioRXiv, 2020)

UCI University of California, Irvine



Differences between tissues can involve novel isoforms

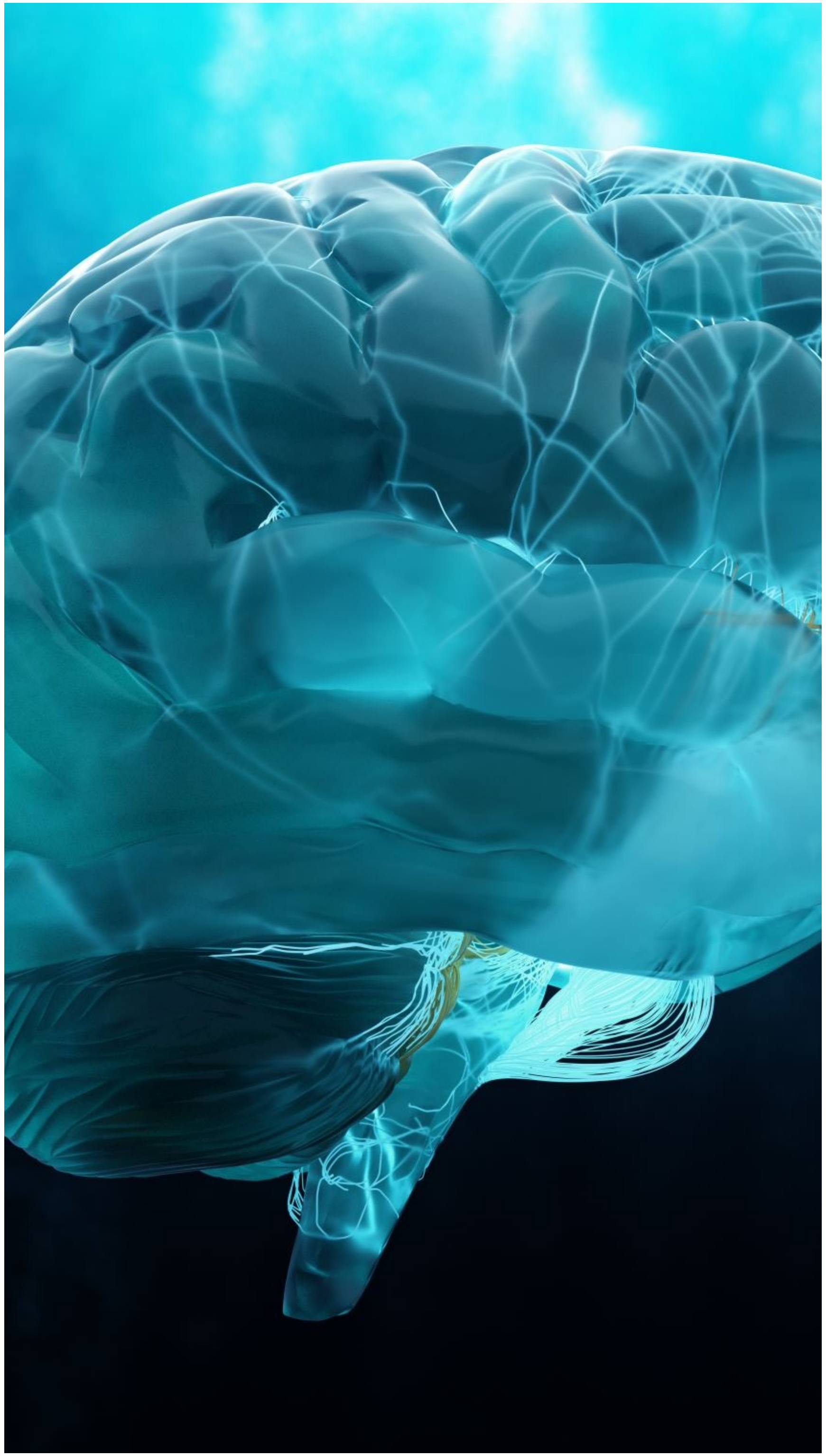


(Wyman and Balderrama-Gutierrez et al, BioRXiv, 2020)

UCI University of California, Irvine

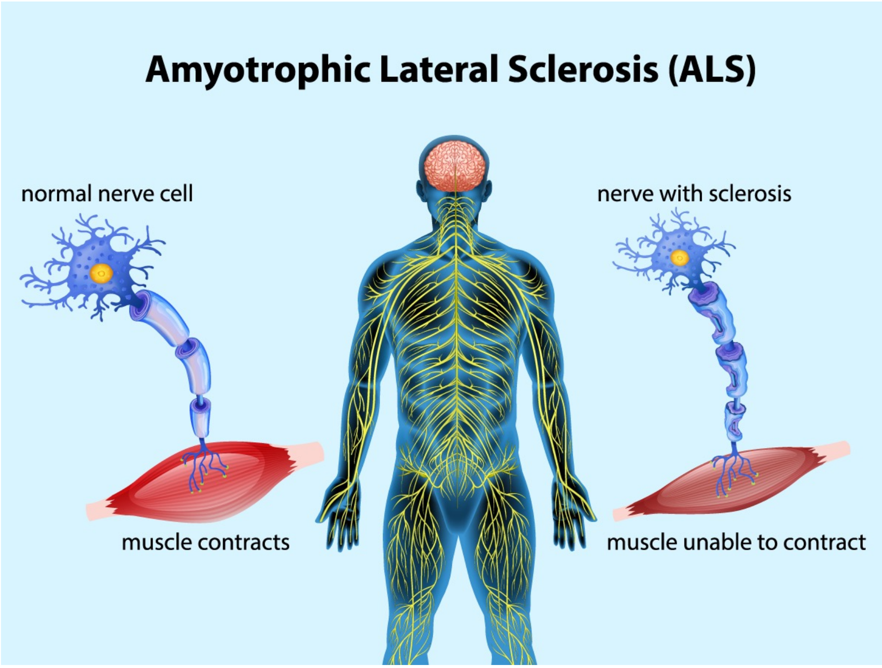
ALS
Amyotrophic lateral sclerosis
"肌萎缩侧索硬化症".

Example



ALS

- ALS, is a **nervous system disease** that affects **nerve cells** in the **brain and spinal cord**. ALS causes **loss of muscle control**. The disease gets worse over time.
- The exact cause of the disease is still **not known**. A small number of cases are inherited.
- ALS often begins with **muscle twitching** and **weakness in an arm or leg**, **trouble swallowing** or slurred speech. Eventually ALS affects control of the muscles needed to move, speak, eat and breathe. There is no cure for this fatal disease.


Amyotrophic Lateral Sclerosis (ALS)
normal nerve cell
nerve with sclerosis
muscle contracts
muscle unable to contract

nature communications

Article https://doi.org/10.1038/s41467-023-41033-y

Divergent single cell transcriptome and epigenome alterations in ALS and FTD patients with C9orf72 mutation

Received: 16 December 2022

Accepted: 21 August 2023

Published online: 15 September 2023

Junhao Li[1,6], Manoj K. Jaiswal[2,6], Jo-Fan Chien[3], Alexey Kozlenkov[2], Jinyoung Jung[2], Ping Zhou[2], Mahammad Gardashli[4], Luc J. Pregent[4], Erica Engelberg-Cook[4], Dennis W. Dickson[4], Veronique V. Belzil[4], Eran A. Mukamel[1] & Stella Dracheva[2,5]

Background

- Aggregates of the RNA-binding transactive response DNA binding 43 protein (TDP-43) accumulate in neurons of almost all ALS cases
- The most common genetic cause is a dynamic hexanucleotide (GGGGCC; G4C2) repeat expansion in the first intron of the C9orf72 gene

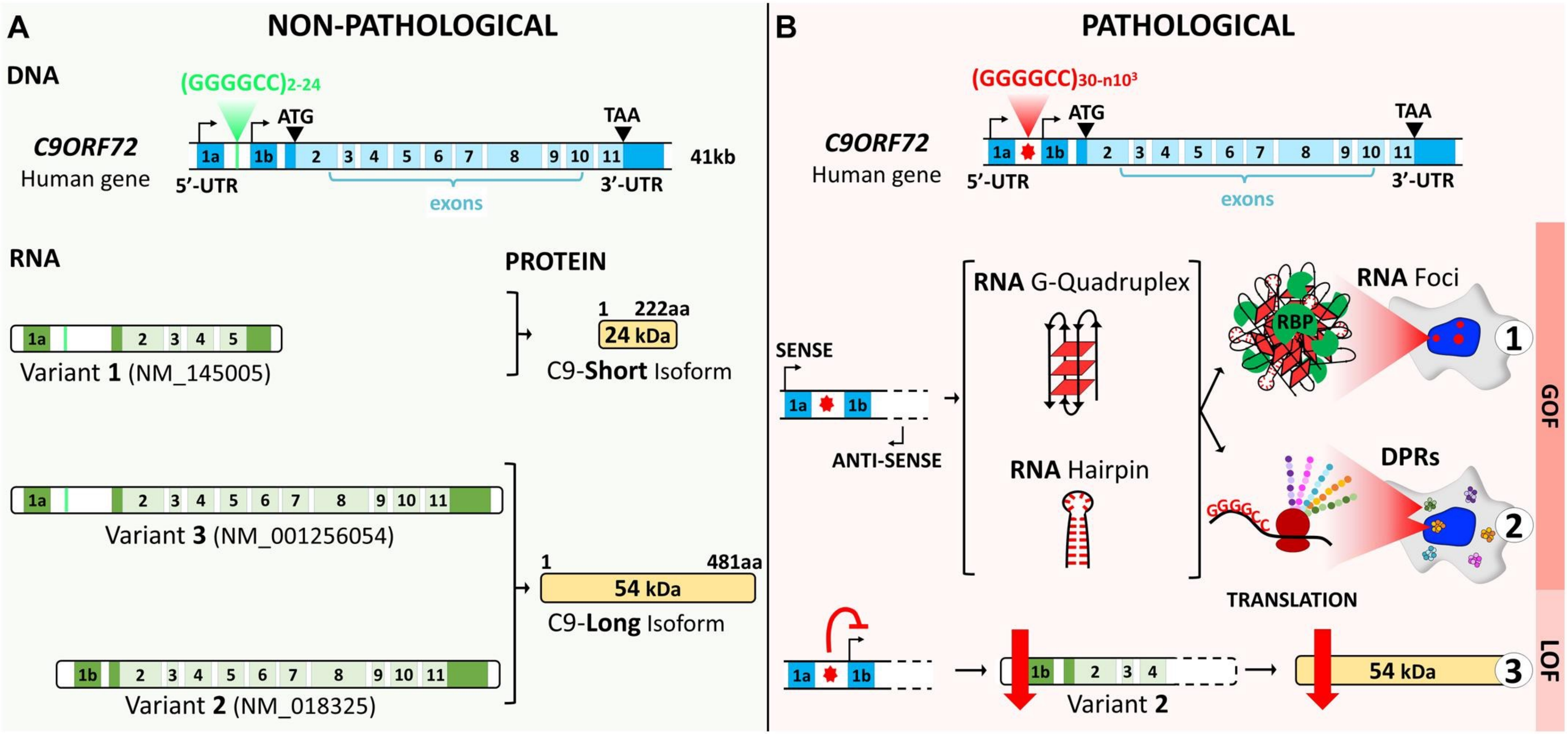
REVIEW article

Front. Cell. Neurosci., 05 May 2021
Sec. Cellular Neuropathology
Volume 15 - 2021 | https://doi.org/10.3389/fncel.2021.661447

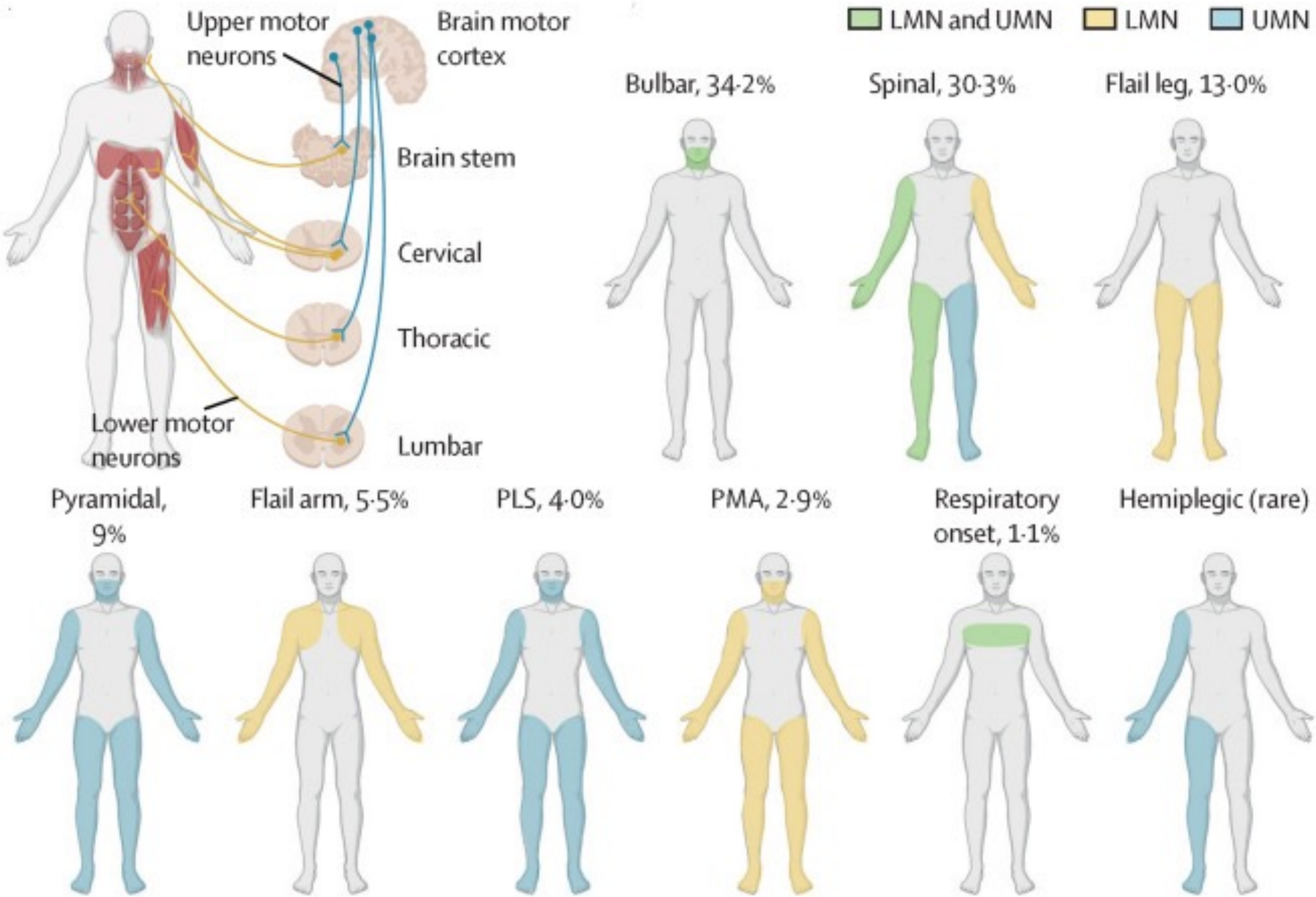
This article is part of the Research Topic
Molecular Mechanisms Underlying C9orf72 Neurodegeneration
View all 8 Articles

C9ORF72: What It Is, What It Does, and Why It Matters

Julie Smeyers[1,2] Elena-Gaia Banchi[1] Morwena Latouche[1,2*]



- Bidirectional transcription of the hexanucleotide repeat expansion (HRE) generates G4C2 sense and G2C4 antisense expanded RNAs.
- These HRE transcripts give rise to G-quadruplex and hairpin structures that can form RNA foci and sequester RNA-binding proteins (RBP)
- Expanded RNAs are also translated through repeat-associated non-ATG (RAN) translation, resulting in the synthesis of dipeptide protein repeats (DPRs)
- Finally, the presence of the HRE inhibits transcription, leading to a decrease in the C9ORF72 protein



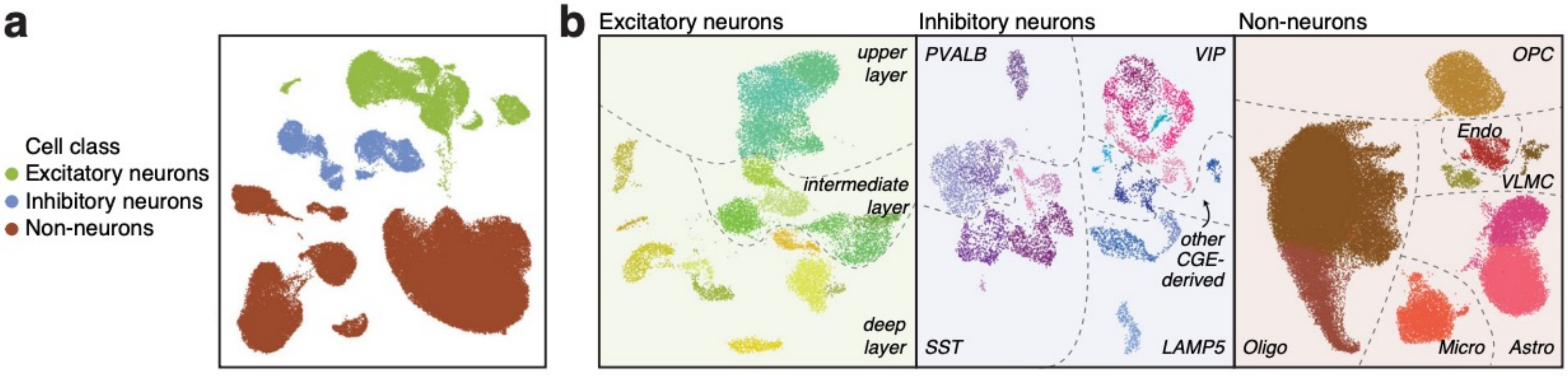
Upper motor neurons
Brain motor cortex
Brain stem
Cervical
Thoracic
Lumbar
Lower motor neurons
LMN and UMN
LMN
UMN
Bulbar, 34·2%
Spinal, 30·3%
Flail leg, 13·0%
Pyramidal, 9%
Flail arm, 5·5%
PLS, 4·0%
PMA, 2·9%
Respiratory onset, 1·1%
Hemiplegic (rare)

Design

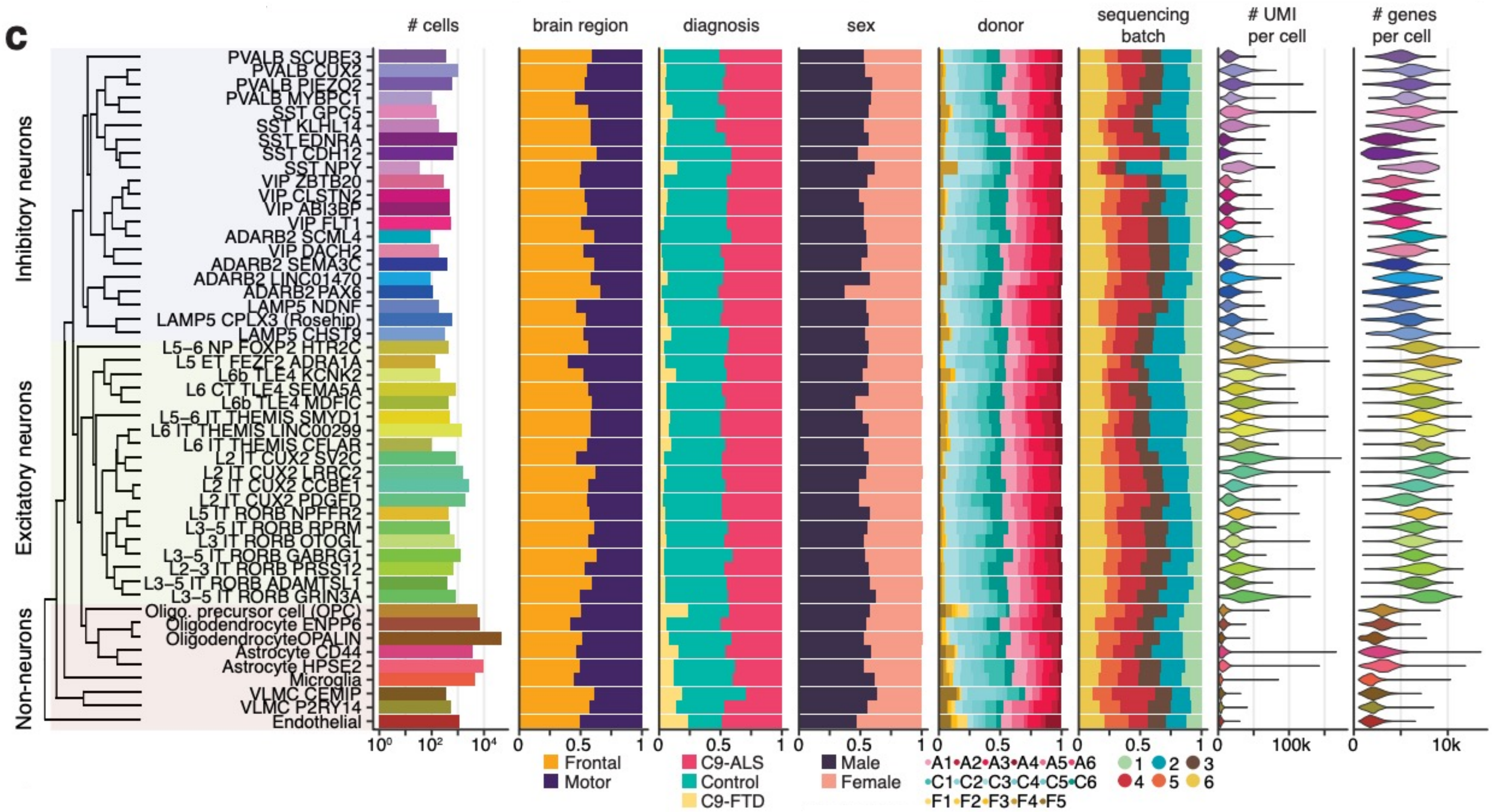
single-nucleus transcriptomic (snRNA-seq) and epigenomic (snATAC-seq) analysis of human brain samples to investigate the cell-type-specific molecular alterations in C9-ALS or C9-FTD donors

two key cortical regions, the motor and frontal cortices, in an attempt to uncover regional differences.

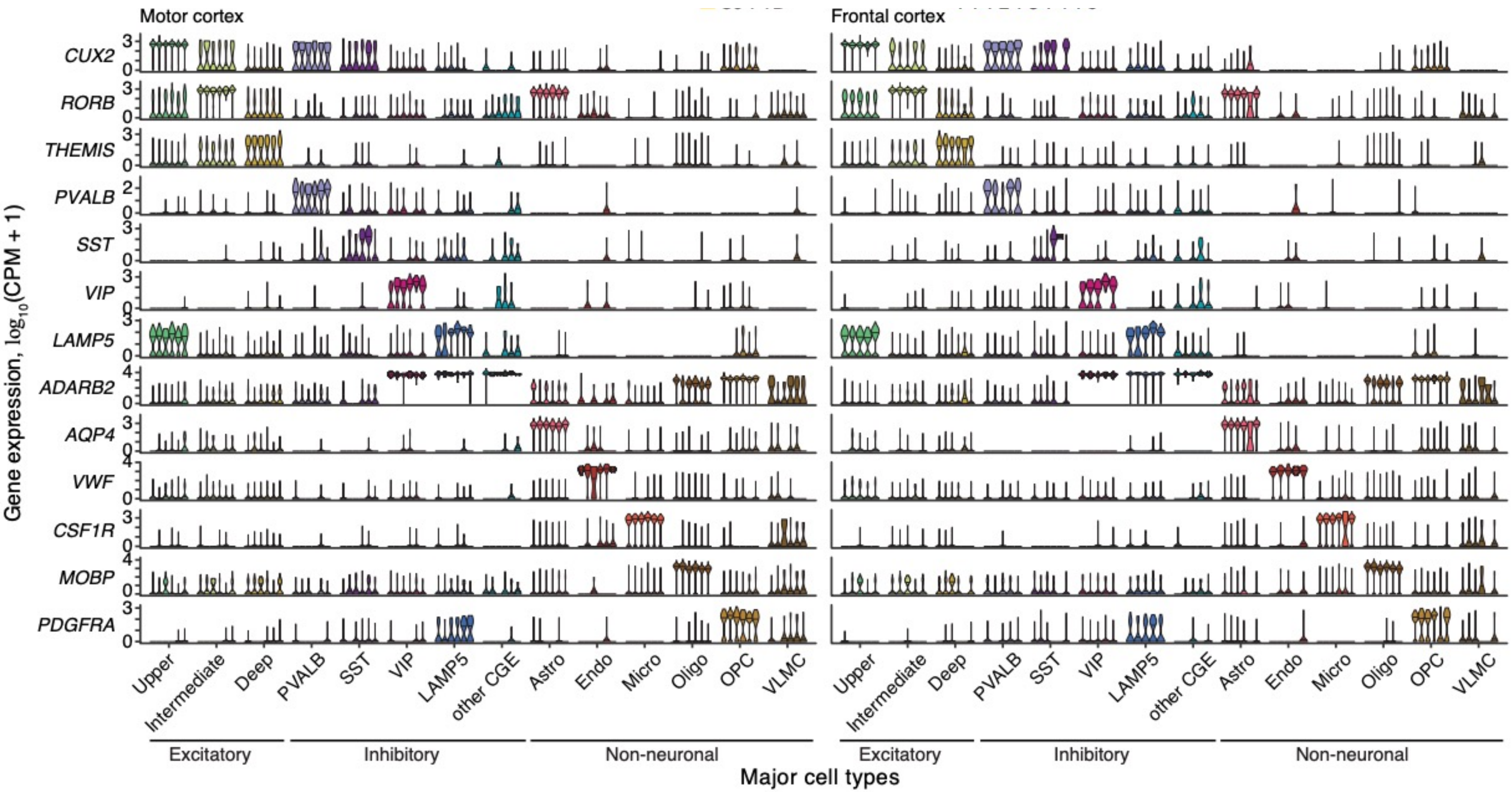
Single nucleus transcriptomes in C9-ALS and C9-FTD specimens identify fine-grained cortical cell types



- n = 105,120 cells from 17 donors
- Nuclei were first classified into three cell classes
- then subpopulations in each class were identified

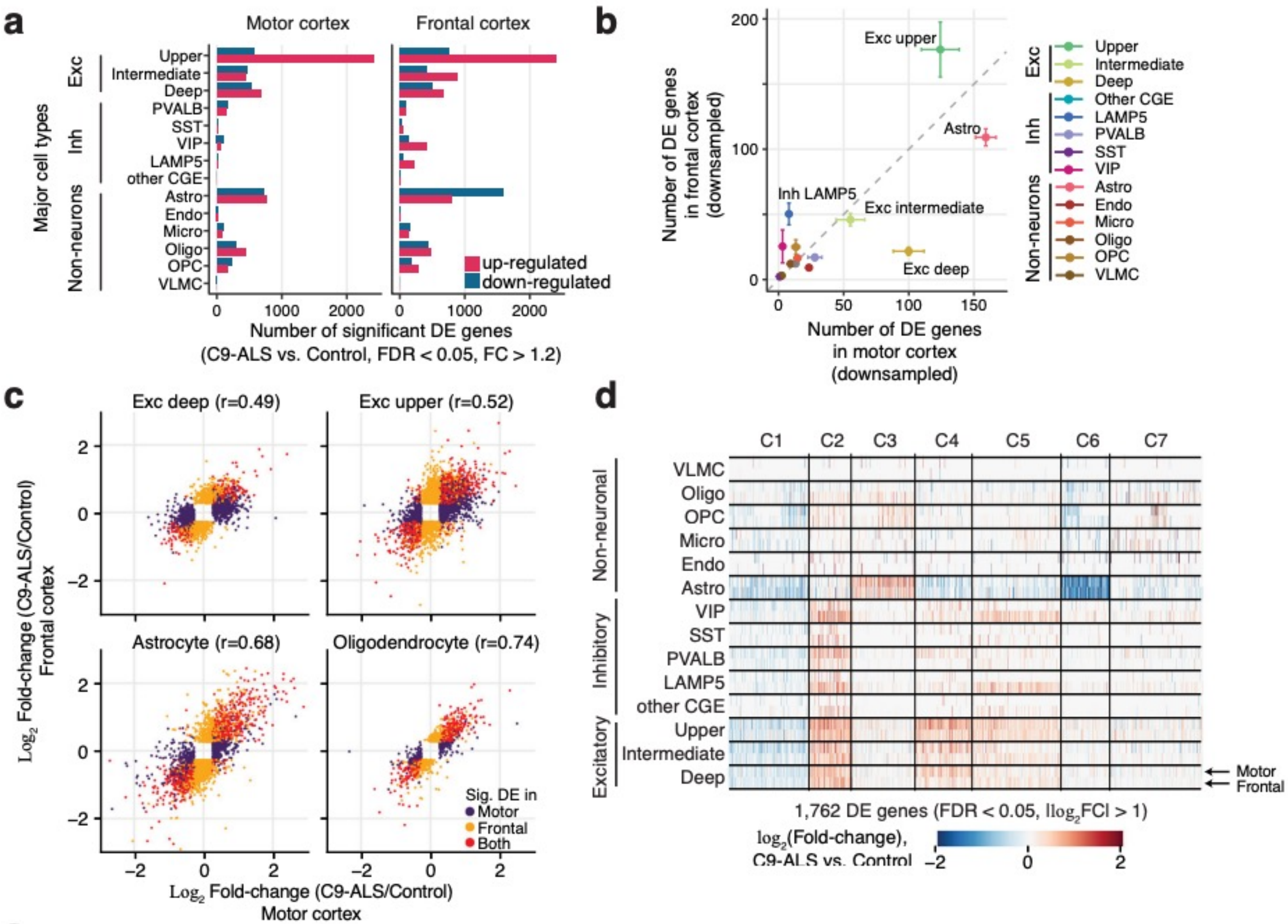


- Neuronal and non-neuronal cell types were distributed across brain regions, donor diagnosis groups, sex, and sequencing batches. UMI unique molecular identifier.

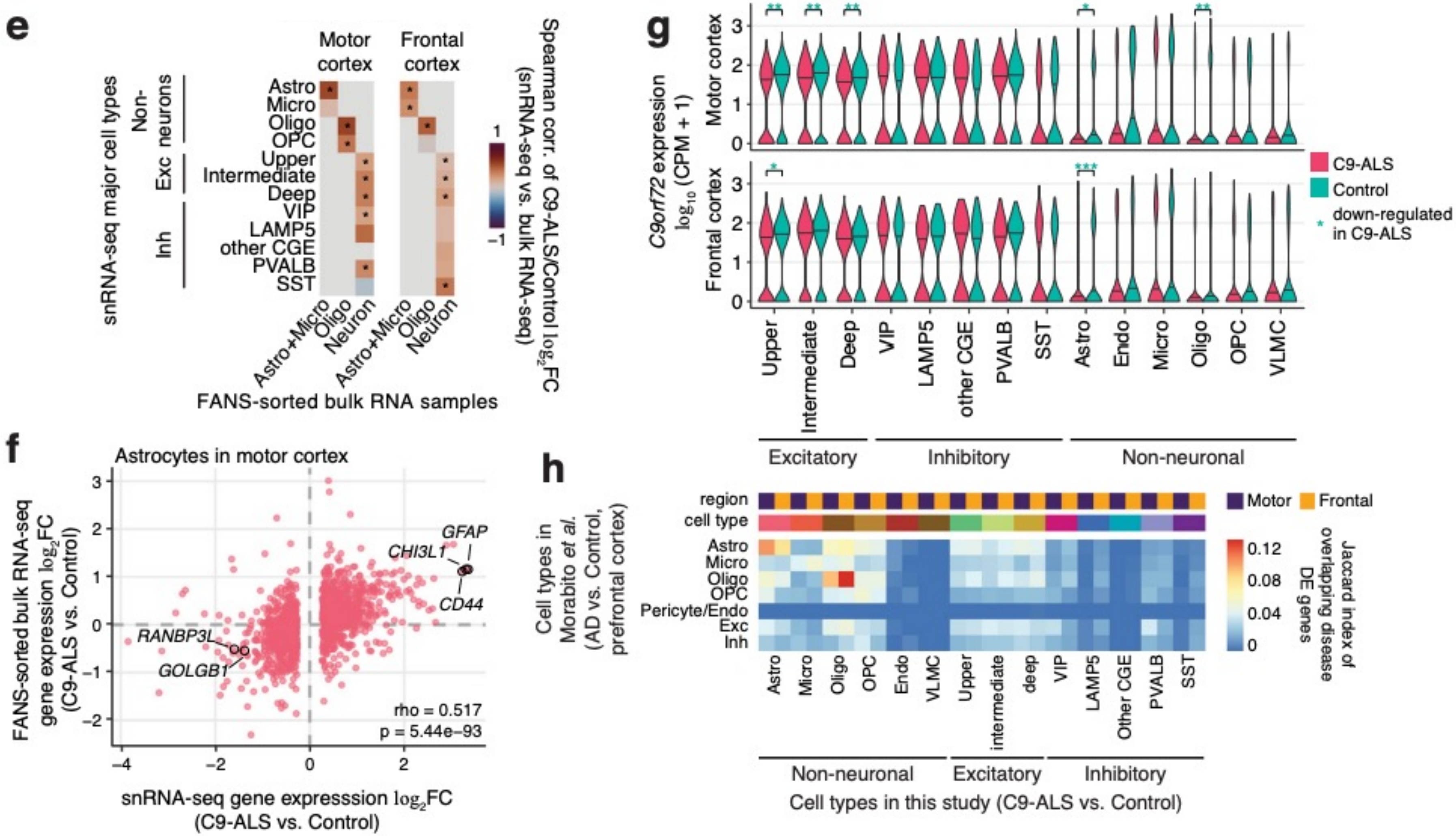


- Violin plots show marker gene expression in major cell types in each of the 6 control donors.

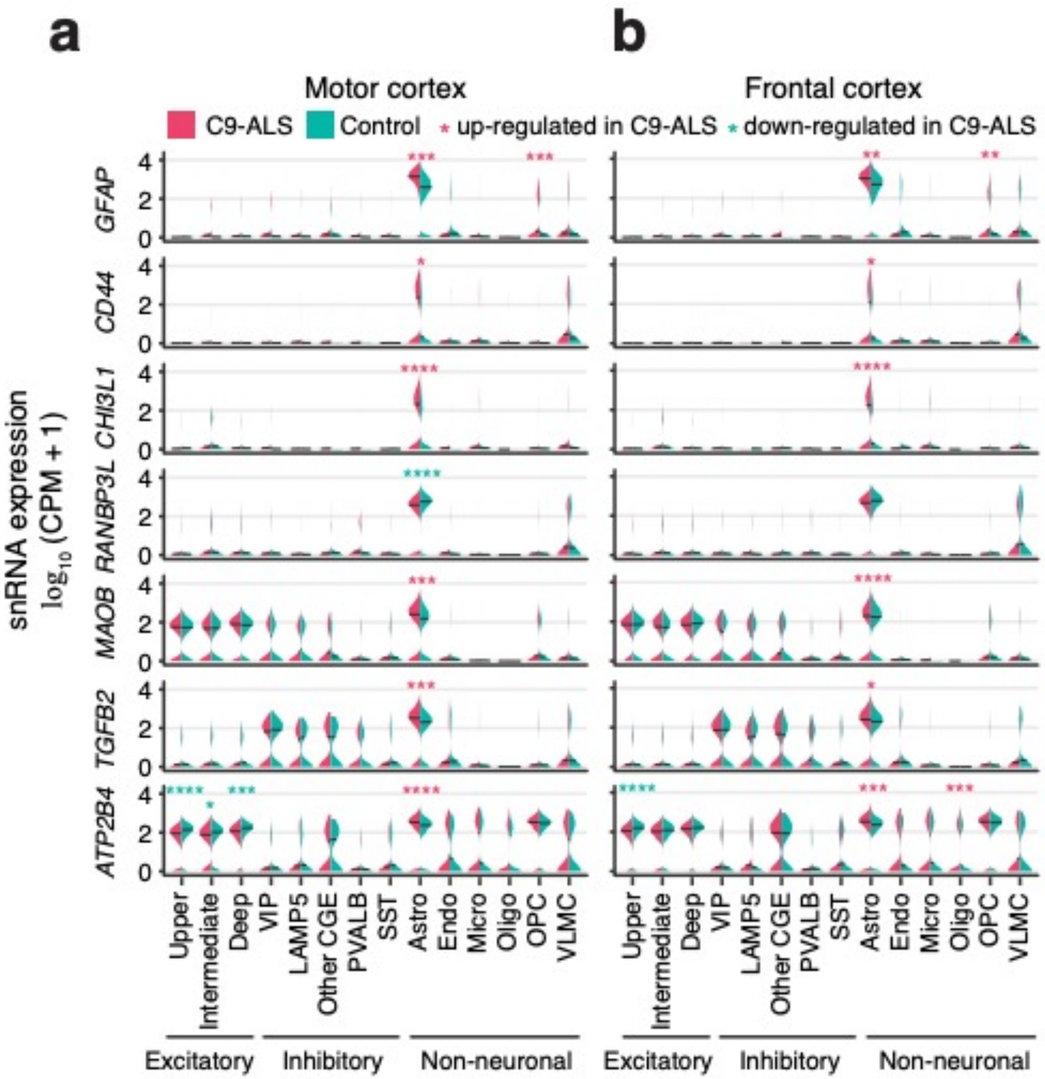
Dysregulation of gene expression in C9-ALS is concentrated in astrocytes and excitatory neurons



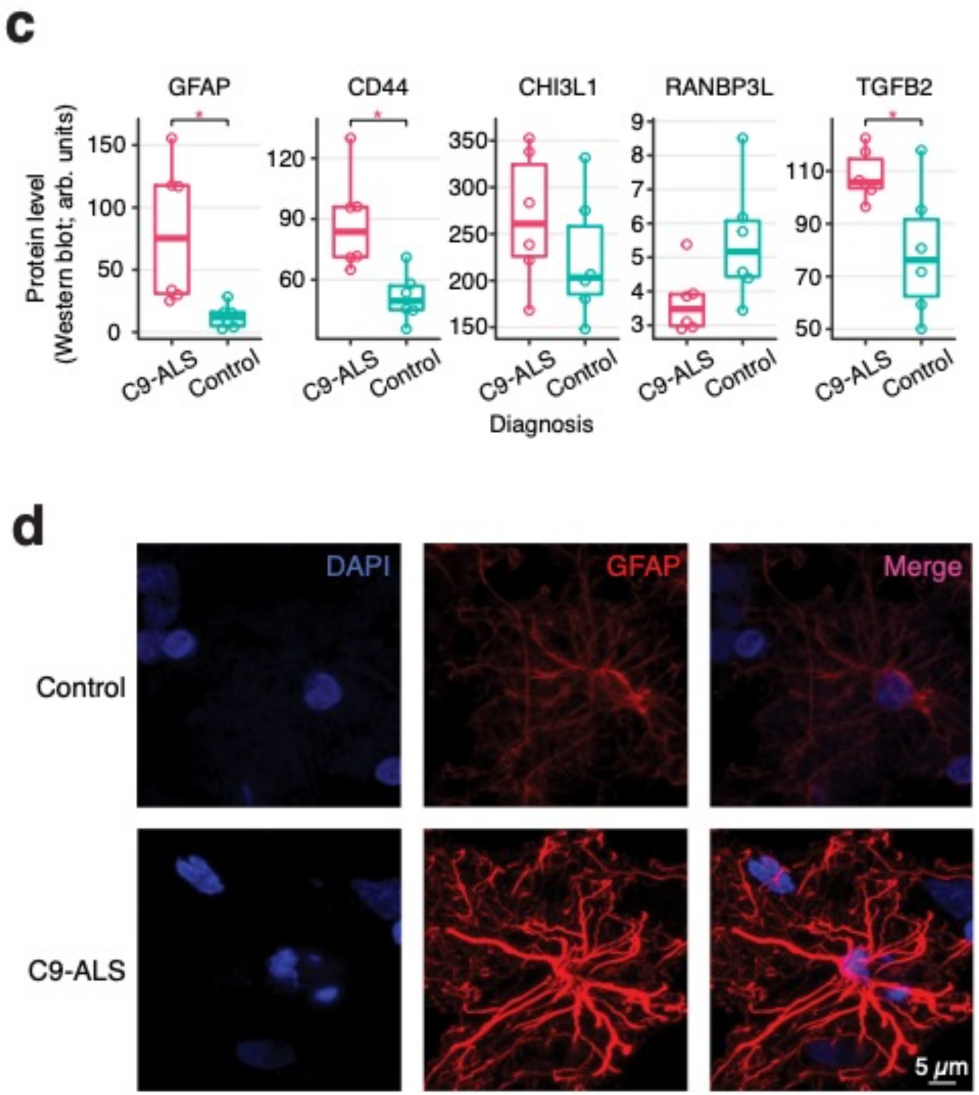
Dysregulation of gene expression in C9-ALS is concentrated in astrocytes and excitatory neurons



Dysregulation of gene expression in C9-ALS astrocytes suggests activation and structural remodeling

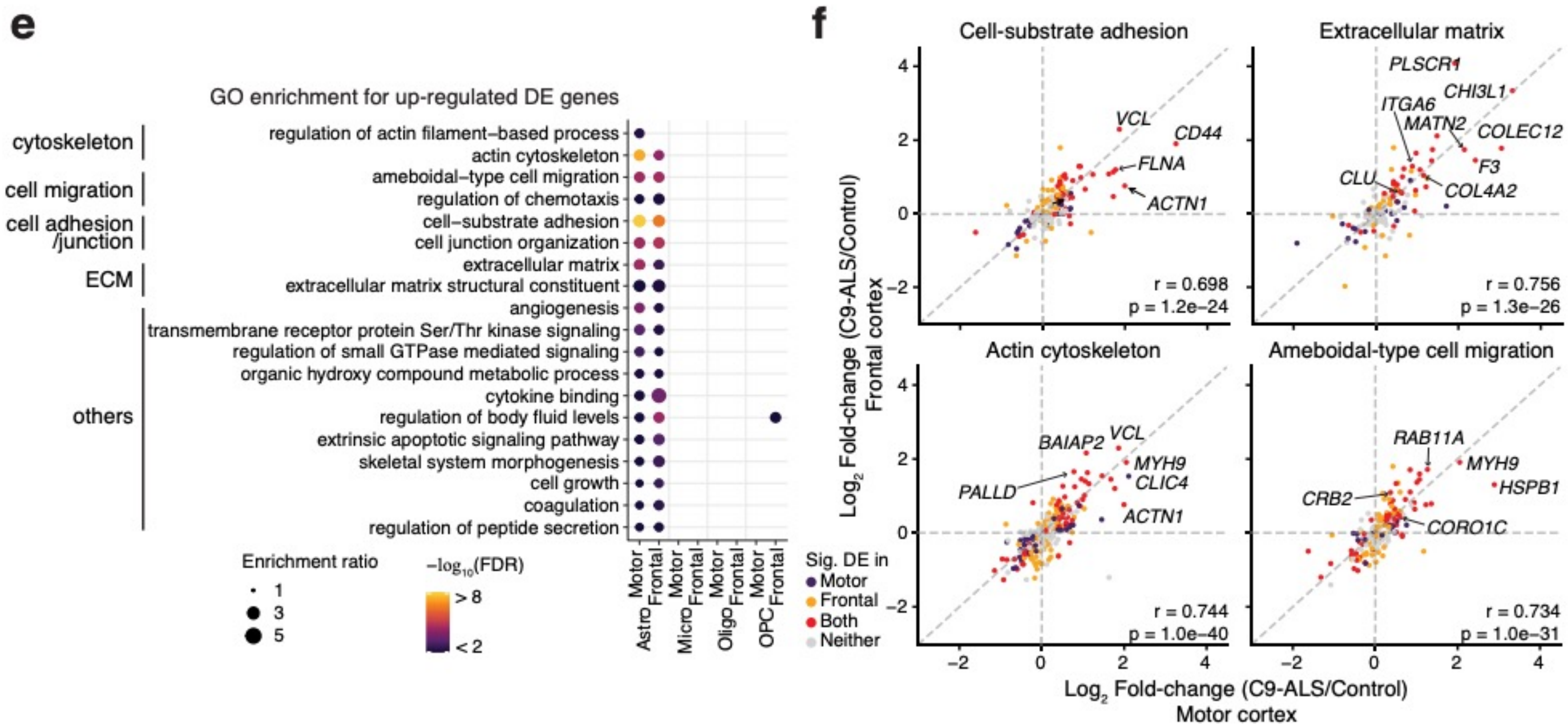


Dysregulation of gene expression in C9-ALS astrocytes suggests activation and structural remodeling



- immunofluorescence images of the GFAP-positive astrocytes in motor cortex (red)

Dysregulation of gene expression in C9-ALS astrocytes suggests activation and structural remodeling



Heterochromatin anomalies and double-stranded RNA accumulation underlie *C9orf72* poly(PR) toxicity

Yong-Jie Zhang[1,2], Lin Guo[3], Patrick K. Gonzales[4], Tania F. Gendron[1,2], Yanwei Wu[1], Karen Jansen-West[1], Aliesha D. O'Raw[1], Sarah R. Pickles[1], Mercedes Prudencio[1,2], Yari Carlomagno[1], Mariam A. Gachechiladze[5], Connor Ludwig[6*], Ruilin Tian[6], Jeannie Chew[1,2], Michael DeTure[1,2], Wen-Lang Lin[1], Jimei Tong[1], Lillian M. Daughrity[1], Mei Yue[1], Yuping Song[1], Jonathan W. Andersen[1], Monica Castanedes-Casey[1], Aishe Kurti[1], Abhishek Datta[7†], Giovanna Antognetti[7], Alexander McCampbell[8], Rosa Rademakers[1,2], Björn Oskarsson[9], Dennis W. Dickson[1,2], Martin Kampmann[6], Michael E. Ward[5], John D. Fryer[1,2], Christopher D. Link[4], James Shorter[3], Leonard Petrucelli[1,2‡]

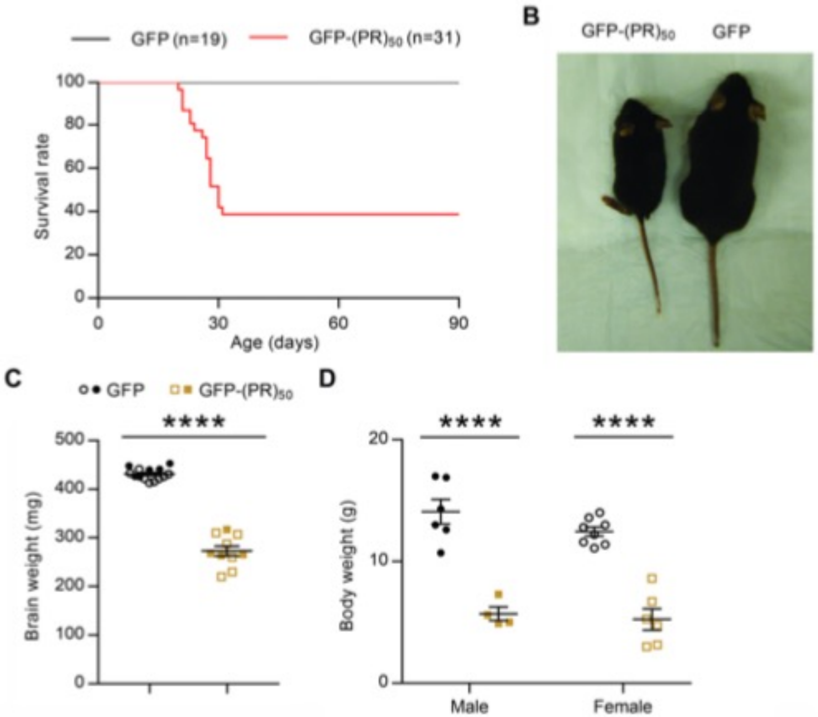


Fig. S1. Expression of GFP-(PR)$_{50}$ in mouse brain was extremely toxic. (**A**) ~60% of GFP-(PR)$_{50}$ mice died by 4 weeks of age. (**B**) GFP-(PR)$_{50}$ mice that died prematurely were markedly smaller than age-matched GFP mice. (**C**) Mean brain weight of GFP-(PR)$_{50}$ mice that died prematurely (n =10) and age-matched GFP mice (n = 14). (**D**) Mean body weight of GFP-(PR)$_{50}$ mice that died prematurely (male, n = 4; female, n = 6) and age-matched GFP mice (male, n = 6; female, n = 8). Data are presented as mean ± SEM. Male mice are represented by solid symbols, while female mice are represented by empty symbols. In (**C**) and (**D**): **** P < 0.0001, two-tailed unpaired t test.

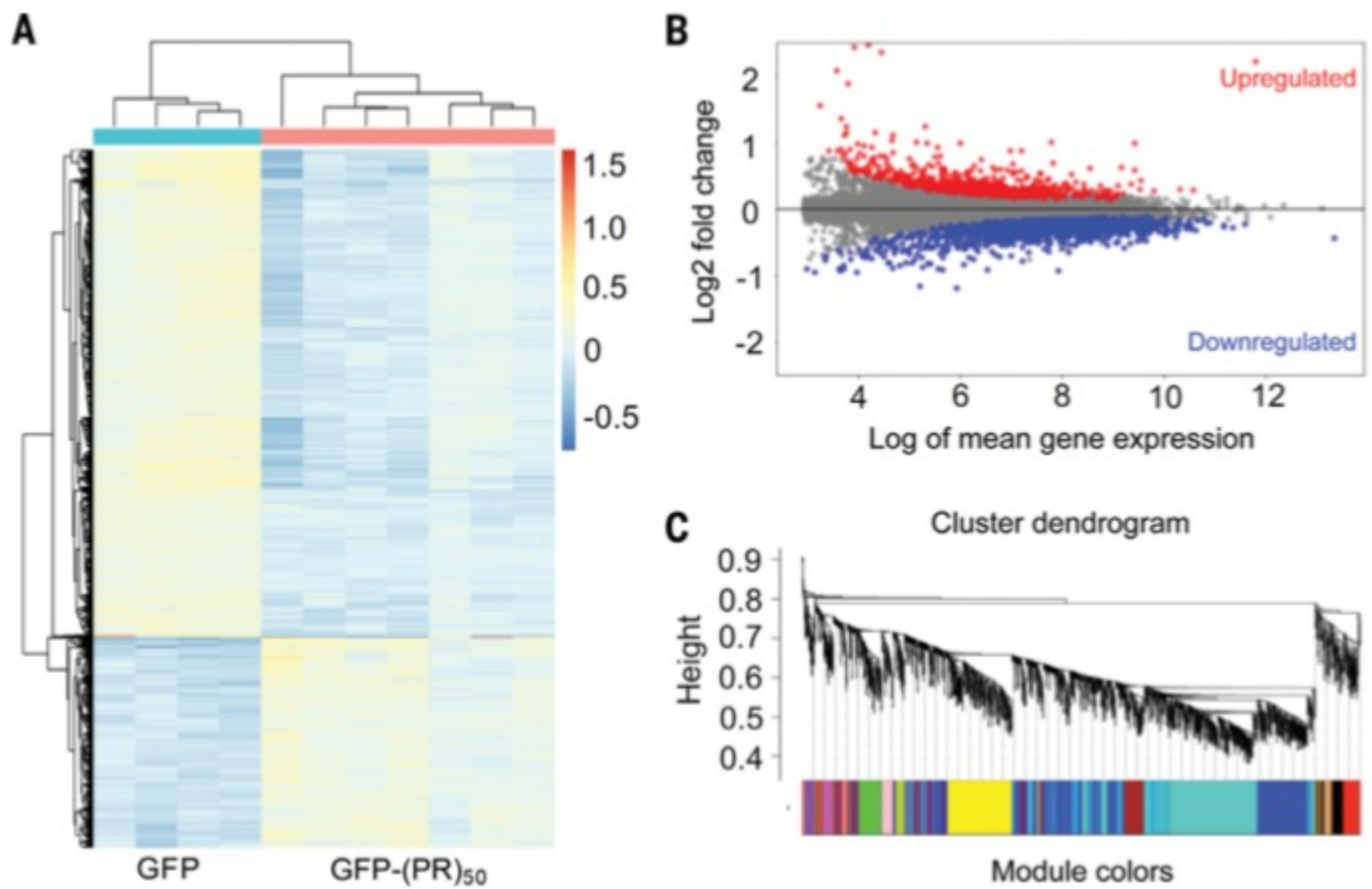
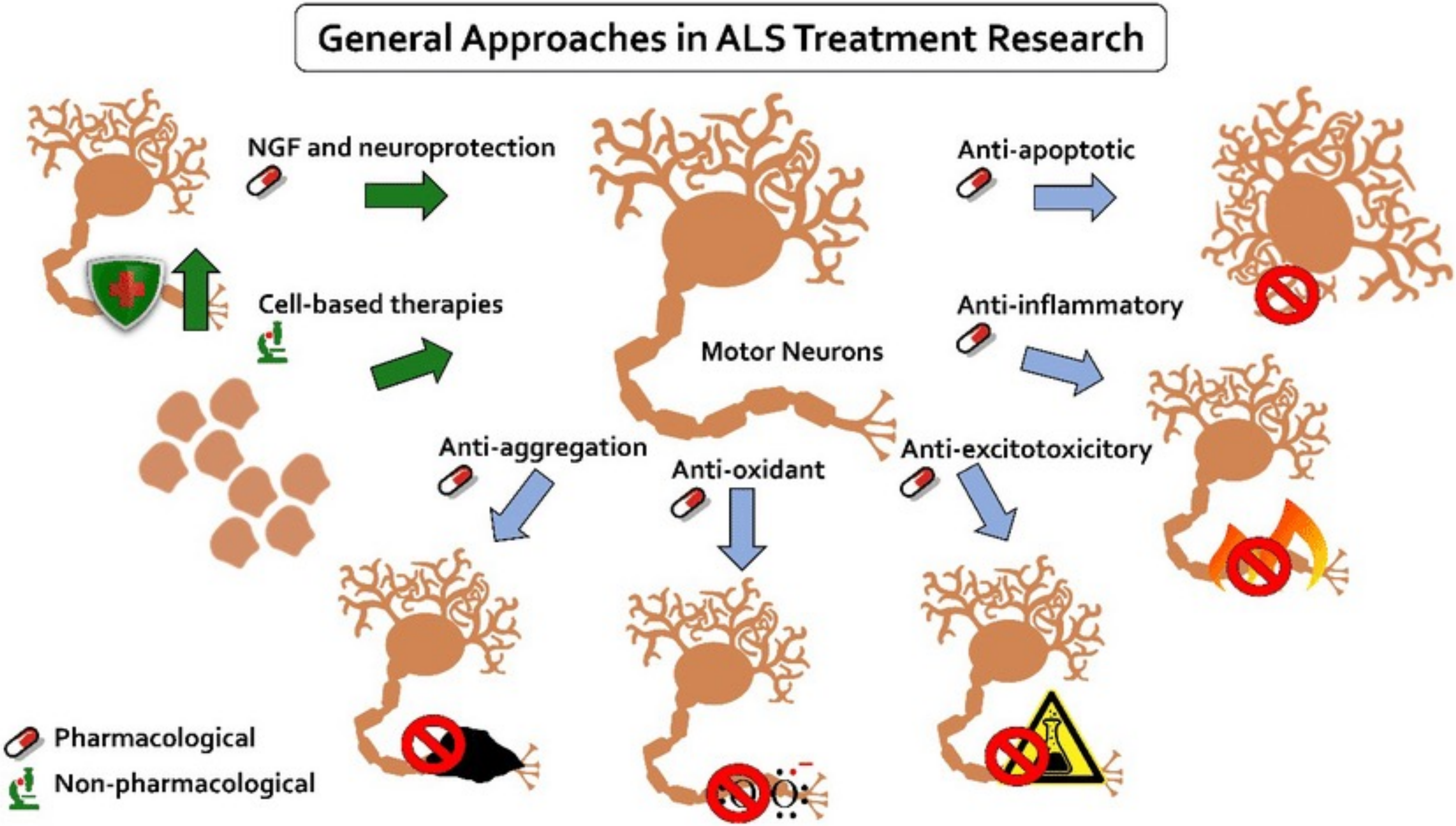


Fig. 4. Transcriptome alterations were identified in the brains of mice expressing GFP-(PR)$_{50}$. (**A**) Hierarchical clustering of the 1000 most variable genes between 3-month-old GFP mice (n = 4) and GFP-(PR)$_{50}$ mice (n = 7). (**B**) MA plots of up- and down-regulated genes (FDR < 0.01) in the cortices and hippocampi of 3-month-old mice expressing GFP-(PR)$_{50}$ (n = 7) compared with those of GFP controls (n = 4). The MA plot is based on the Bland-Altman plot, where M represents the log$_2$ fold change (y axis) and A represents the log of mean gene expression (x axis). (**C**) Gene modules identified in brains of 3-month-old mice expressing GFP-(PR)$_{50}$ (n = 7 mice) through weighted gene coexpression correlation network analyses using differentially expressed genes.


General Approaches in ALS Treatment Research
NGF and neuroprotection
Cell-based therapies
Anti-apoptotic
Anti-inflammatory
Motor Neurons
Anti-aggregation
Anti-oxidant
Anti-excitotoxicitory
Pharmacological
Non-pharmacological